# Multimodal Machine Learning

## Lecture 3.1: Multimodal Representation Fusion

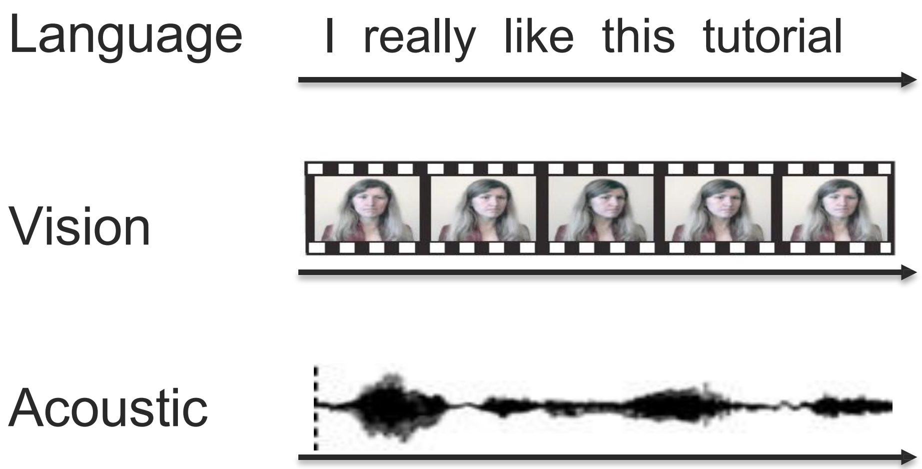
Dam Quang Tuan

# Lecture Objectives

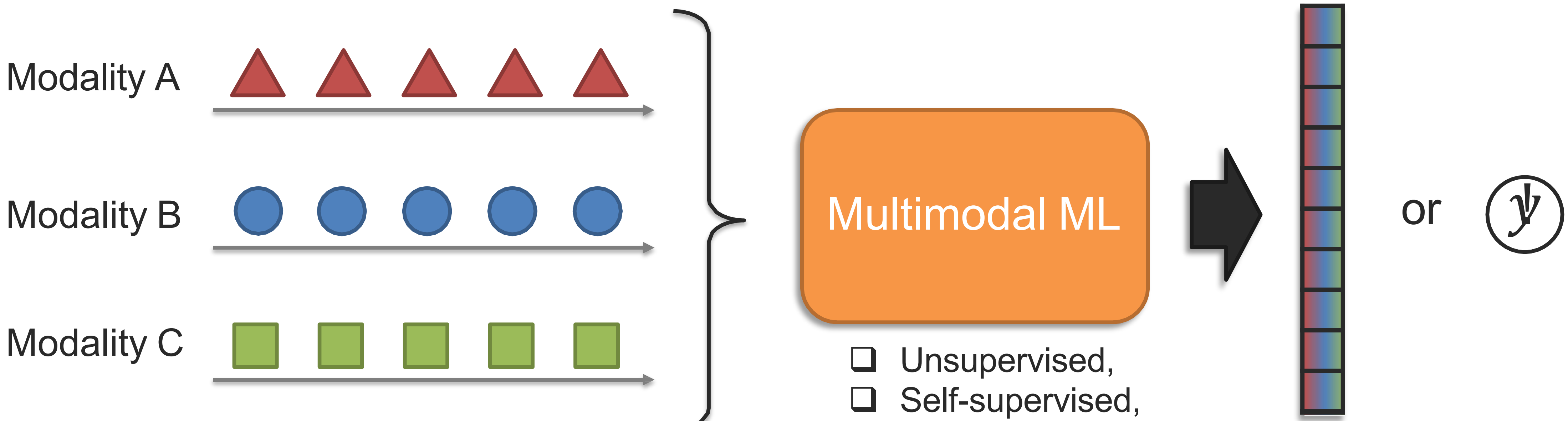
- Multimodal representations
  - Cross-modal interactions
- Representation fusion
  - Additive and multiplicative fusion
    - Tensor and polynomial fusion
  - Gated fusion
    - Modality-shift fusion
    - Dynamic fusion
  - Fusion on raw modalities
  - Heterogeneity-aware fusion
- Measuring non-additive interactions

# Multimodal Representation

# Multimodal Machine Learning



# Multimodal Machine Learning




- ❑ Unsupervised,
- ❑ Self-supervised,
- ❑ Supervised,
- ❑ Reinforcement,

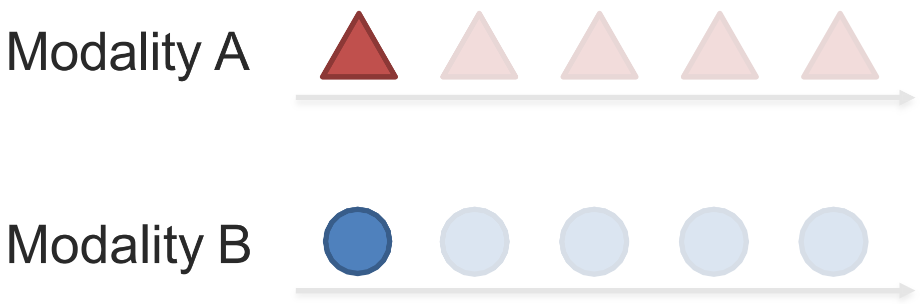
…

# Challenge 1: Representation

**Definition:** Learning representations that reflect cross-modal interactions between individual elements, across different modalities

➡ This is a core building block for most multimodal modeling problems!

**Individual elements:**




*It can be seen as a "local" representation*

or

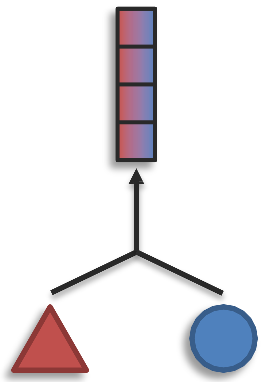
*representation using holistic features*

# Challenge 1: Representation

**Definition:** Learning representations that reflect cross-modal interactions between individual elements, across different modalities

## Sub-challenges:

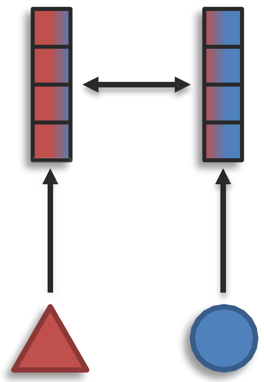
### Fusion



# modalities > # representations

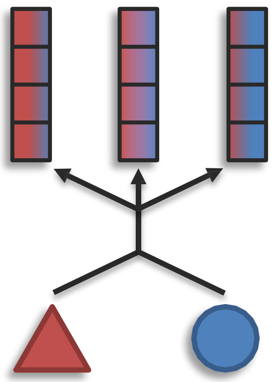
Today

### Coordination



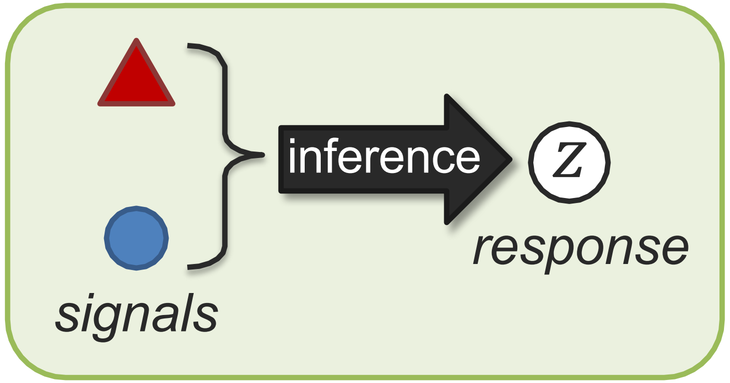
# modalities = # representations

### Fission



# modalities < # representations

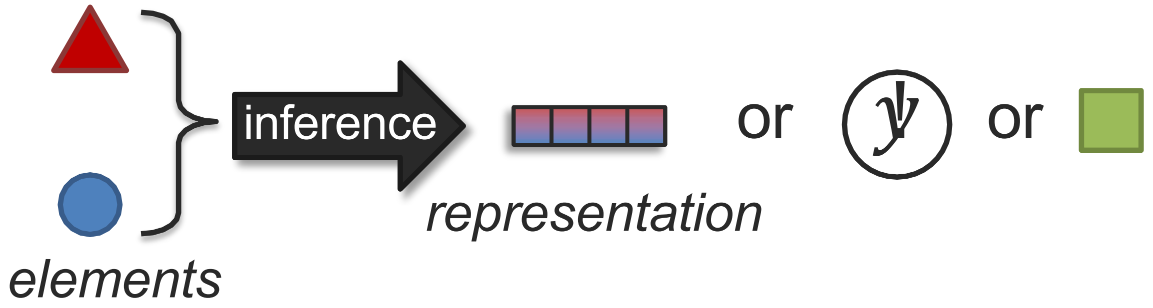
# Cross-modal Interactions




Interactions happen during inference!

"Inference" examples:

- Representation fusion
- Prediction task
- Modality translation



# Interacting Modalities

Interactions: How multimodal information changes when modalities are combined for a response.

***Is this indoors?***

inference

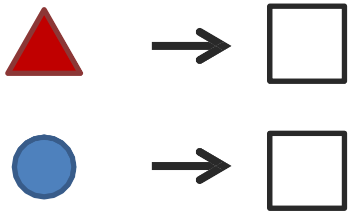
*Yes!*

*A teacup on the right of a laptop in a clean room.*

inference

*Yes!*

Redundant



# Interacting Modalities

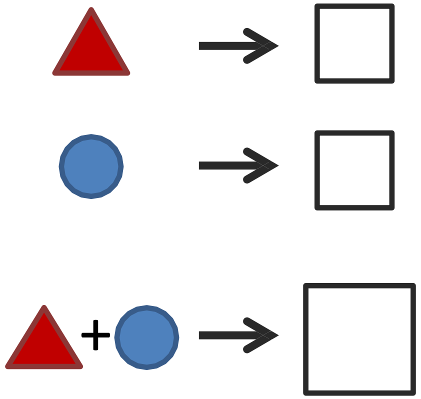
**Interactions:** How multimodal information changes when modalities are combined for a response.

***Is this indoors?***

*A teacup on the right of a laptop in a clean room.*

inference → ***Yes!***

Redundant



Enhancement

# Interacting Modalities

**Interactions:** How multimodal information changes when modalities are combined for a response.

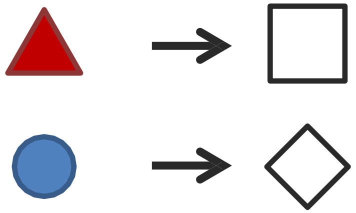
***Is this a living room?***

inference → *Yes!*

*A teacup on the right of a laptop in a clean room.*

inference → *No, probably study room.*

Non-redundant



# Interacting Modalities

**Interactions:** How multimodal information changes when modalities are combined for a response.

***Is this a living room?***

*A teacup on the right of a laptop in a clean room.*

inference → ***Yes!***

Non-redundant

Dominance

# Interacting Modalities

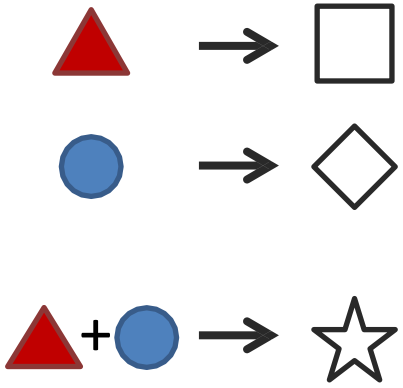
**Interactions:** How multimodal information changes when modalities are combined for a response.

***Should I work here?***

inference

*Maybe? Comfy sofa but table's too small.*

Non-redundant



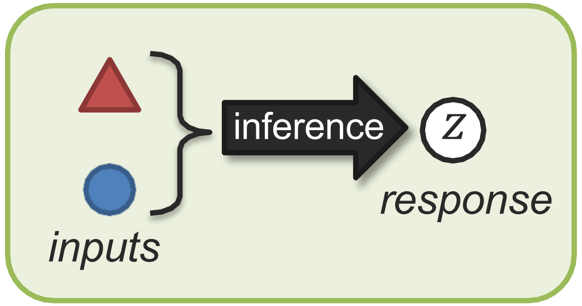

*A teacup on the right of a laptop in a clean room.*

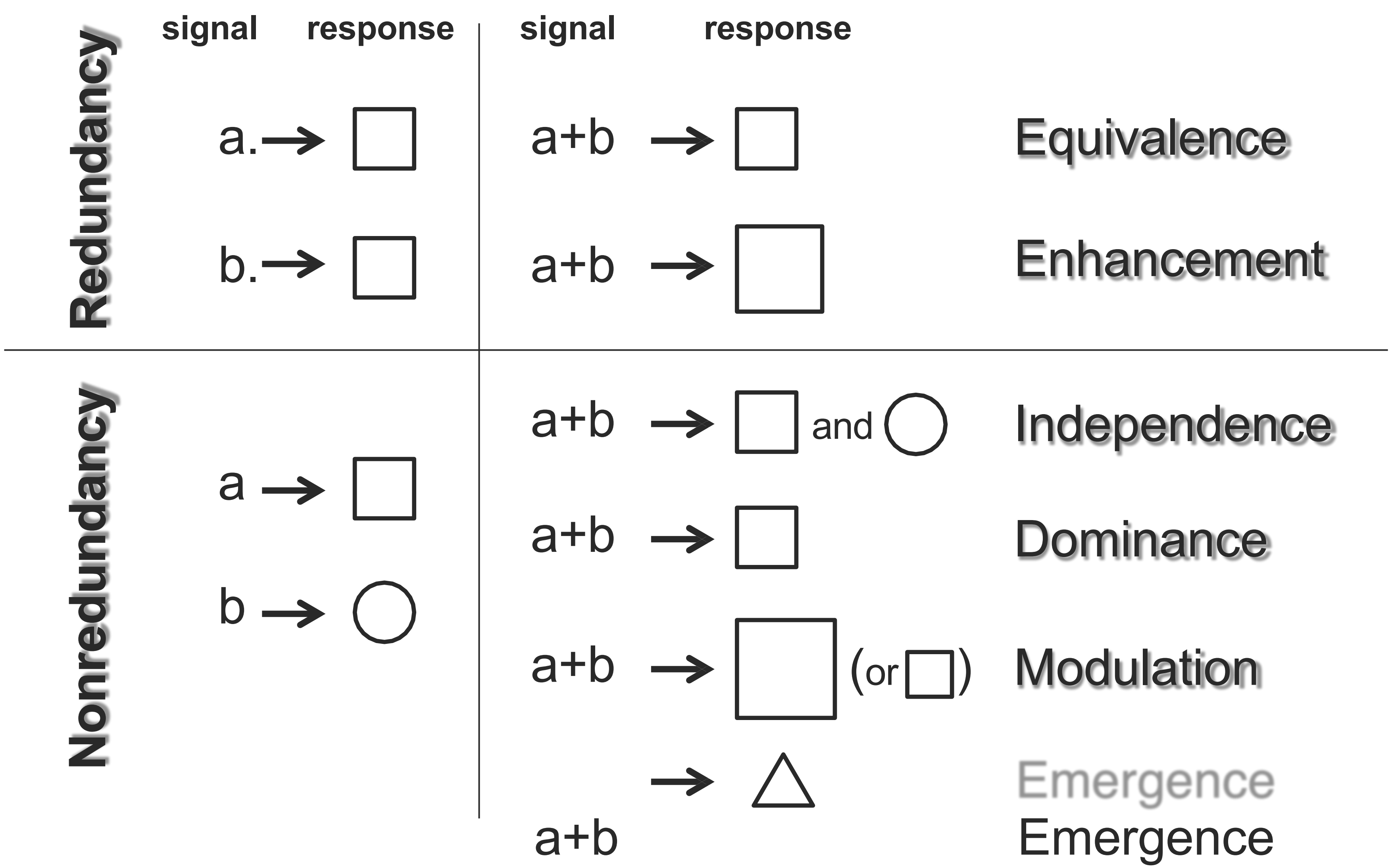
inference

*Maybe? Clean and there's tea.*

Emergence

# Taxonomy of Interaction Responses – A Behavioral Science View



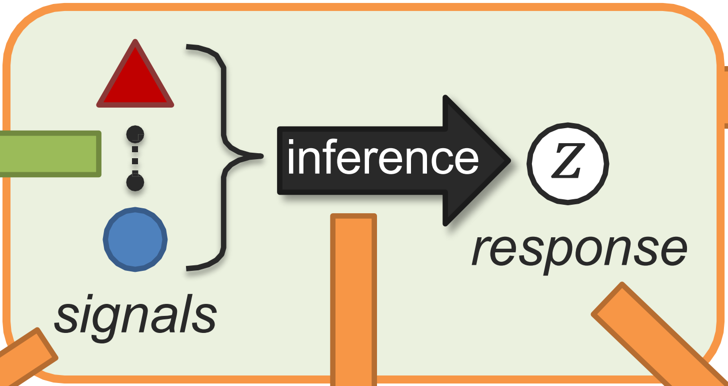




## Multimodal Communication



Partan and Marler (2005). *Issues in the classification of multimodal communication signals*. American Naturalist, 166(2)

# Cross-modal Interactions – A Taxonomy




## 5 Connections
- Association
- Dependency
- Correspondence
- Relationship
- ...

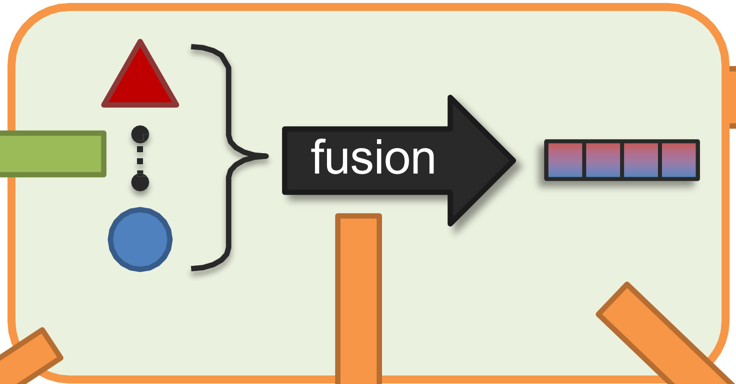
## 4 Context
- Structure context
- Task relevance
- Context dependence
- ...

## 2 Modalities
- Unimodal
- Bimodal
- Trimodal
- High-modal,
- ...

## 3 Interactions
- Additive
- Multiplicative
- Polynomial
- Gated
- Nonlinear
- …

## 1 Responses
- Redundancy
- Non-redundancy
- Dominance
- Emergence
- …

# Cross-modal Interactions – Representation Fusion




Next week

### 5 Connections

- Association
- Dependency
- Correspondence
- Relationship

...

### 4 Context

- Structure context
- Task relevance
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...

### 2 Modalities

- Unimodal
- Bimodal
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### 3 Interactions

- Additive
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…

Today

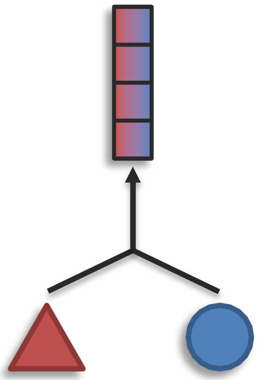
### 1 Responses

- Redundancy
- Non-redundancy
- Dominance
- Emergence

…

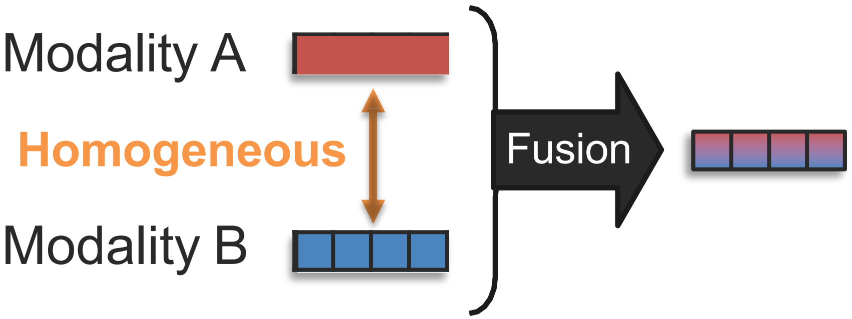
# Representation Fusion

# Sub-Challenge 1a: Representation Fusion

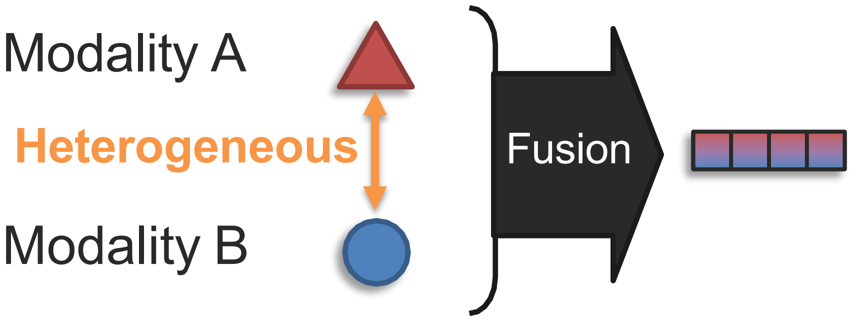


**Definition:** Learn a joint representation that models cross-modal interactions between individual elements of different modalities

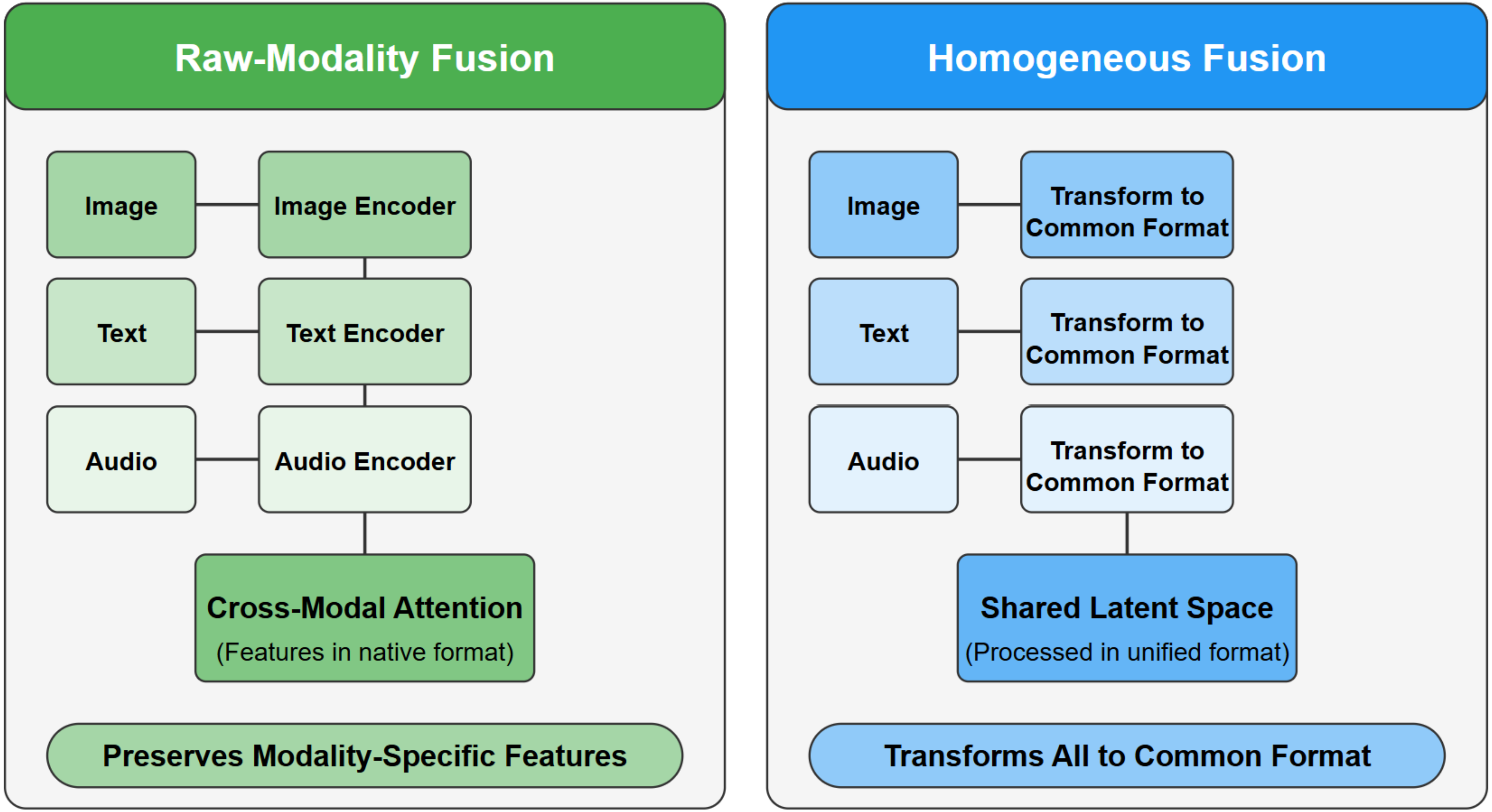
Basic fusion:




Raw-modality fusion:

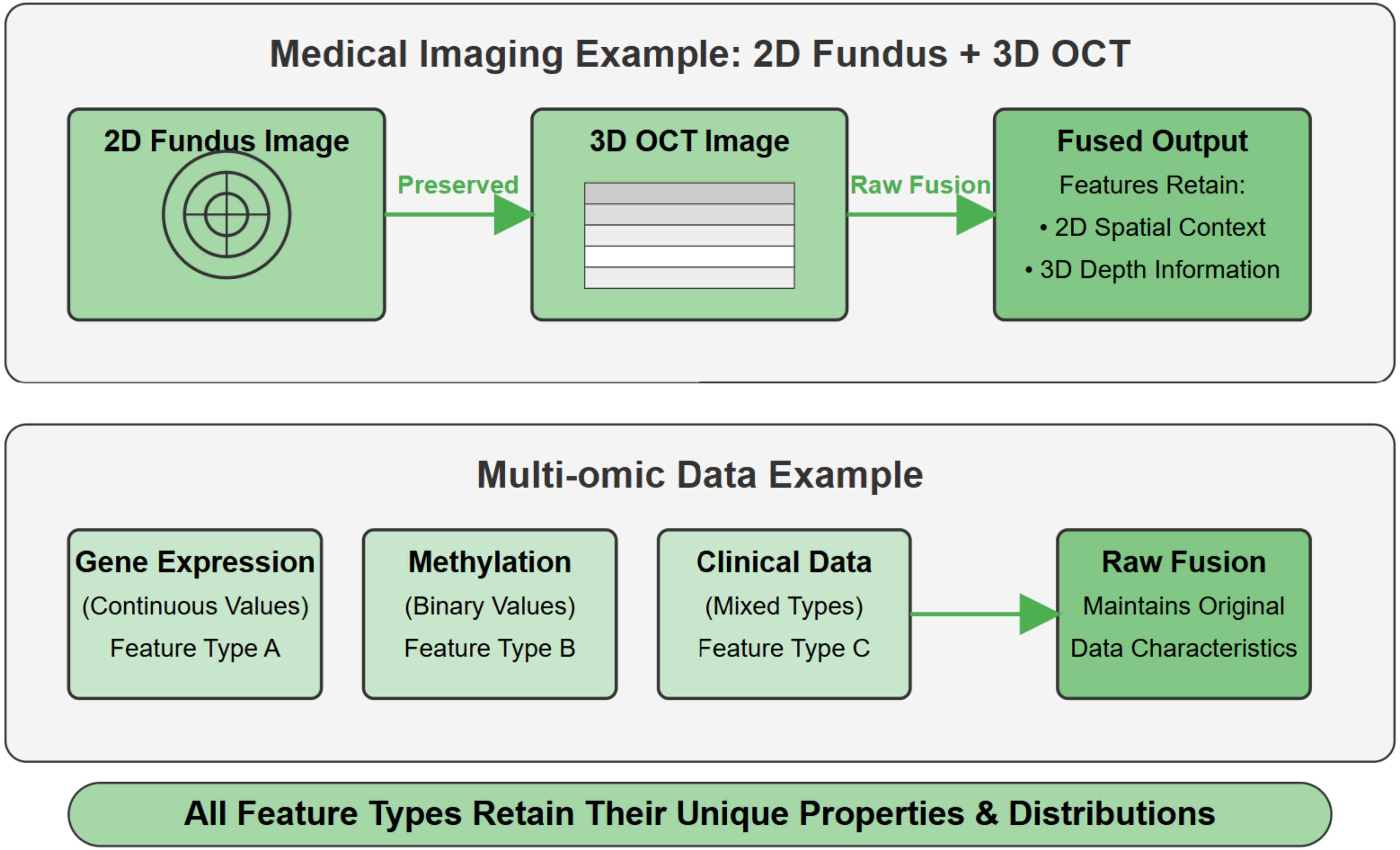


# Raw-Modality vs. Homogeneous Fusion Approaches



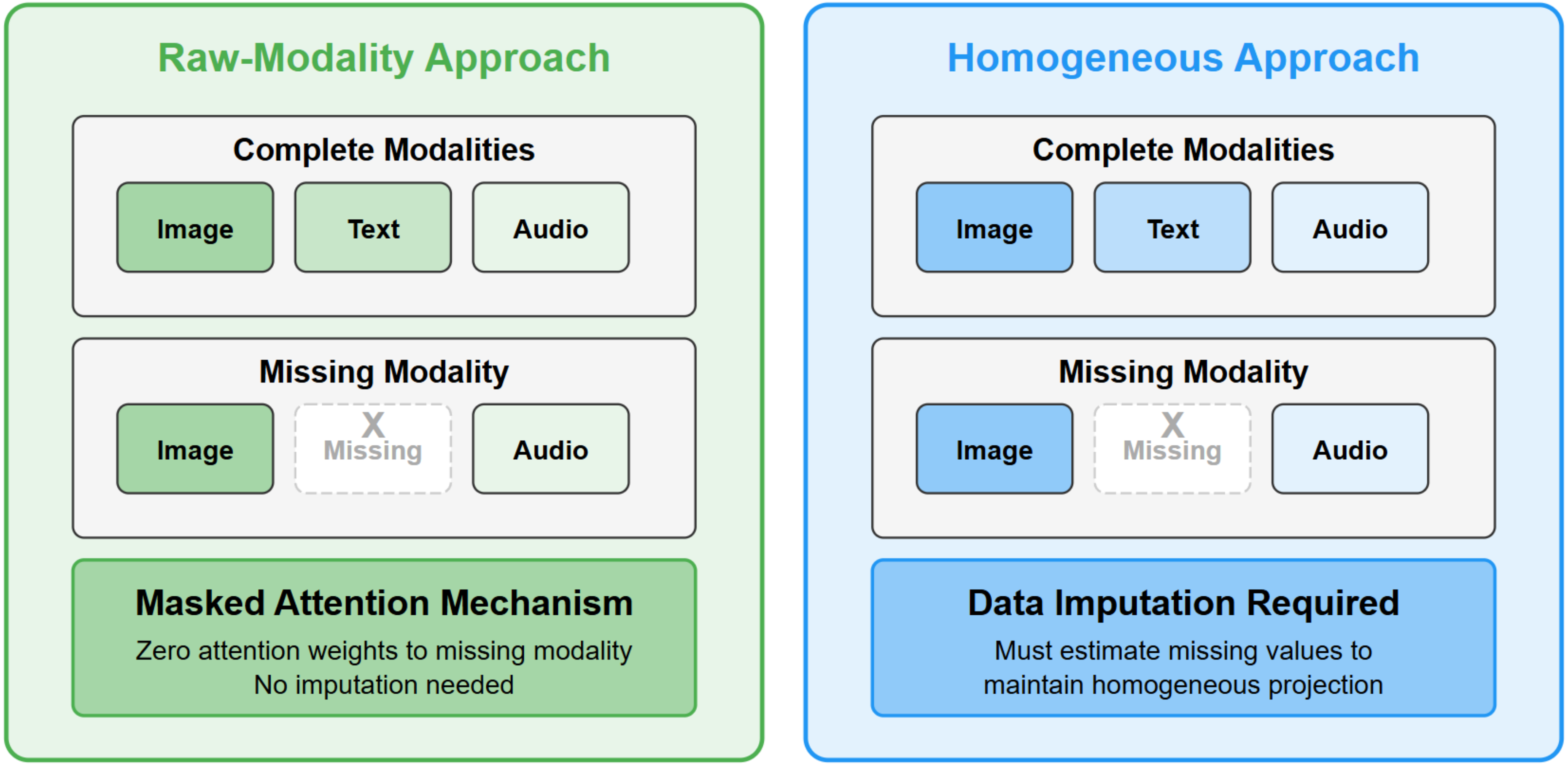
# Preservation of Modality-Specific Features

No Loss of Critical Information



# Adaptability to Missing Modalities

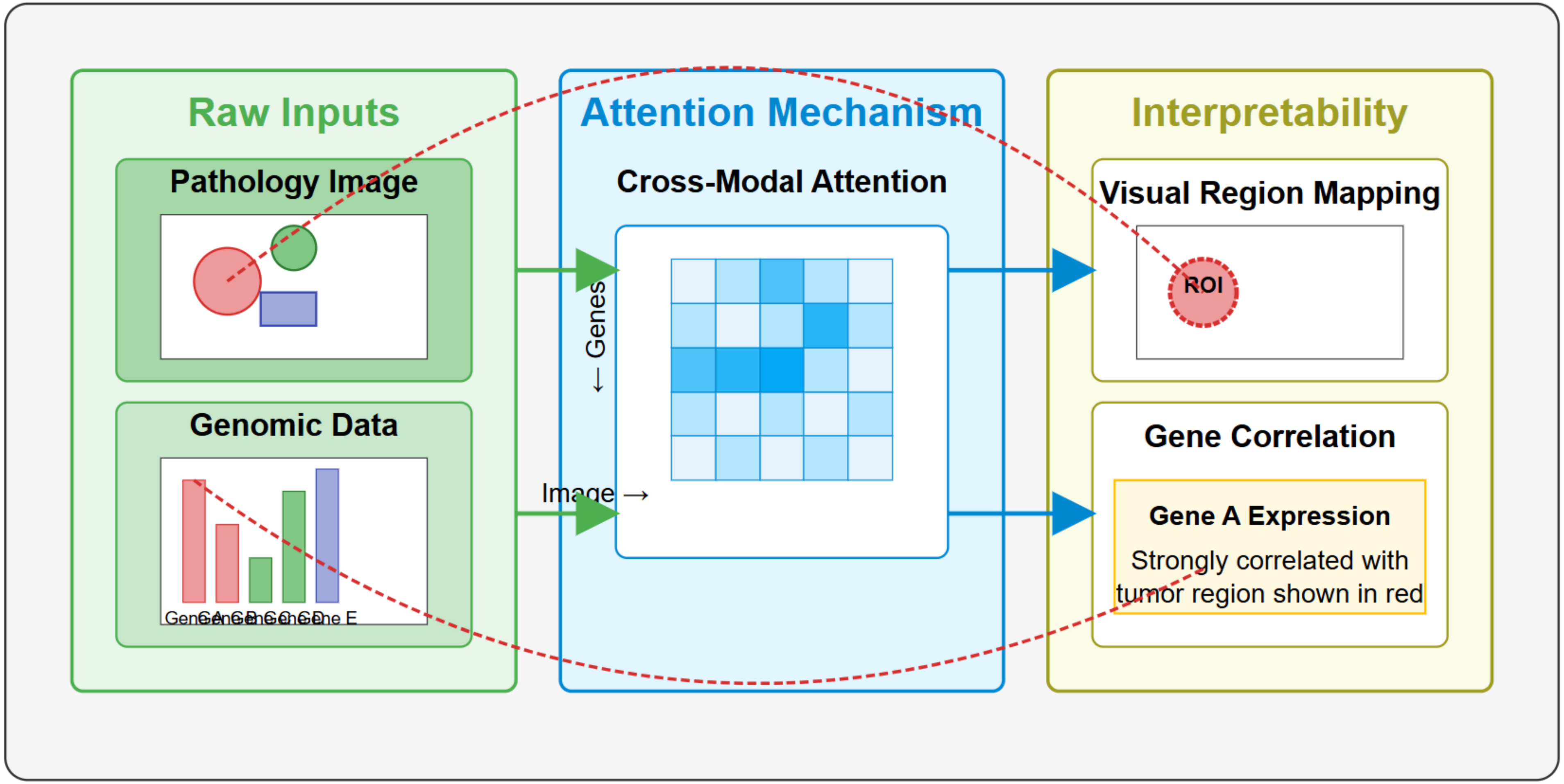
Raw-Modality Fusion vs. Homogeneous Fusion




**Raw-Modality Fusion:** Maintains >90% accuracy with 40% missing data

# Enhanced Interpretability

Preserves Original Feature Spaces




**Direct visualization of how specific image regions relate to genomic features**

# When to Use Raw-Modality Fusion

Summary of Key Scenarios and Benefits

| Scenario | Reason to Choose Raw-Modality Fusion |
| --- | --- |
| Heterogeneous Data Formats | Handles data with different structures (e.g., text + image) more effectively |
| Preservation of Features | Retains modality-specific details (critical in medical imaging, genomics, etc.) |
| Missing Data | Flexible handling of incomplete inputs via masking or attention mechanisms |
| Interpretability | Clear insights into cross-modal interactions at the raw feature level |
| Low Computational Resources | Avoids heavy transformations or shared latent space overhead |

**Raw-Modality Fusion excels when flexibility and preserving original data characteristics are paramount**

# When to Use Homogeneous Fusion

Key Scenarios and Benefits of Homogeneous Approaches

| Scenario | Reason to Choose Homogeneous Fusion |
|---|---|
| **Unified Representation** | When a common representation space is beneficial for downstream tasks |
| **Similar Modality Structures** | When modalities share similar underlying structures (e.g., multiple image types) |
| **Pre-trained Embeddings** | When leveraging pre-trained embeddings that already map into vector spaces |
| **Complete Data Availability** | When all modalities are consistently available with little to no missing data |
| **Sufficient Computational Resources** | When transformation overhead is not a concern and advanced projection models are available |

**Homogeneous Fusion excels when creating a unified representation is more important than preserving modality-specific features**

# Example: HMFI for 3D Object Detection

*Li et al., "Homogeneous Multi-modal Feature Fusion and Interaction for 3D Object Detection", ECCV 2022*

## Motivation

Camera images and LiDAR point clouds have very different structures:

- Camera: dense 2D pixel grids (appearance, color, texture)
- LiDAR: sparse 3D point clouds (geometry, depth, shape)

Direct fusion is hard because the representations are heterogeneous.

**Key idea:** Transform both modalities into a shared 3D voxel space before fusion.

## Three-stage pipeline

### 1. Structure alignment

Image Voxel Lifter Module (IVLM) converts 2D image features into 3D voxel space using depth bins from the point cloud.

### 2. Feature fusion

Query Fusion Module (QFM) fuses the now-homogeneous voxel features from both modalities via transformer attention.

### 3. Object-level interaction

Voxel Feature Interaction Module (VFIM) aligns features at the proposal level with a similarity objective.

# HMFI Architecture Overview

## Sub-Challenge 1a: Representation Fusion

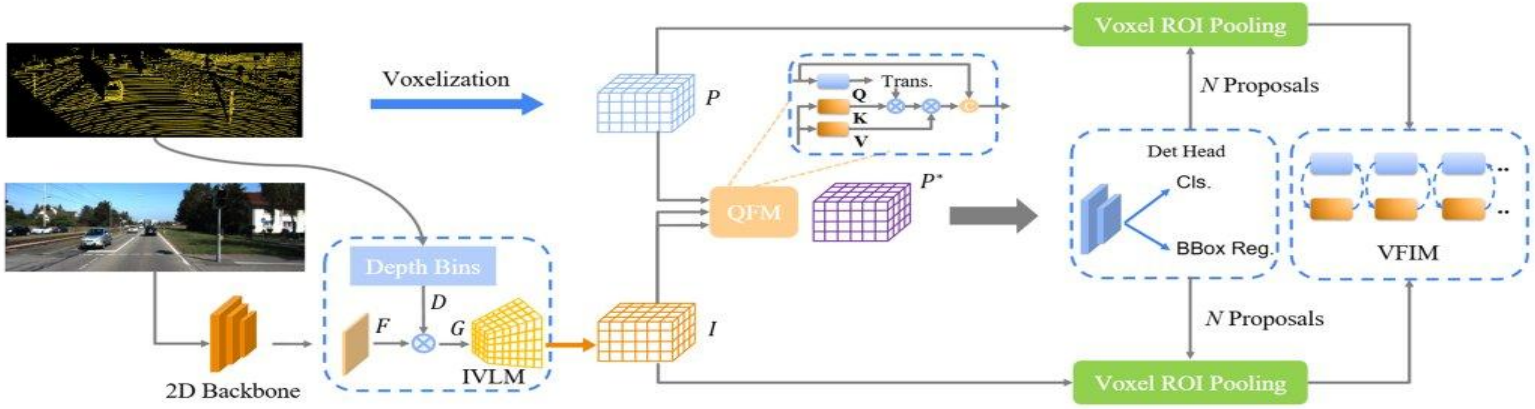



Fig. 2: The architecture of HMFI. Each image is processed by a 2D backbone network and fed into an image voxel lifter module (IVLM) to produce a homogeneous structure based on the depth bins transformed by the point cloud. Then, the processed homogeneous image and point cloud features are fused by the query fusion mechanism (QFM). Next, a voxel-based object detector is employed on fused features to produce 3D detection results. Finally, the voxel feature interaction module (VFIM) conducts feature interaction at object-level based on the detection results to improve semantic consistency in these two homogeneous cross-modal features.



Li et al., "Homogeneous Multi-modal Feature Fusion and Interaction for 3D Object Detection", ECCV 2022

# Multi-Stage Fusion & Key Takeaways

## Sub-Challenge 1a: Representation Fusion

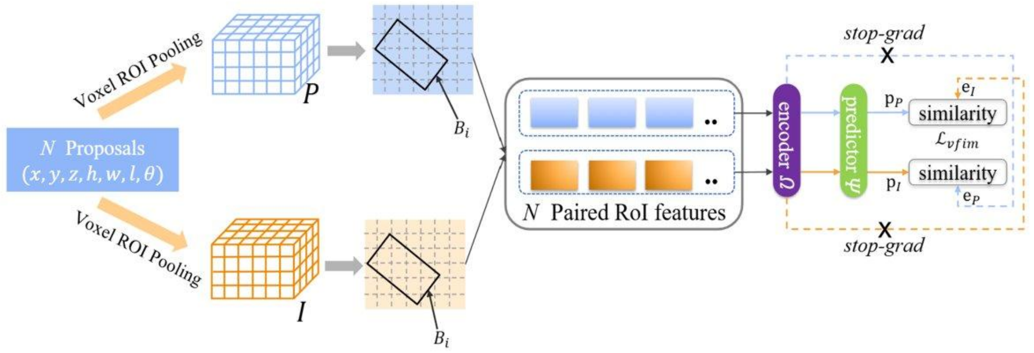



Fig. 4: The voxel feature interaction module. We use the voxel RoI pooling to extract features in these two homogeneous features according to the predicted proposals and form the paired RoI feature set. Then we adopt feature interaction between each pair of RoI features based on the symmetry similarity constraint loss to improve the object-level semantic consistency between two homogeneous representations

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*Fig. 4: Voxel Feature Interaction Module (VFIM)*

### Key takeaways

**Fusion can be multi-stage**

Global fusion (QFM) generates proposals; object-level fusion (VFIM) refines them.

**Structure alignment before fusion**

Making modalities structurally compatible (IVLM) is often harder than the fusion operator itself.

**Focused interaction is easier to learn**

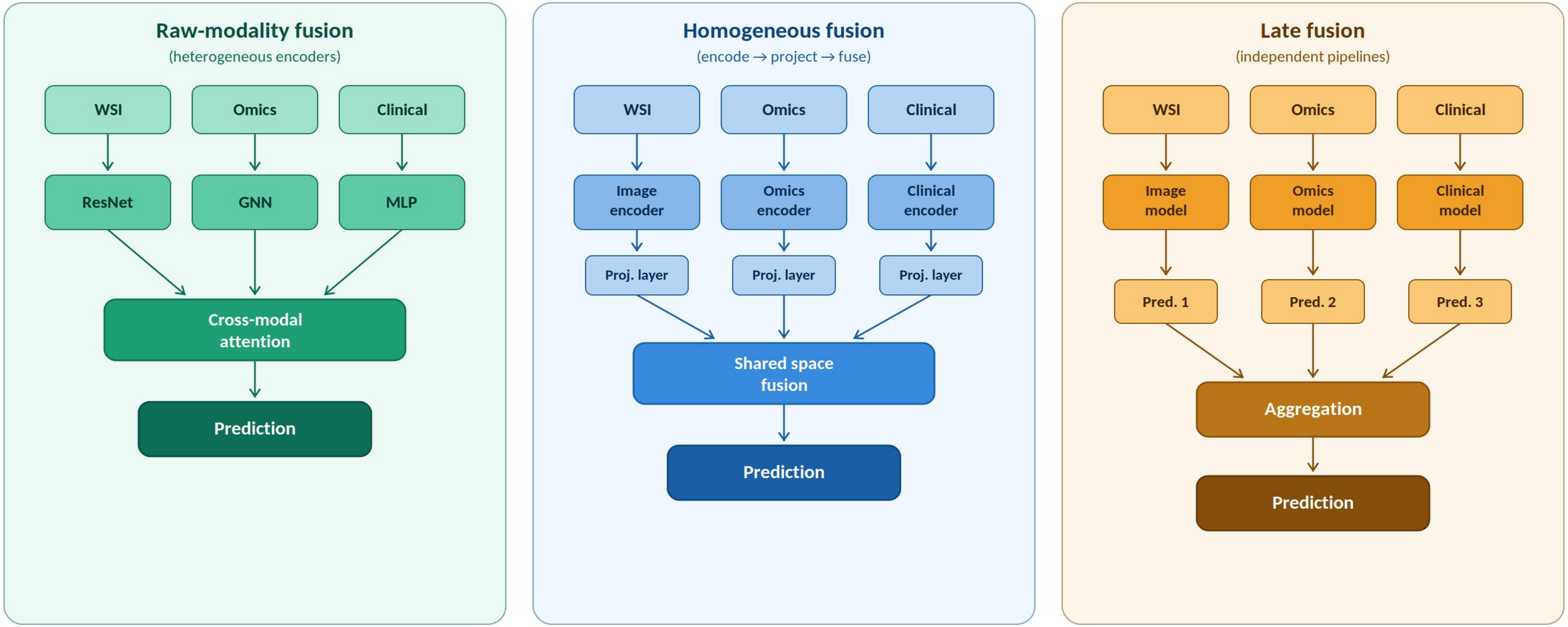
Object proposals narrow the search space for cross-modal alignment.

**Similarity as supervision**

VFIM uses symmetric similarity loss to enforce cross-modal consistency — no extra labels needed.

# Sub-Challenge 1a: Representation Fusion

## Raw-modality fusion vs. other approaches




**Key attributes** (Raw-modality fusion)

- Modality-specific encoders (ResNet, GNN, ...)
- Cross-modal interaction on rich features
- Hardest to optimize, highest potential
- Can mask missing modalities in attention

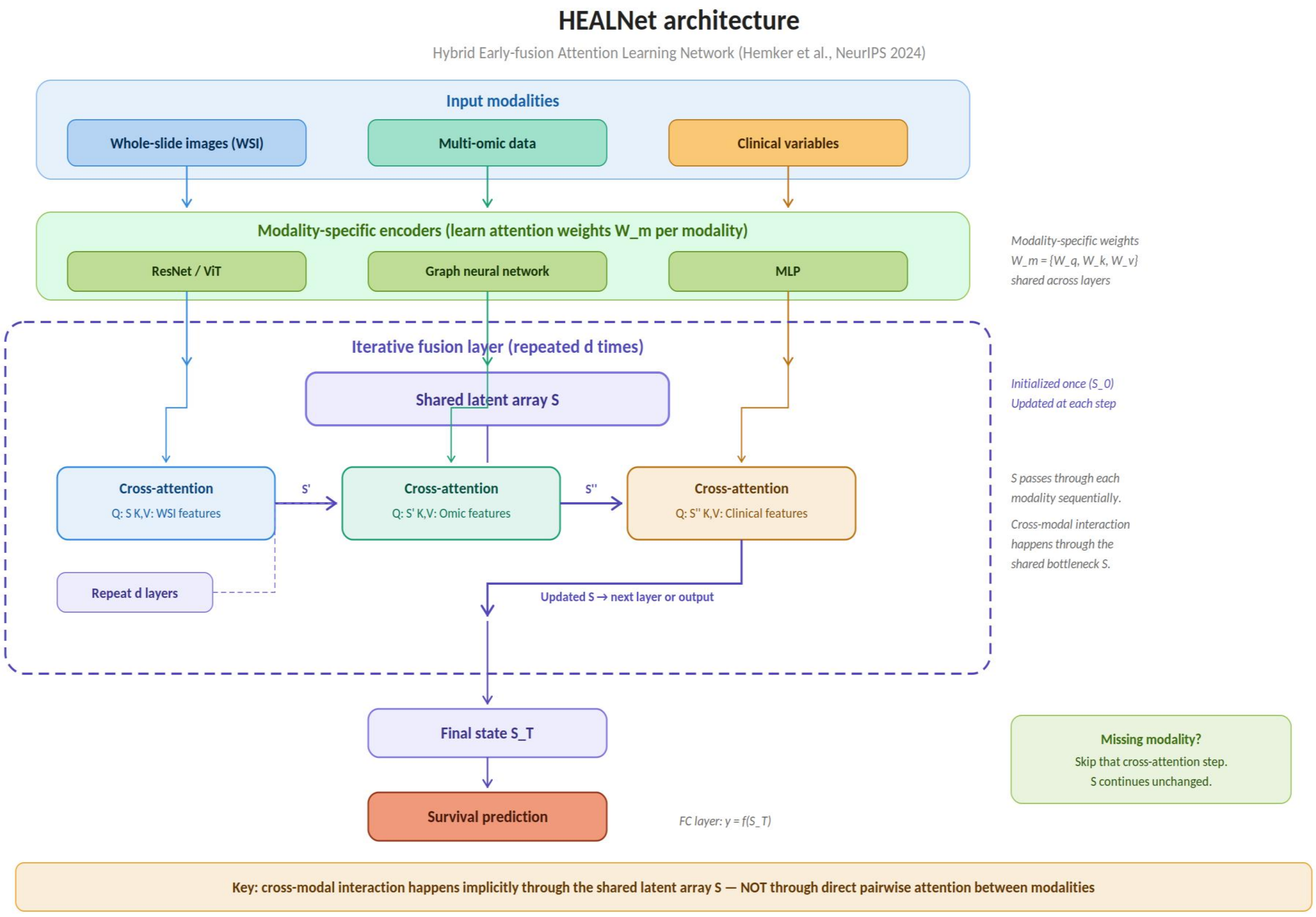
**Key attributes** (Homogeneous fusion)

- Each modality encoded then projected
- Fusion on aligned, homogeneous embeddings
- Practical middle ground
- Modular: swap encoders independently

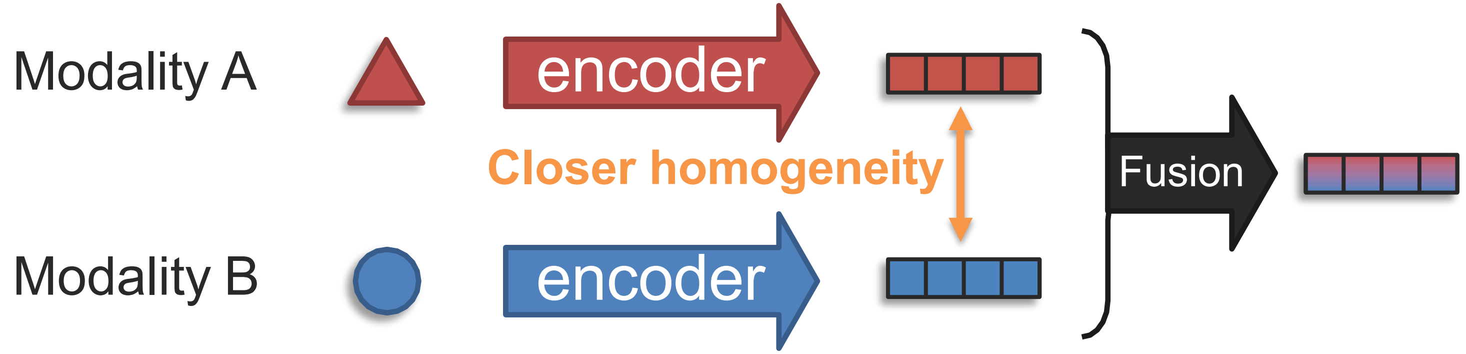
**Key attributes** (Late fusion)

- Fully independent modality pipelines
- Decision-level aggregation only
- Most robust to missing modalities
- No joint feature-level reasoning

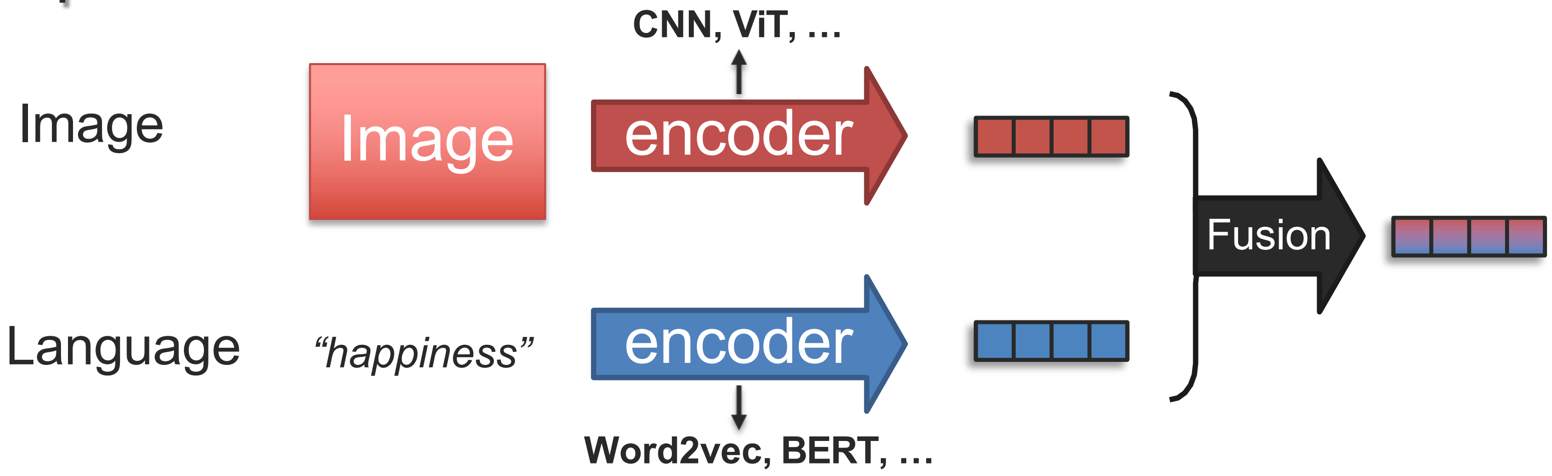
# Sub-Challenge 1a: Representation Fusion



# Fusion with Unimodal Encoders




Example:

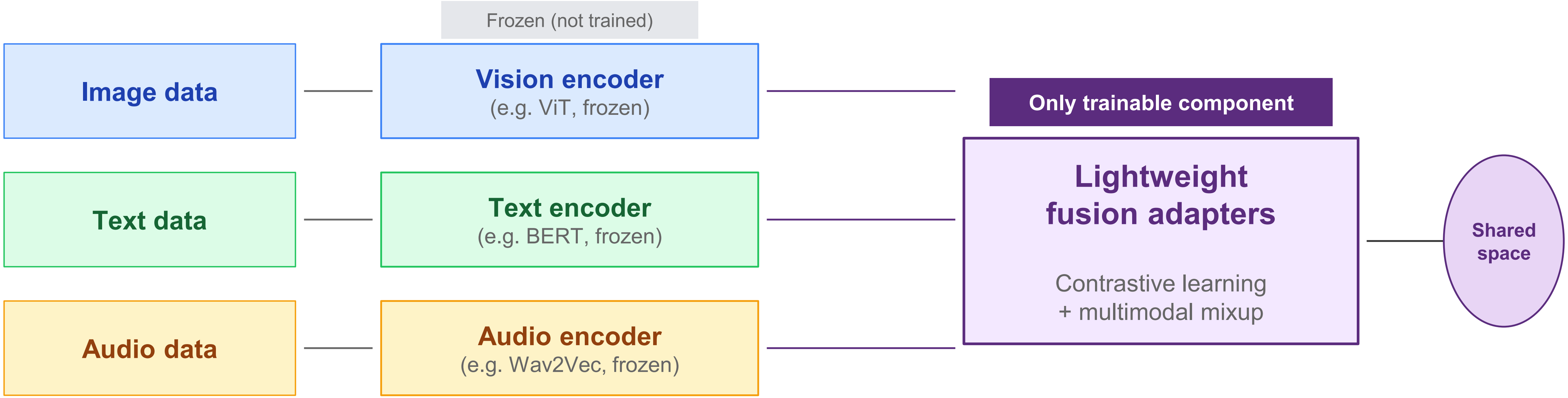



Unimodal encoders can be jointly learned with fusion network, or pre-trained

# FuseMix: Data-Efficient Multimodal Fusion

*Vouitsis et al., "Data-Efficient Multimodal Fusion on a Single GPU", CVPR 2024*

**Core idea:** Keep pre-trained unimodal encoders frozen. Train only lightweight adapters to align latent spaces via contrastive learning + multimodal mixup.



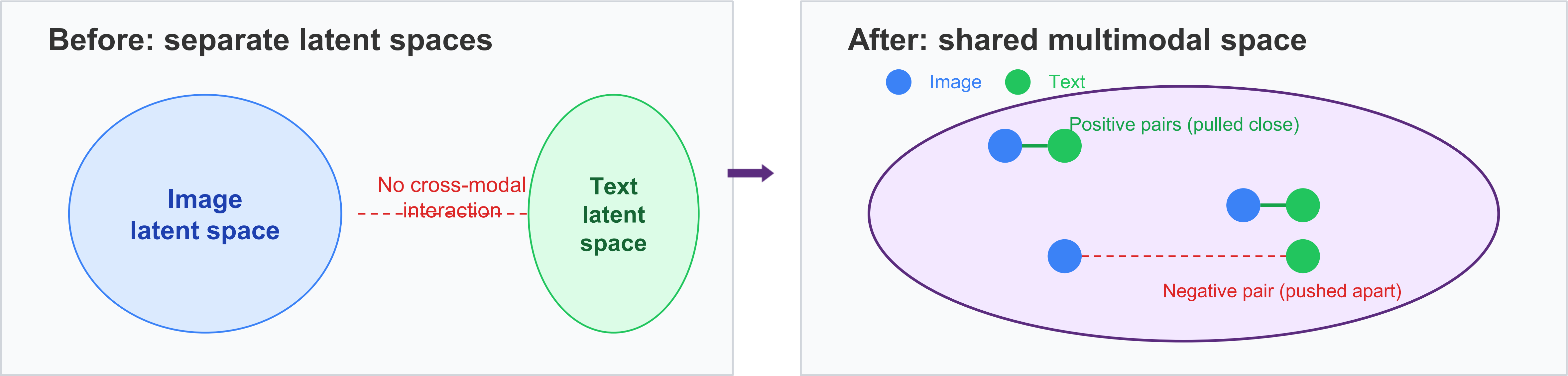

**600x** fewer GPU days

**80x** fewer data pairs

**1 GPU** is enough

# Contrastive Latent Space Alignment

*How FuseMix creates a shared multimodal representation space*




## How it works

**1. Contrastive objective:** For each image-text pair, pull their adapter outputs close; push non-matching pairs apart (InfoNCE loss).

**2. Multimodal mixup:** Interpolate between latent representations of different pairs to create synthetic training examples — key to data efficiency.

# Computational Efficiency

*FuseMix achieves competitive performance with orders-of-magnitude less compute and data*

| Method | GPU days | Training pairs | Hardware | Flickr30K R@1 |
|---|---|---|---|---|
| **FuseMix** | **~1** | **~50K** | **1 GPU** | **Competitive** |
| CLIP | ~600 | ~400M | 256 V100s | Strong |
| Other SotA | 1000+ | 100M+ | Multi-node | Strong |

### How does FuseMix achieve this?

**Frozen encoders:**
No backprop through large models

**Lightweight adapters:**
Only ~1-2M trainable parameters

**Multimodal mixup:**
Synthetic augmentation reduces data needs

**Contrastive learning:**
Efficient alignment without generative overhead

### Teaching lesson

Two separate questions in multimodal ML:

**1. Can the model represent cross-modal interaction?**
(expressiveness of the fusion operator)

**2. Can we train it under realistic constraints?**
(data efficiency, compute budget, single GPU)

*FuseMix argues question 2 deserves much more attention than it usually gets.*

# Plug-and-Play Versatility

*The same principle works across modality pairs and downstream tasks*

## Image-text retrieval

Outperforms CLIP on Flickr30K with significantly less training data.

**Setup:**

ViT (frozen) + BERT (frozen) + adapter

## Audio-text retrieval

Swap image encoder with audio encoder (e.g. Wav2Vec). Keep the same adapter architecture and training procedure.

**Setup:**

Wav2Vec (frozen) + BERT (frozen) + adapter

## Generative applications

Once audio and text share a space, text-to-image generators can be repurposed for audio-to-image generation with no retraining.

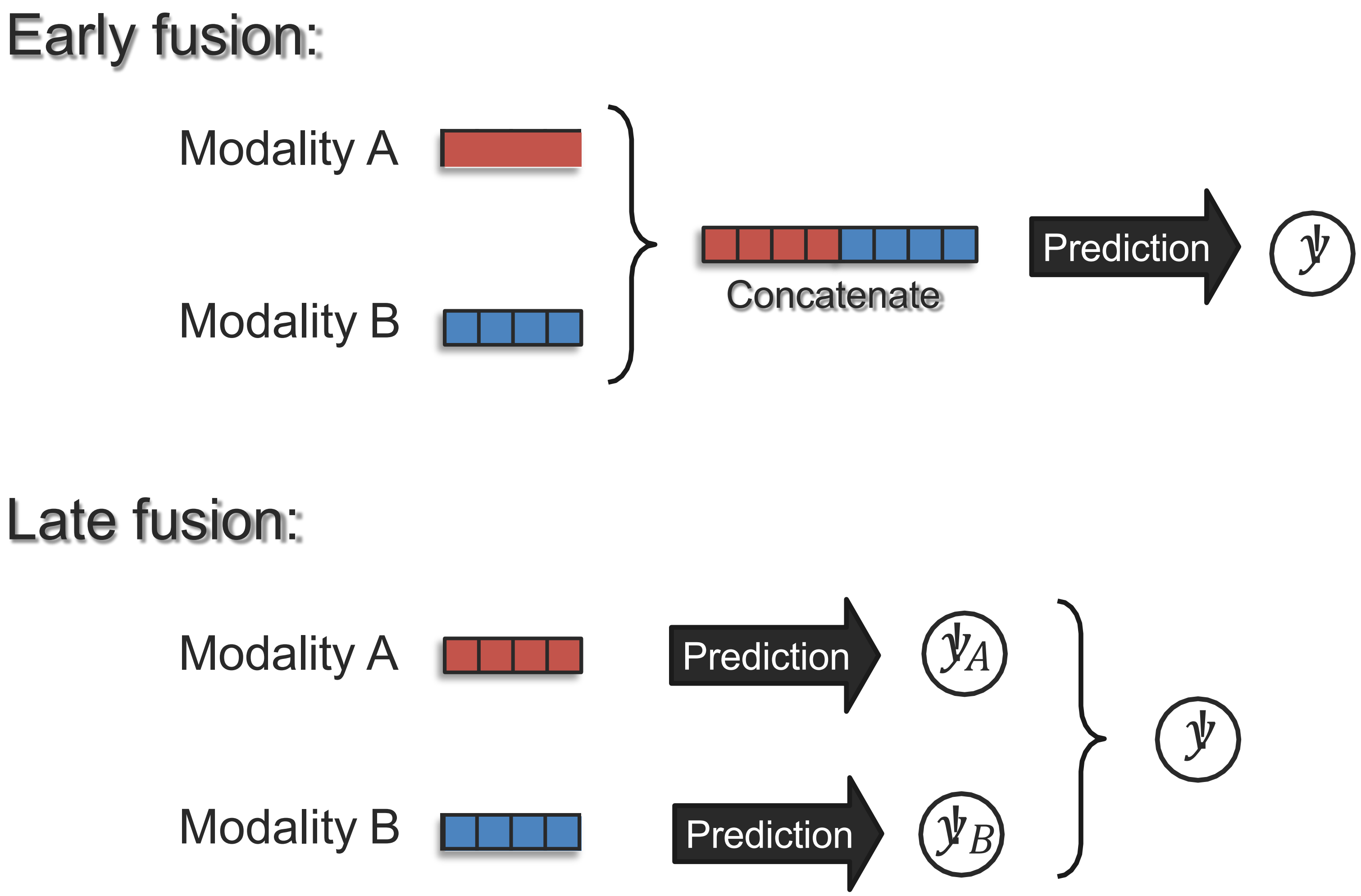
**Key:**

Aligned spaces enable zero-shot transfer to new tasks

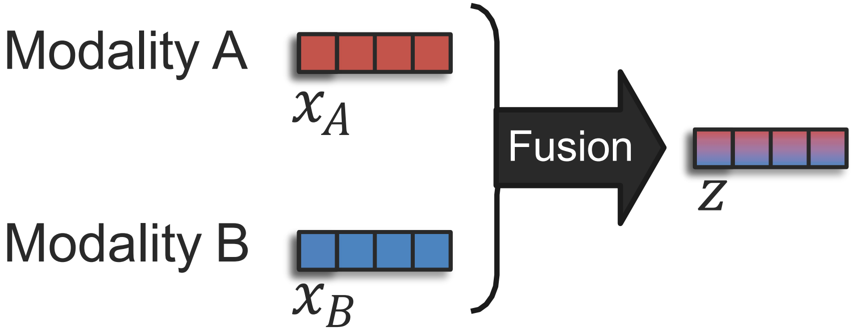
## Key takeaway

A good fusion idea should transfer across tasks and modality pairings. If the mechanism captures a useful principle — here, efficient latent alignment — it should generalize with minimal redesign. **FuseMix demonstrates this: same architecture, same training, different encoders.**

# Early and Late Fusion – A historical View



# Basic Concepts for Representation Fusion (aka, Basic Fusion)




**Goal:** Model *cross-modal interactions* between the multimodal elements

Let's study the univariate case first

(only 1-dimensional features)

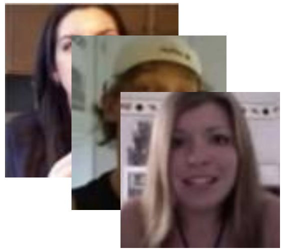
Linear regression:

$$z = \underbrace{w_0}_{\substack{\text{intercept} \\ \text{(bias term)}}} + \underbrace{w_1 x_A + w_2 x_B}_{\substack{\text{Additive} \\ \text{terms}}} + \underbrace{w_3 (x_A \times x_b)}_{\substack{\text{Multiplicative} \\ \text{term}}} + \underbrace{\epsilon}_{\substack{\text{error} \\ \text{(residual term)}}}$$

# Linear Regression

Linear regression is used to test research hypotheses, over a whole dataset

300 book reviews



$y$: audience score

$x_A$: percentage of smiling

$x_B$: professional status
(0=non-critic, 1=critic)

***H1:** Does smiling reveal what the audience score was?*

***H2:** Does the effect of smiling depend on professional status?*

Linear regression:

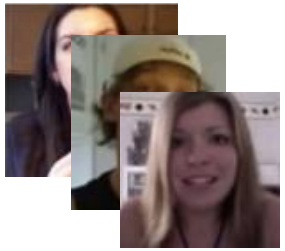
$$y = \underset{\substack{\downarrow \\ \text{intercept} \\ \text{(bias term)}}}{w_0} + \underbrace{w_1 x_A + w_2 x_B}_{\substack{\text{Additive} \\ \text{terms}}} + \underbrace{w_3 (x_A \times x_b)}_{\substack{\text{Multiplicative} \\ \text{term}}} + \underset{\substack{\downarrow \\ \text{error} \\ \text{(residual term)}}}{\epsilon}$$

- $w_0$: average score when $x_A$ and $x_B$ are zero
- $w_1$: effect from $x_A$ variable only
- $w_2$: effect from $x_B$ variable only
- $w_3$: effect from $x_A$ and $x_B$ interaction only
- $\epsilon$: residual not modeled by $w_0$, $w_1$, $w_2$ or $w_3$

# Linear Regression

Linear regression is used to test research hypotheses, over a whole dataset

300 book reviews



$y$: audience score

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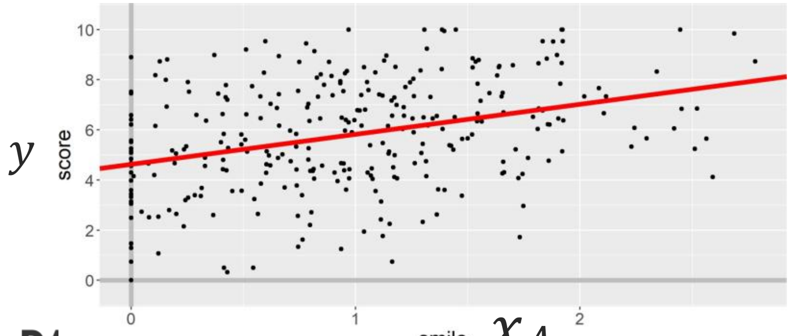
***H1:** Does smiling reveal what the audience score was?*

***H2:** Does the effect of smiling depend on professional status?*

Linear regression:

$$z = w_0 + \boxed{w_1} x_A + \epsilon$$

slope




Confidence interval: *"95% confident that $w$ parameter is contained within this interval"*

| | Estimate | 95% CI |
|---|---|---|
| $w_0$ | 4.63 | [4.20, 5.06] |
| $w_1$ | 1.20 | [0.83, 1.57] |

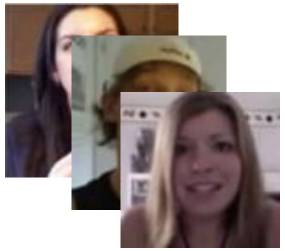
Confidence interval does not contain 0, so effect is significant

p-values would be another way to test hypothesis

# Linear Regression

Linear regression is used to test research hypotheses, over a whole dataset

300 book reviews



$y$: audience score

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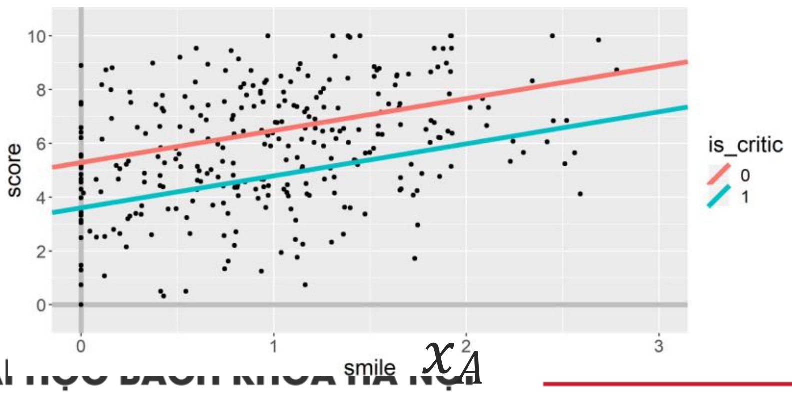
*H1: Does smiling reveal what the audience score was?*

*H2: Does the effect of smiling depend on professional status?*

Linear regression:

$$z = w_0 + \boxed{w_1} x_A + \boxed{w_2} x_B + \epsilon$$

$y$




$x_A$

| | Estimate | 95% CI |
|---|---|---|
| $w_0$ | 5.29 | [4.86, 5.73] |
| $w_1$ | 1.19 | [0.85, 1.53] |
| $w_2$ | −1.69 | [−2.14, −1.24] |

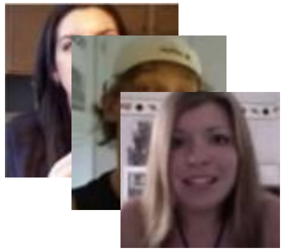
Positive effect

Negative effect

# Linear Regression

Linear regression is used to test research hypotheses, over a whole dataset

300 book reviews



$y$: audience score

$x_A$: percentage of smiling

$x_B$: professional status (0=non-critic, 1=critic)

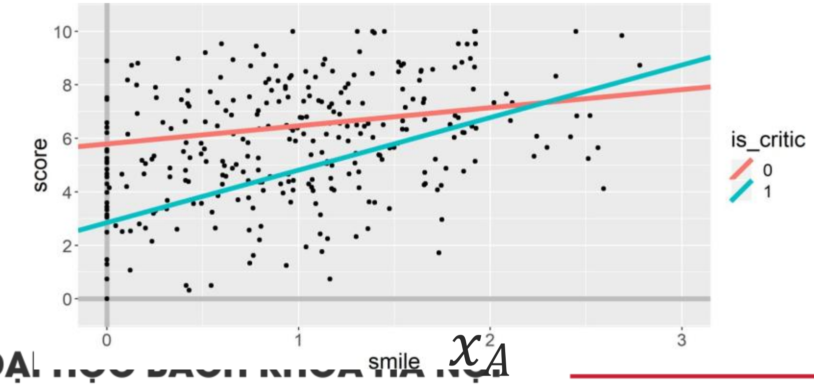
***H1:*** *Does smiling reveal what the audience score was?*

***H2:*** *Does the effect of smiling depend on professional status?*

Linear regression:

$$z = w_0 + \boxed{w_1} x_A + \boxed{w_2} x_B + \boxed{w_3} (x_A \times x_b) + \epsilon$$

$y$

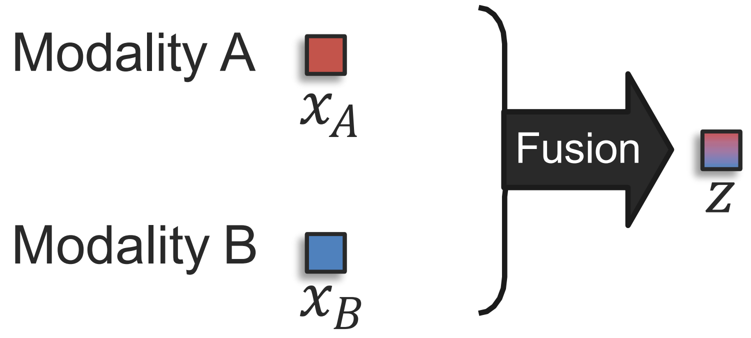



$x_A$

| | Estimate | 95% CI |
|---|---|---|
| $w_0$ | 5.79 | [5.29, 6.29] |
| $w_1$ | 0.68 | [0.25, 1.11] |
| $w_2$ | −2.94 | [−3.73, −2.15] |
| $w_3$ | 1.29 | [0.61, 1.97] |

**Multiplicative interaction!**

# Basic Concepts for Representation Fusion (aka, Basic Fusion)




- **Goal:** Model cross-modal interactions
  - between the multimodal elements
    - (only 1-dimensional features)

Linear regression:

$$z = \underset{\text{intercept (bias term)}}{w_0} + \underbrace{w_1 x_A + w_2 x_B}_{\text{Additive terms}} + \underbrace{w_3 (x_A \times x_b)}_{\text{Multiplicative term}} + \underset{\text{error (residual term)}}{\epsilon}$$

1 Additive terms:

$$z = w_1 x_A + w_2 x_B + \epsilon$$

2 Multiplicative "interaction" term:

$$z = w_3 (x_A \times x_b) + \epsilon$$

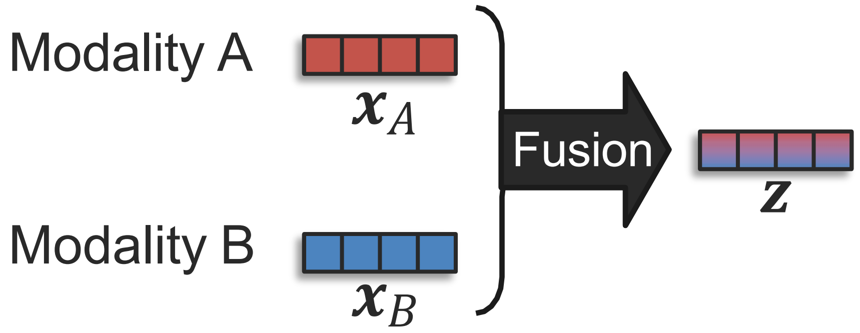
3 Additive and multiplicative terms:

$$z = w_1 x_A + w_2 x_B + w_3 (x_A \times x_b) + \epsilon$$

# Additive Fusion

Back to multivariate case!

(multi-dimensional features)

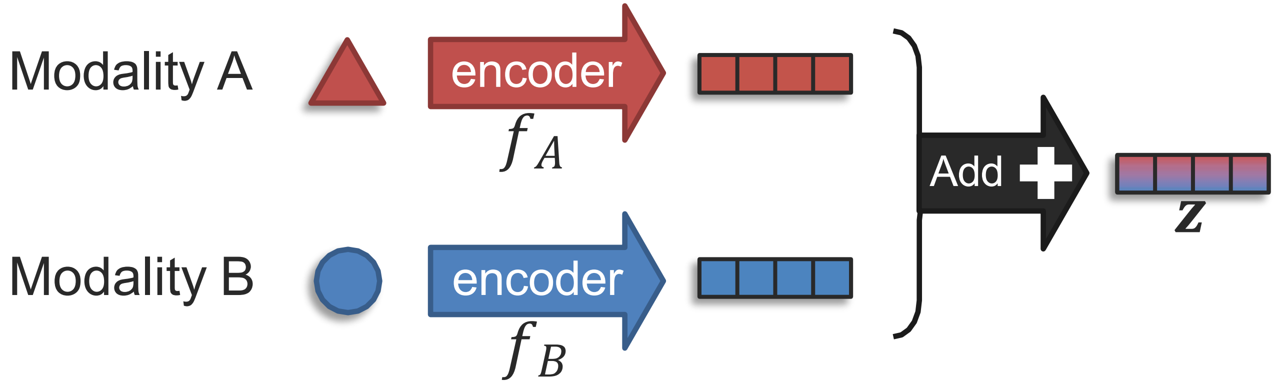



Additive fusion:

$$\boldsymbol{z} = \boldsymbol{w}_1\boldsymbol{x}_A + \boldsymbol{w}_2\boldsymbol{x}_B$$

1-layer neural network can be seen as additive

With unimodal encoders:



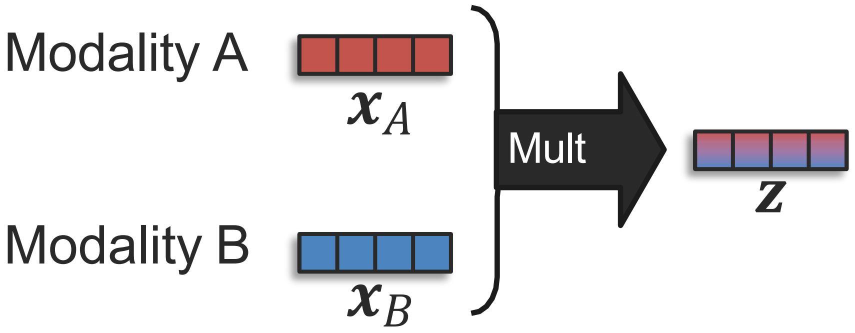

Additive fusion:

$$\boldsymbol{z} = f_A \qquad + f_B(\quad)$$

It could be seen as an ensemble approach

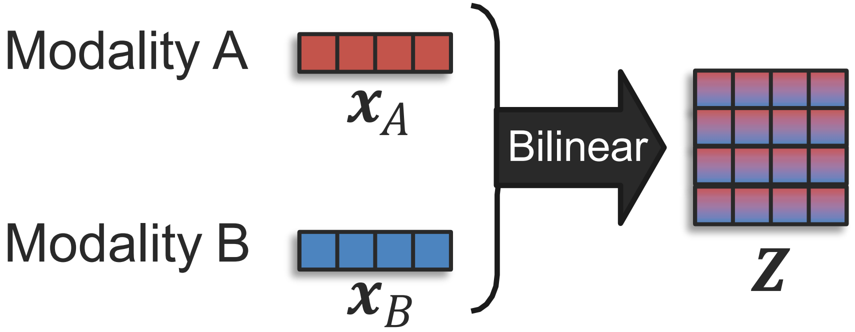
(late fusion)

# Multiplicative Fusion




Simple multiplicative fusion:

$$z = w(x_A \times x_B)$$

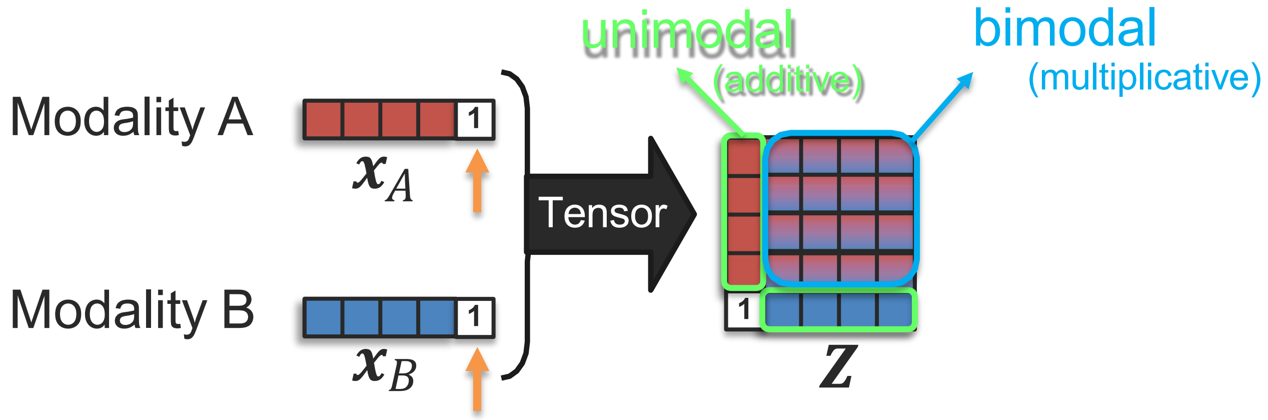



Bilinear Fusion:

$$Z = W(x_A^T / x_B)$$

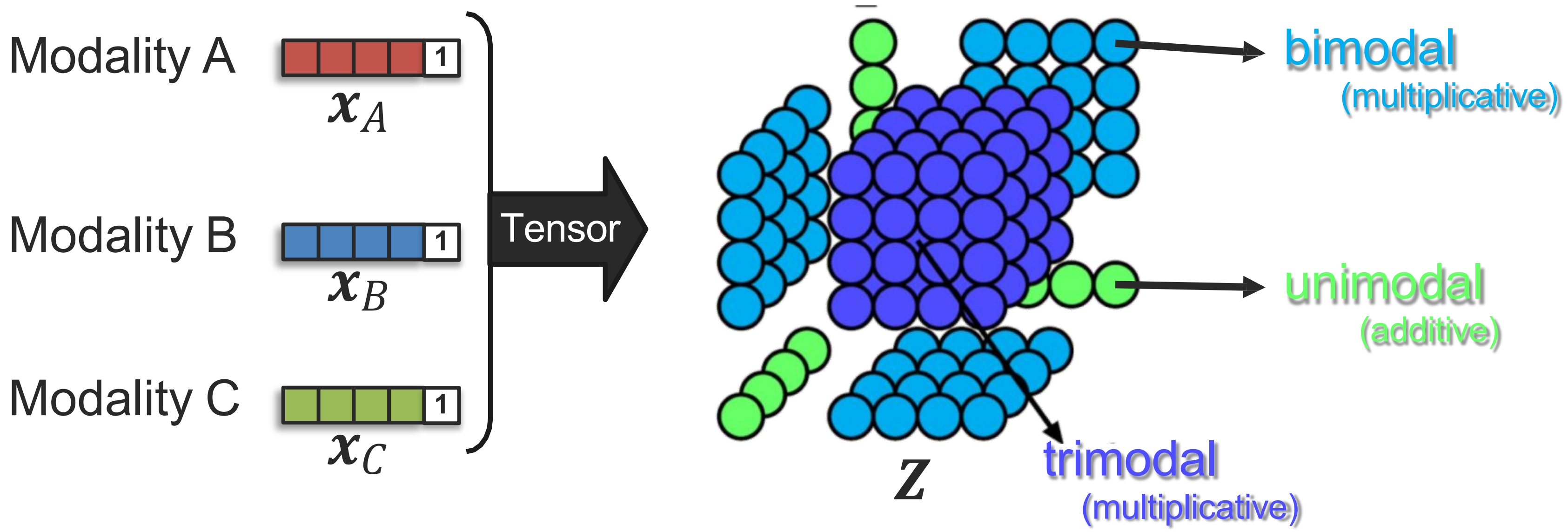
Jayakumar et al., Multiplicative Interactions and Where to Find Them, ICLR 2020

# Tensor Fusion




Tensor Fusion (bimodal):

$$Z = w([x_A \quad 1]^T / [x_B \quad 1])$$




… but the weight matrix may end up quite large!



Zadeh et al., Tensor Fusion Network for Multimodal Sentiment Analysis, EMNLP 2017

# Low-rank Fusion

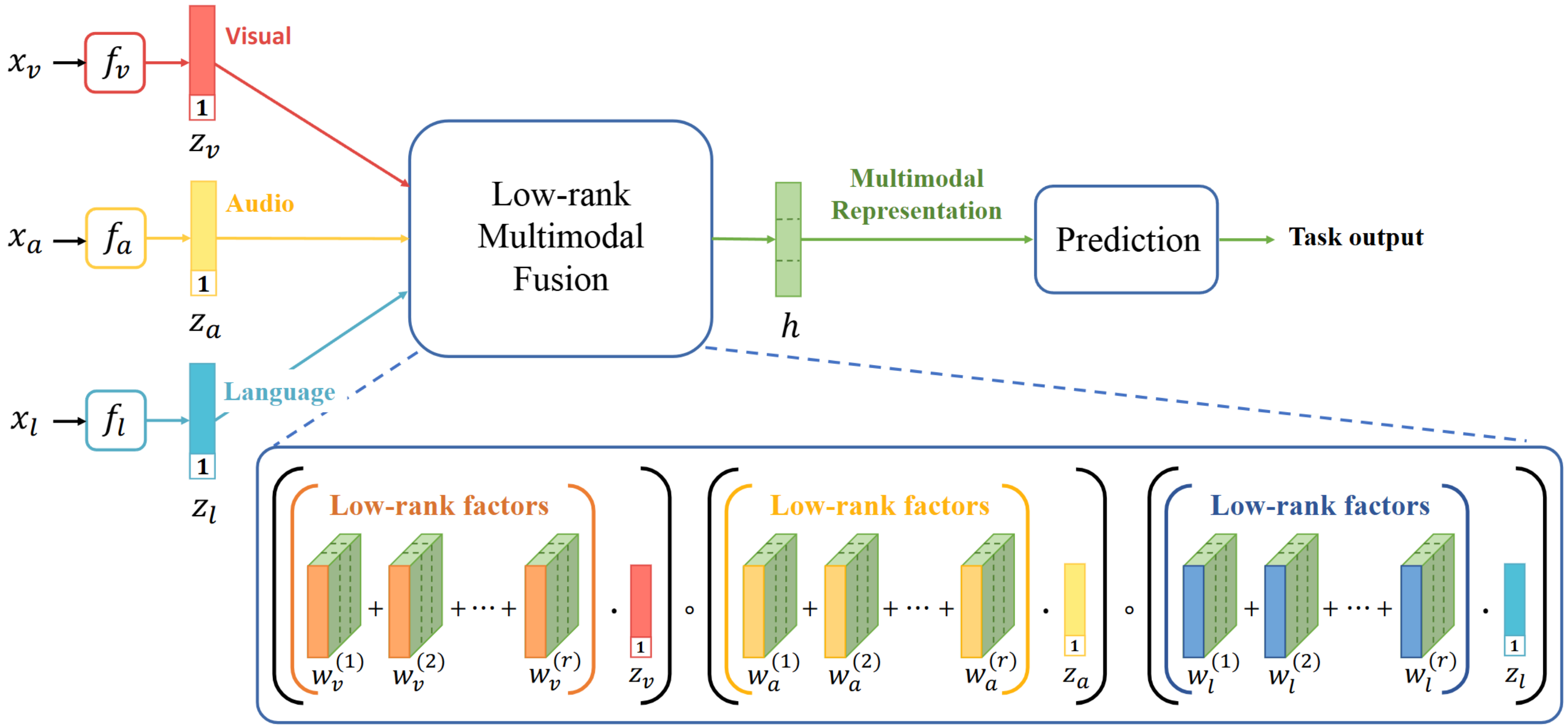
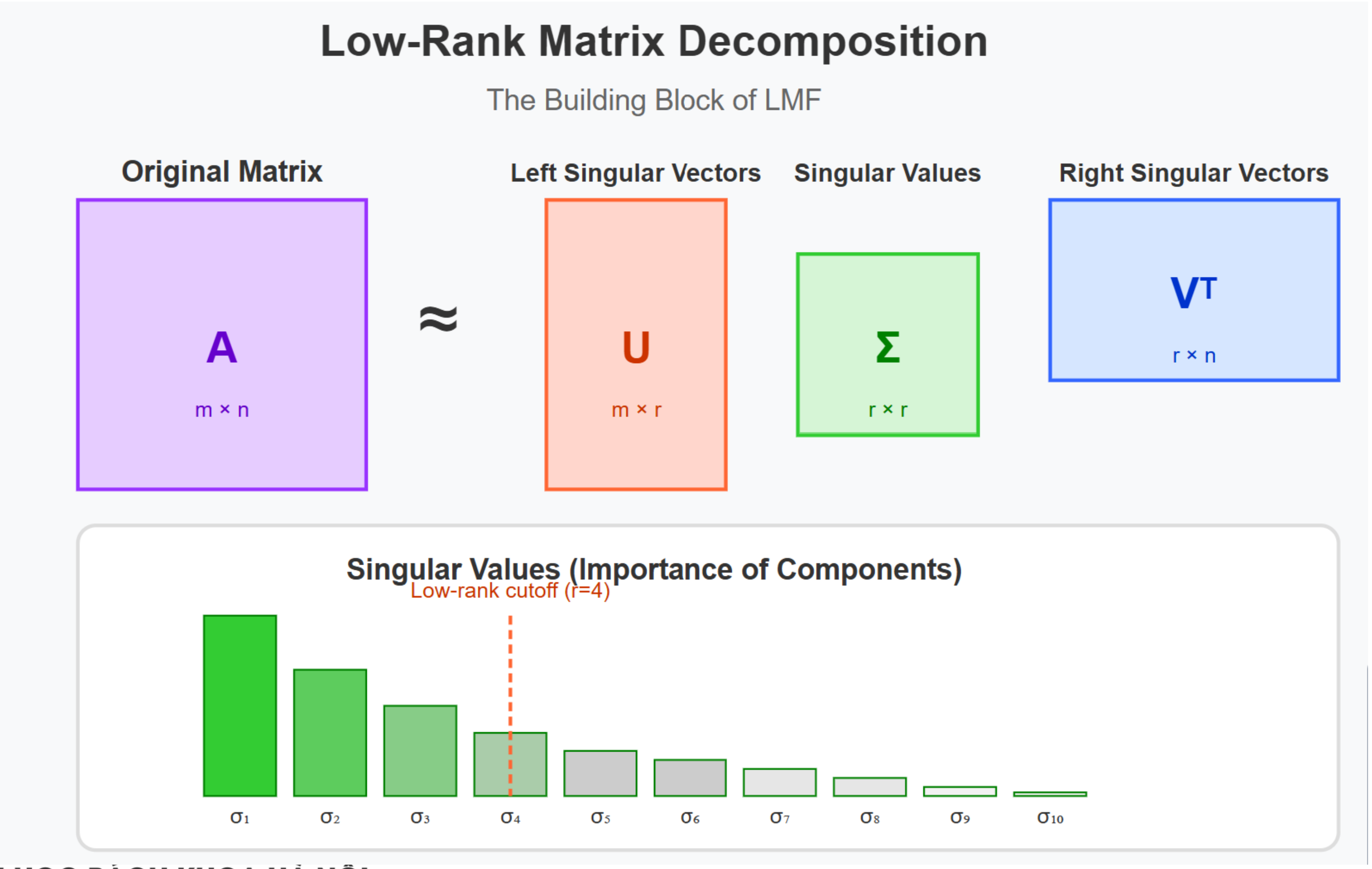



Figure 1: Overview of our Low-rank Multimodal Fusion model structure: LMF first obtains the unimodal representation $z_a, z_v, z_l$ by passing the unimodal inputs $x_a, x_v, x_l$ into three sub-embedding networks $f_v, f_a, f_l$ respectively. LMF produces the multimodal output representation by performing low-rank multimodal fusion with modality-specific factors. The multimodal representation can be then used for generating prediction tasks.

Liu et al., Efficient Low-rank Multimodal Fusion with Modality-Specific Factors, ACL 2018

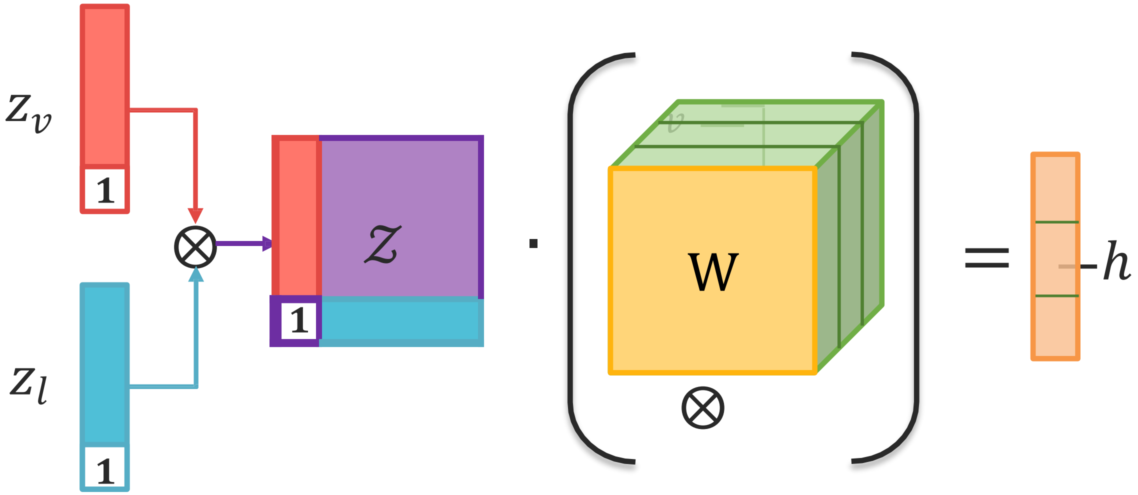
# Low-rank Fusion





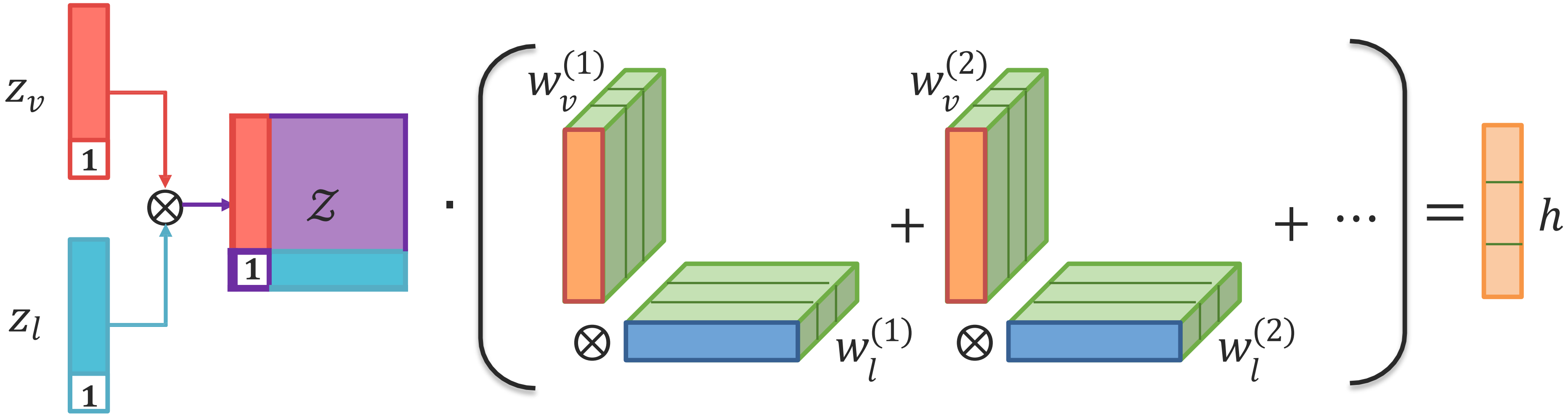

Liu et al., Efficient Low-rank Multimodal Fusion with Modality-Specific Factors, ACL 2018

# Low-rank Fusion




Liu et al., Efficient Low-rank Multimodal Fusion with Modality-Specific Factors, ACL 2018

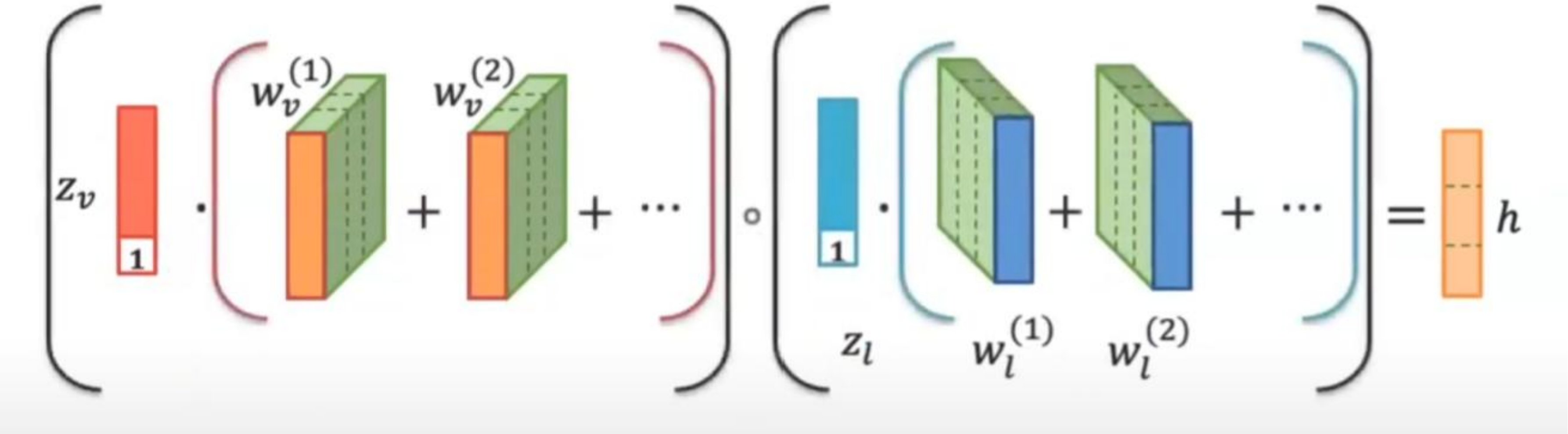
# Low-rank Fusion





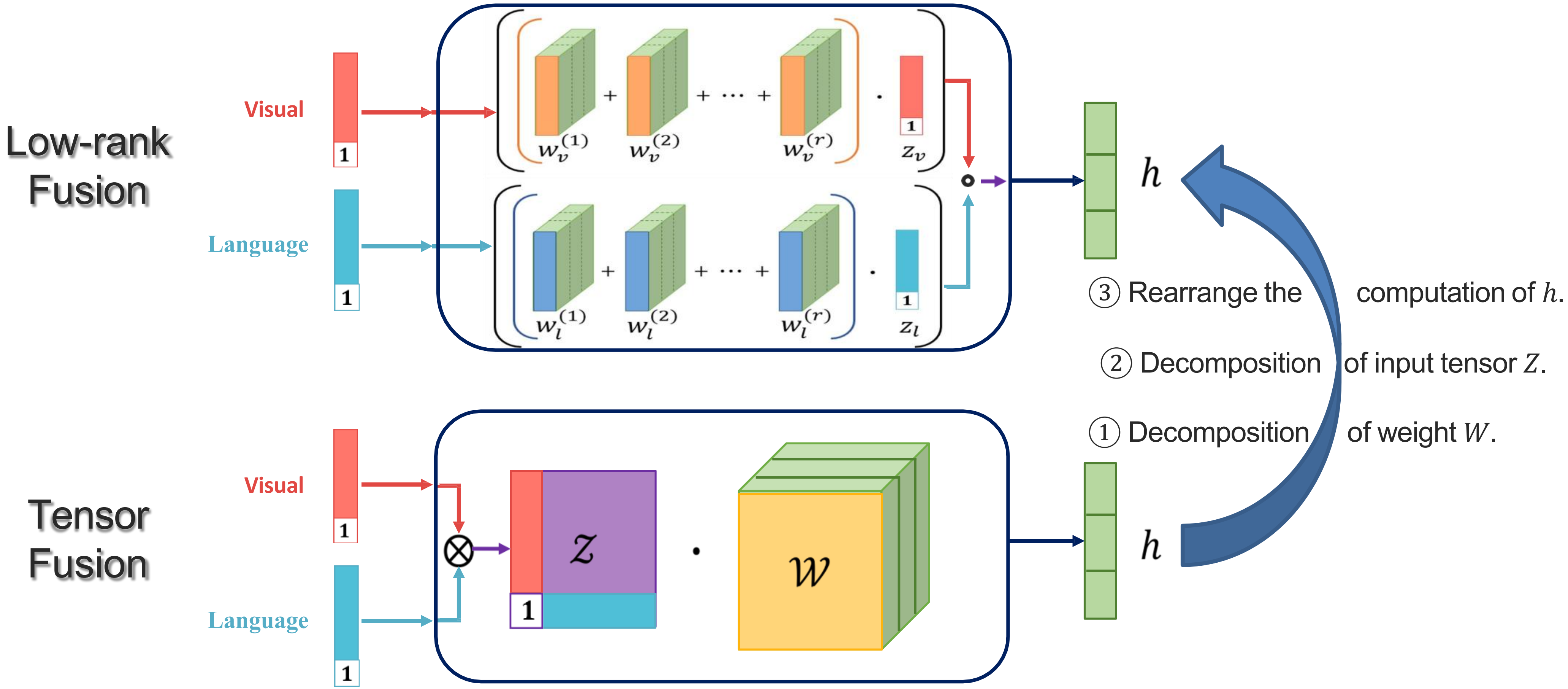

Liu et al., Efficient Low-rank Multimodal Fusion with Modality-Specific Factors, ACL 2018

# Low-rank Fusion




Liu et al., Efficient Low-rank Multimodal Fusion with Modality-Specific Factors, ACL 2018

# Low-rank Fusion

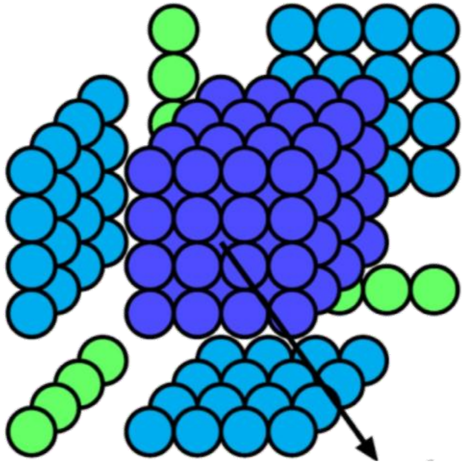




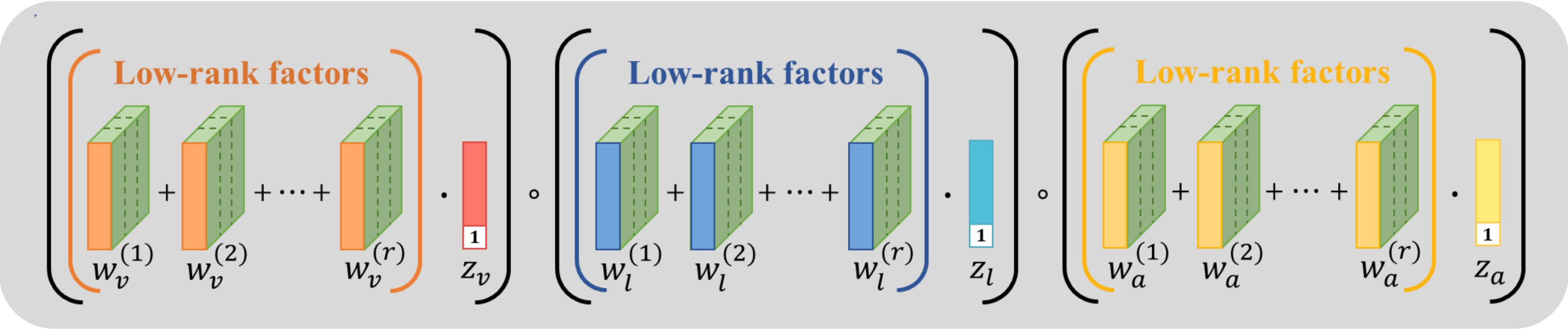

Liu et al., Efficient Low-rank Multimodal Fusion with Modality-Specific Factors, ACL 2018

# Low-rank Fusion with Trimodal Input

Tensor Fusion



## Low-rank Fusion :




Canonical Polyadic Decomposition



Liu et al., Efficient Low-rank Multimodal Fusion with Modality-Specific Factors, ACL 2018

# Going Beyond Additive and Multiplicative Fusion

Additive interaction:

$$z = w_1 x_A + w_2 x_B$$

First-order polynomial

Additive and multiplicative interaction:

$$z = w_1 x_A + w_2 x_B + w_3 \; x_A \times x_B$$

Second-order polynomial

Trimodal fusion (e.g., tensor fusion):

$$z = \underbrace{w_1 x_A + w_2 x_B + w_3 x_C}_{\text{Unimodal terms (first-order)}} + \underbrace{w_4 \; x_A \times x_C + w_5 (x_A \times x_C) + w_6 (x_B \times x_C)}_{\text{Bimodal terms (second-order)}} + \underbrace{w_7 (x_A \times x_B \times x_C)}_{\text{Trimodal terms (third-order)}}$$

Can we add higher-order interaction terms?

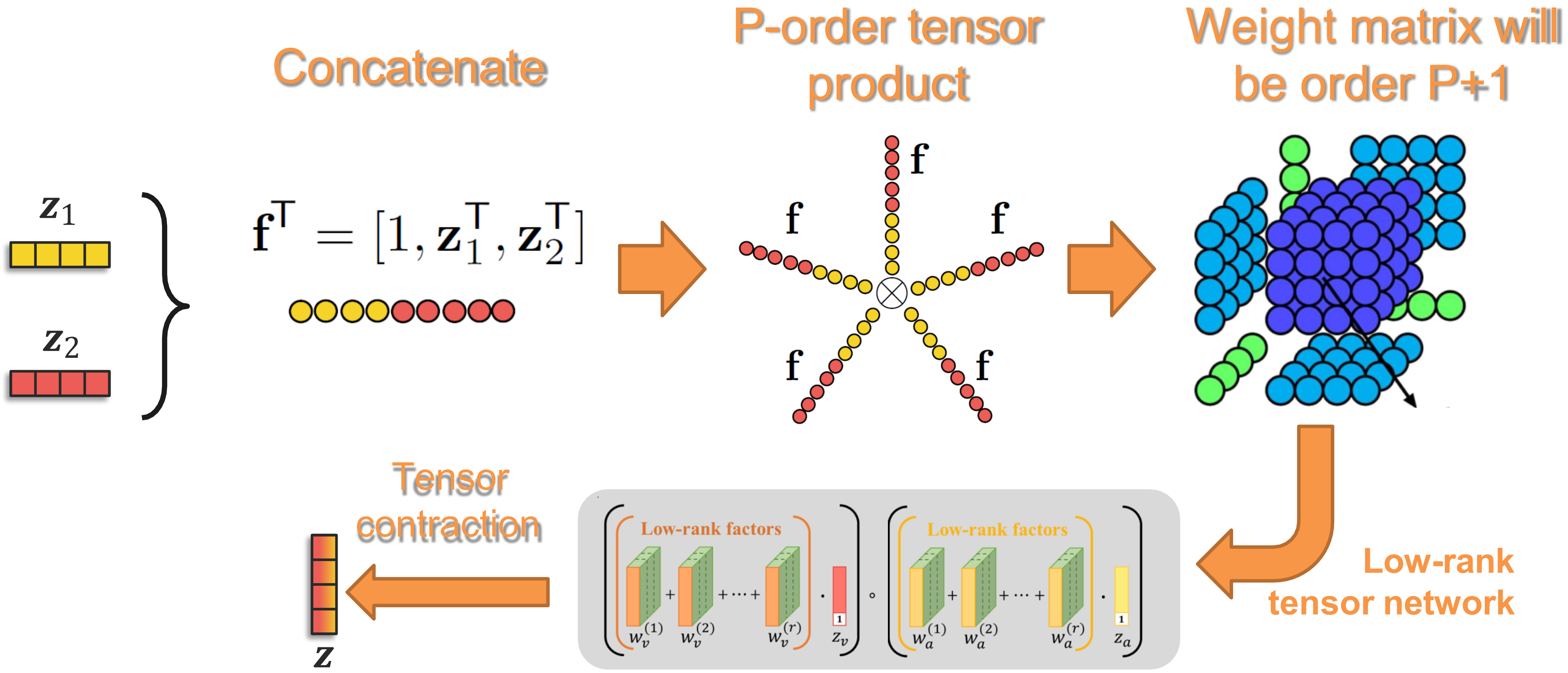
For example:

$$+ w_8 (x_A^2 \times x_B^2 \times x_C^2)$$

$$+ w_9 (x_A^3 \times x_B)$$

$$+ w_{10} (x_B^3 \times x_C^3)$$

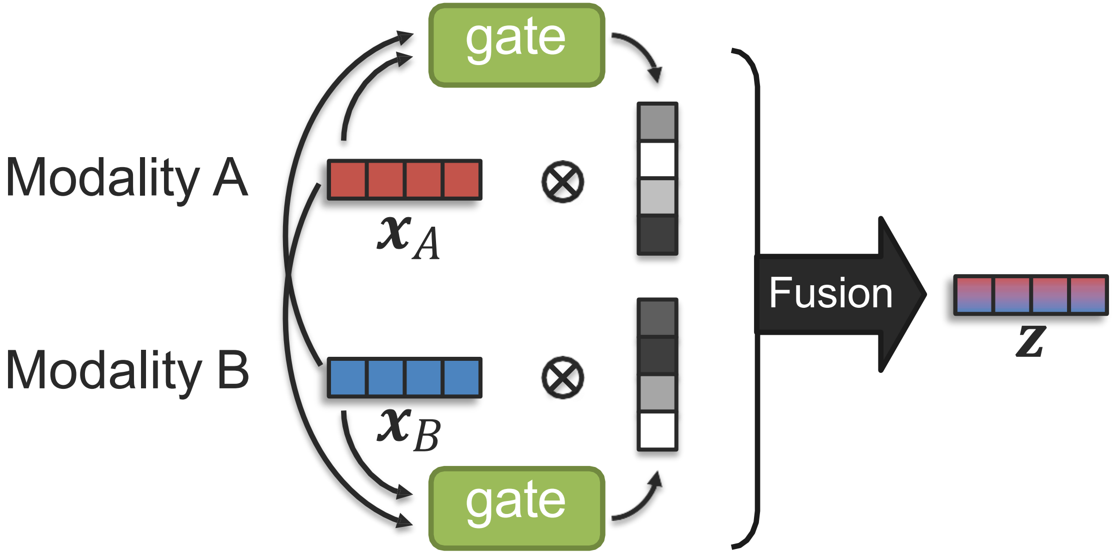
# High-Order Polynomial Fusion






Hou et al., Deep Multimodal Multilinear Fusion with High-order Polynomial Pooling, Neurips 2019

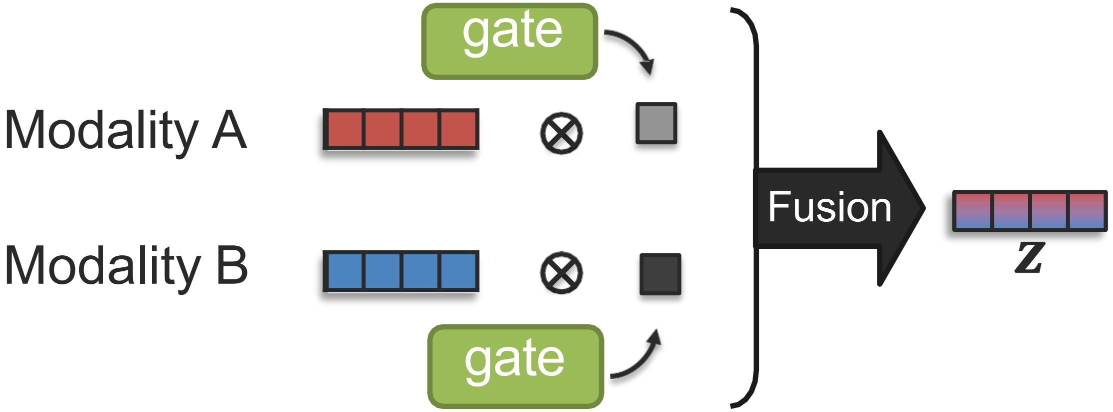
# Gated Fusion




Example with additive fusion:

$$z = g_A(x_A, x_B) * x_A + g_B(x_A, x_B) * x_B$$

➡ $g_A$ and $g_B$ can be seen as attention functions

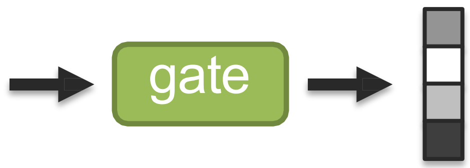



Gating output can be one weight for the whole modality



[Arevalo et al., Gated Multimodal Units for information fusion, ICLR-workshop 2017]

# Gating Module (aka, attention module)

Input 


What should it be?

Target modality

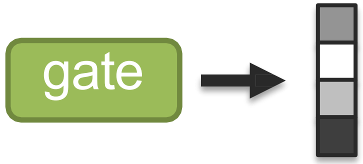
Other modality

All modality

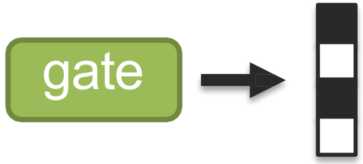
*"Neural network designed to mask unwanted signal from propagating forward"* (gating)

…or with a more positive view:

*"Neural network designed to select preferable signal to move forward"* (attention)

Soft attention 


Easier to compute derivative (gradient)

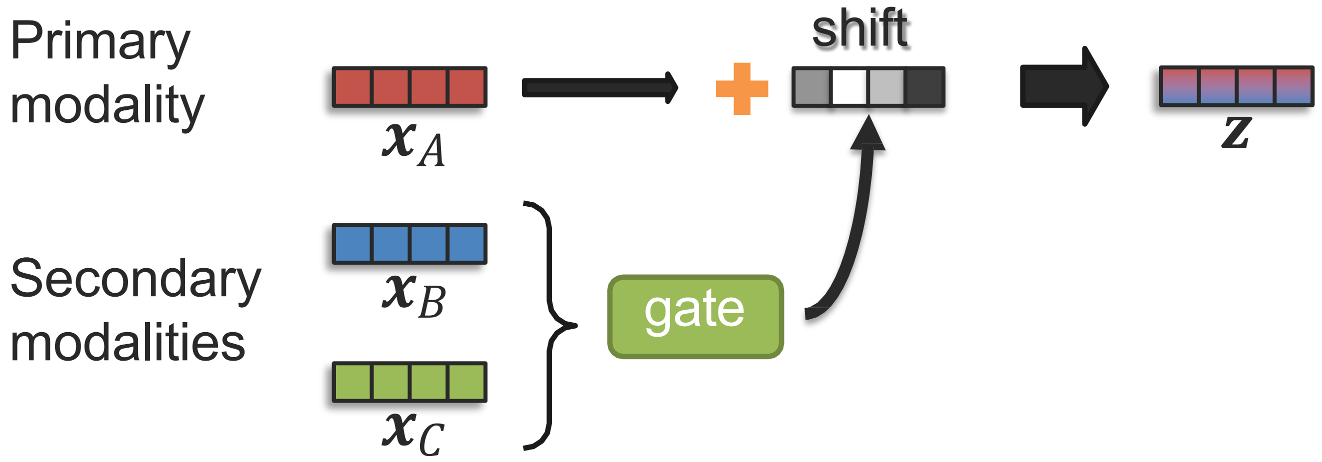
Hard attention 


Derivative is harder (e.g., use reinforcement learning)



[Chen et al., Multimodal Sentiment Analysis with Word-level Fusion and Reinforcement Learning, ICML 2017]

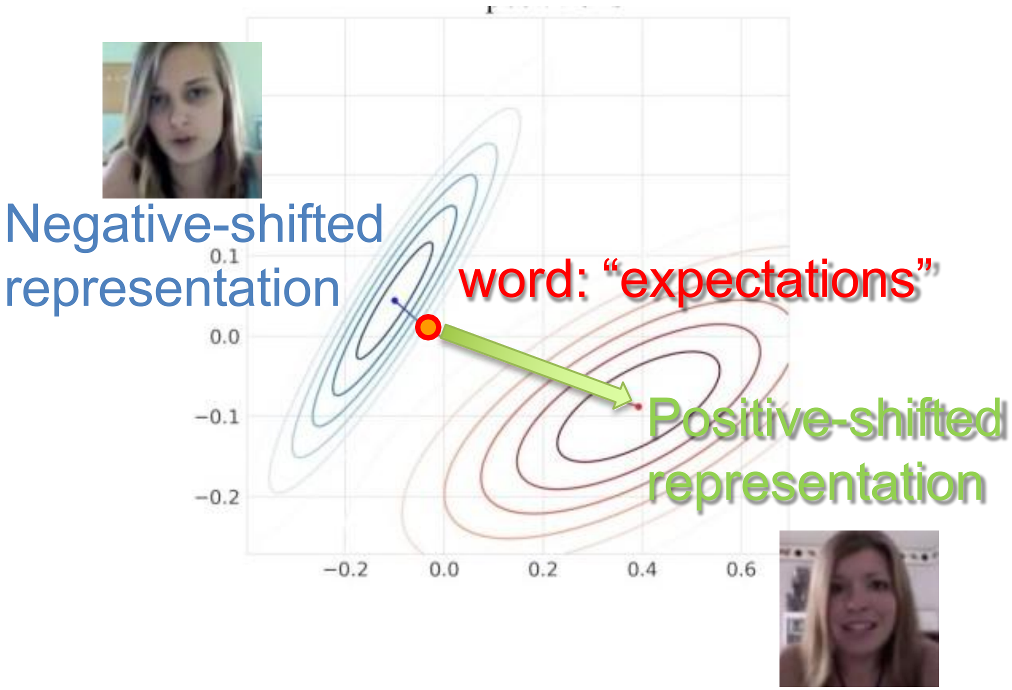
# Modality-Shifting Fusion




Example with language modality:

Primary modality: language

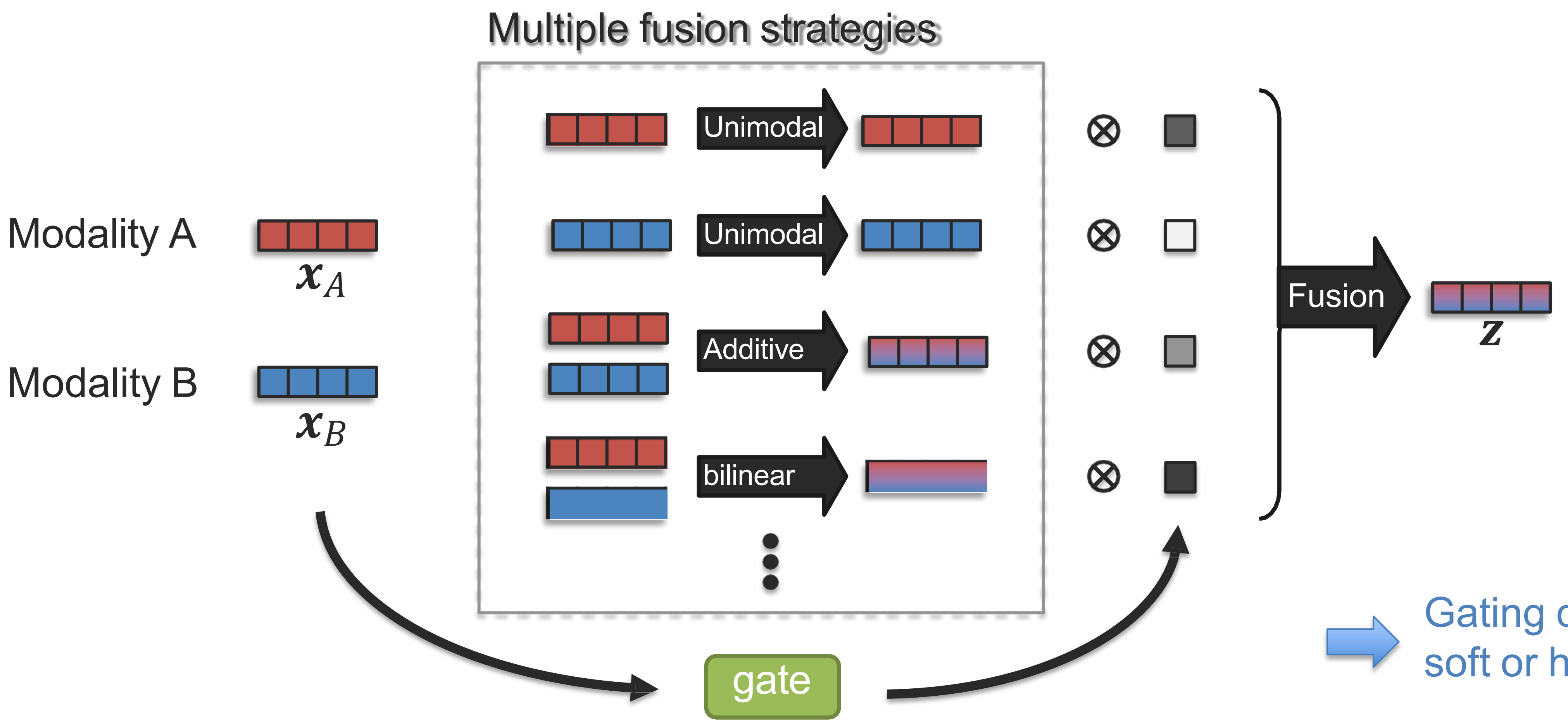
Secondary modalities: acoustic and visual






[Wang et al., Words Can Shift: Dynamically Adjusting Word Representations Using Nonverbal Behaviors, AAAI 2019]
[Rahman et al., Integrating Multimodal Information in Large Pretrained Transformers, ACL 2020]

# Mixture of Fusions

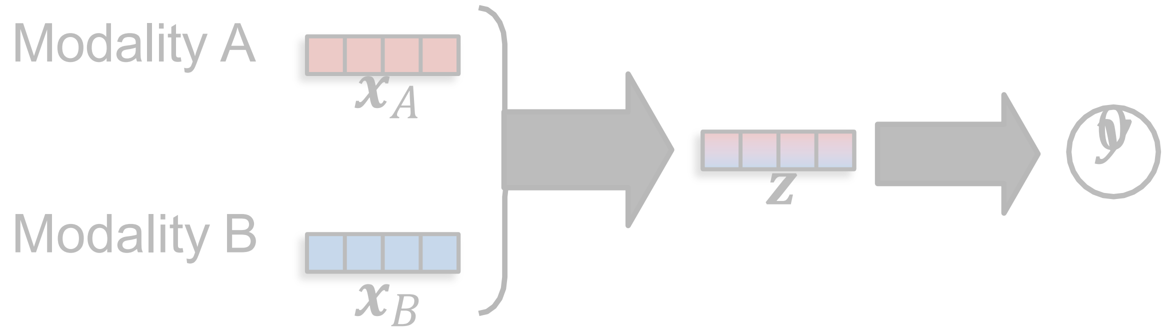


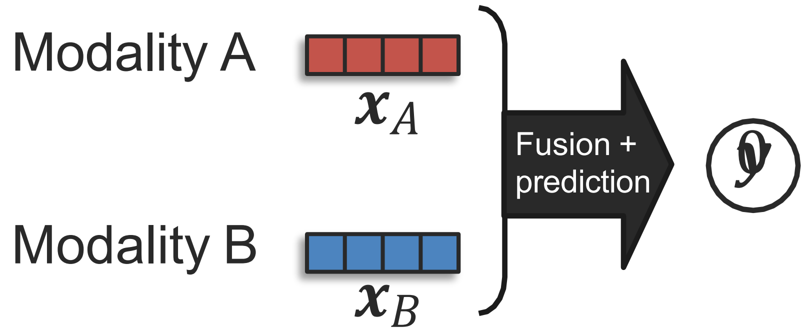

Gating can be with soft or hard attention



[Zadeh et al., Multimodal Language Analysis in the Wild: CMU-MOSEI Dataset and Interpretable Dynamic Fusion Graph, ACL 2018]
[Xu et al., MUFASA: Multimodal Fusion Architecture Search for Electronic Health Records, AAAI 2021]

# Nonlinear Fusion




Nonlinear fusion:

$$\hat{y} = f(\boldsymbol{x}_A, \boldsymbol{x}_B) \in \mathbb{R}^d$$

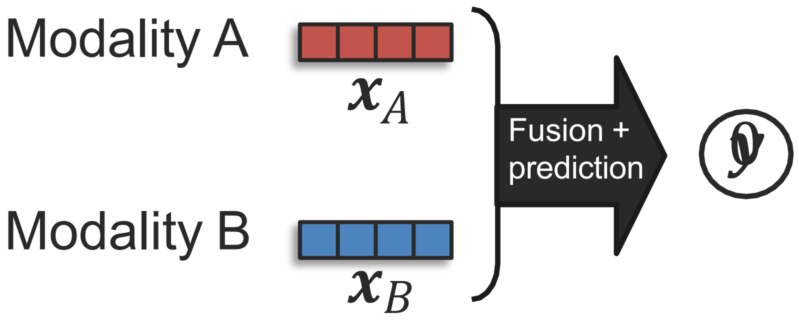
where $f$ could be a multi-layer perceptron or any nonlinear model

This could be seen as *early fusion*:

$$\hat{y} = f([\boldsymbol{x}_A, \boldsymbol{x}_B])$$

… but will our neural network learn the nonlinear interactions?

# Measuring Non-Additive Interactions



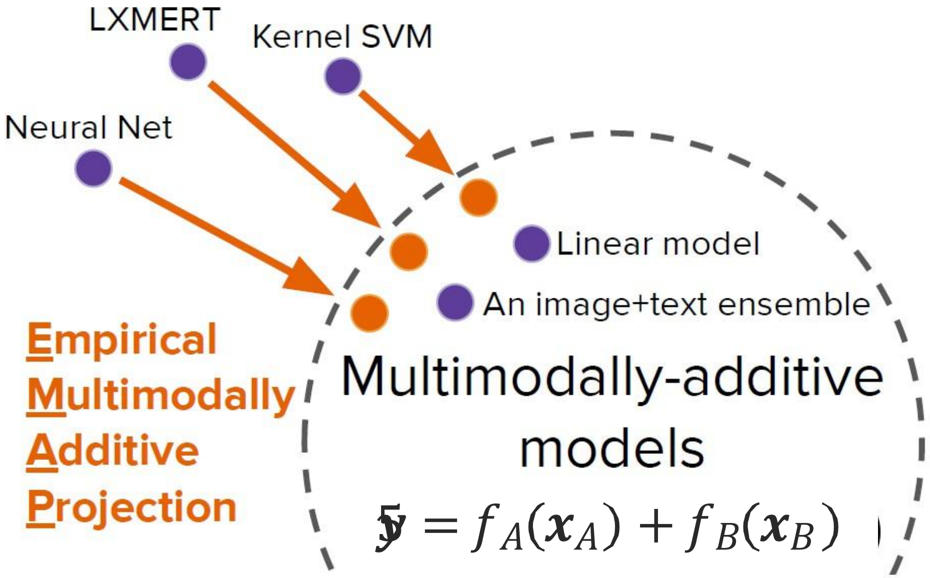

Nonlinear fusion:

$$\hat{y} = f(x_A, x_B)$$

Projection?

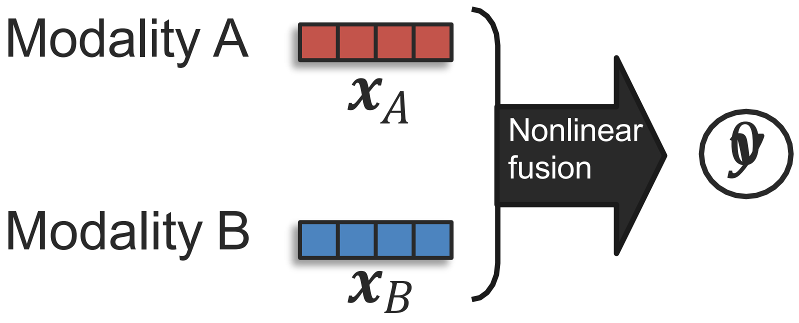
Additive fusion:

$$\hat{y} = f_A(x_A) + f_B(x_B)$$




Hessel and Lee, Does my multimodal model learn cross-modal interactions? It's harder to tell than you might think!, EMNLP 2020 ⟶ introduced the EMAP method

# Measuring Non-Additive Interactions




Nonlinear fusion:

$$y = f(\boldsymbol{x}_A, \boldsymbol{x}_B)$$

Projection?

Additive fusion:

$$y' = f_A(\boldsymbol{x}_A) + f_B(\boldsymbol{x}_B)$$

Projection from nonlinear to additive (using EMAP):

$$\hat{y}(\boldsymbol{x}_A, \boldsymbol{x}_B) = \underbrace{\mathop{\mathrm{E}}_{\boldsymbol{x}_B}\left[f(\boldsymbol{x}_A, \boldsymbol{x}_B)\right]}_{f_A(\boldsymbol{x}_A)} + \underbrace{\mathop{\mathrm{E}}_{\boldsymbol{x}_A}\left[f(\boldsymbol{x}_A, \boldsymbol{x}_B)\right]}_{f_B(\boldsymbol{x}_B)}$$

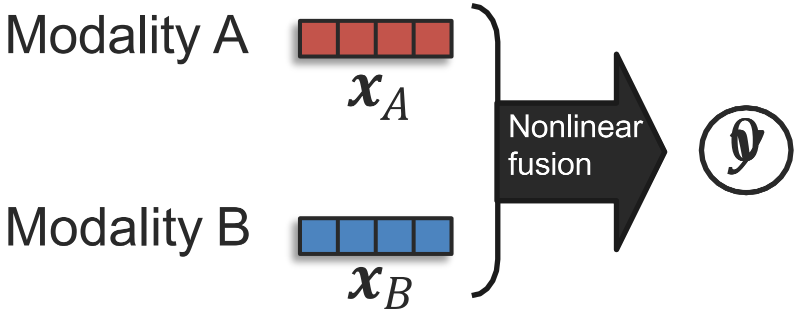
Modality A + Modality B

Additive fusion
(approximation)



[Hessel and Lee, Does my multimodal model learn cross-modal interactions? It's harder to tell than you might think!, EMNLP 2020]

# Measuring Non-Additive Interactions




Nonlinear fusion:

$$\hat{y} = f(x_A, x_B)$$

EMAP projection

Additive fusion:

$$\hat{y}^{@} = f_A(x_A) + f_B(x_B) + \mu$$

| | | I-INT | I-SEM | I-CTX | T-VIS | R-POP | T-ST1 | T-ST2 |
|---|---|---|---|---|---|---|---|---|
| Nonlinear | Neural Network | 90.4 | 69.2 | 78.5 | 51.1 | 63.5 | 71.1 | 79.9 |
| Polynomial | Polykernel SVM | **91.3** | **74.4** | **81.5** | 50.8 | – | 72.1 | **80.9** |
| Nonlinear | FT LXMERT | 83.0 | 68.5 | 76.3 | **53.0** | 63.0 | 66.4 | 78.6 |
| Nonlinear | ↳ + Linear Logits | 89.9 | 73.0 | 80.7 | **53.4** | **64.1** | **75.5** | 80.3 |
| Additive | Linear Model | 90.4 | 72.8 | 80.9 | 51.3 | 63.7 | **75.6** | 76.1 |
| | Best Model | **91.3** | **74.4** | **81.5** | **53.4** | *64.2* | 75.5 | **80.9** |
| Additive | ↳ + EMAP | **91.1** | **74.2** | **81.3** | 51.0 | *64.1* | **75.9** | **80.7** |

Always a good baseline!

Differences are small!!!!



[Hessel and Lee, Does my multimodal model learn cross-modal interactions? It's harder to tell than you might think!, EMNLP 2020]

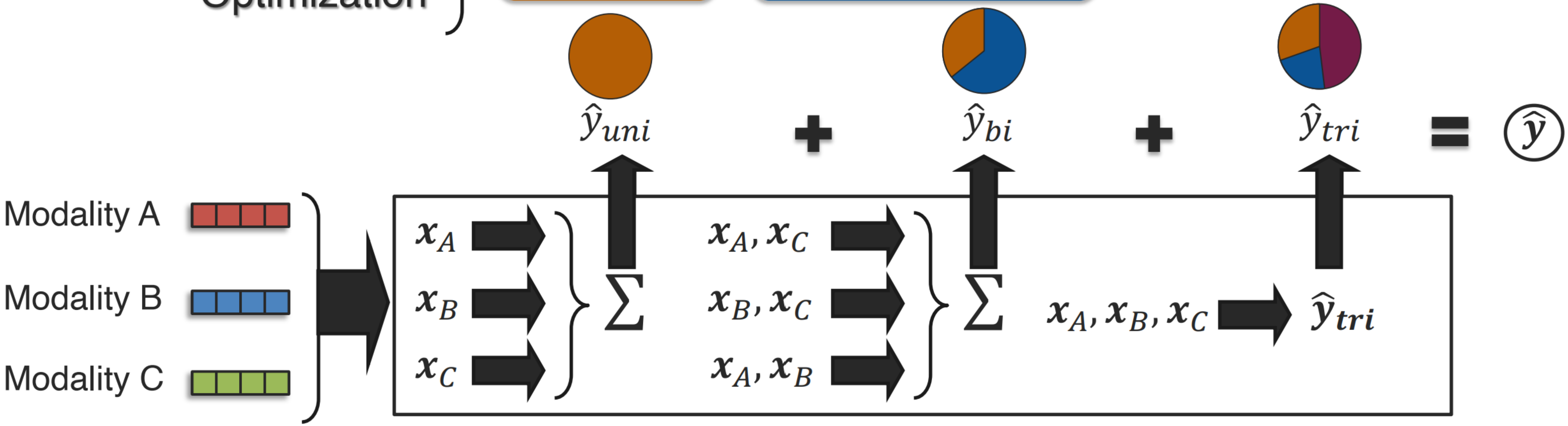
# Learning Non-additive Bimodal and Trimodal Interactions

**Idea:** prioritize simpler interactions

**Unimodal** (additive) — residual → **Bimodal** (non-additive) — residual → **Trimodal** (non-additive)

Multimodal Residual Optimization

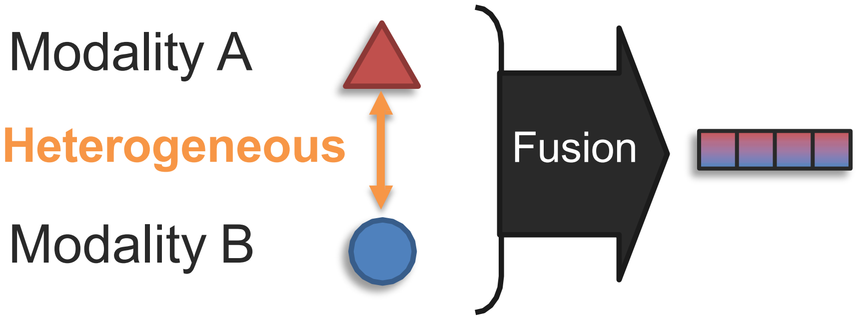
$$\boxed{\mathcal{L}(y, \hat{y}_{uni})} + \boxed{\mathcal{L}(\boxed{y - \hat{y}_{uni}}, \hat{y}_{bi})} + \mathcal{L}(\boxed{y - \hat{y}_{uni} - \hat{y}_{bi}}, \hat{y}_{tri})$$



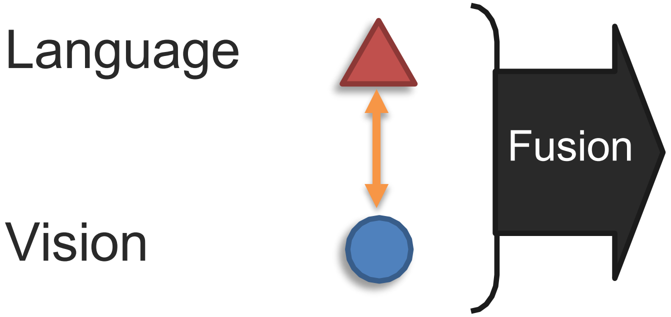

[Wortwein et al., Beyond Additive Fusion: Learning Non-Additive Multimodal Interactions, Findings-EMNLP 2022]

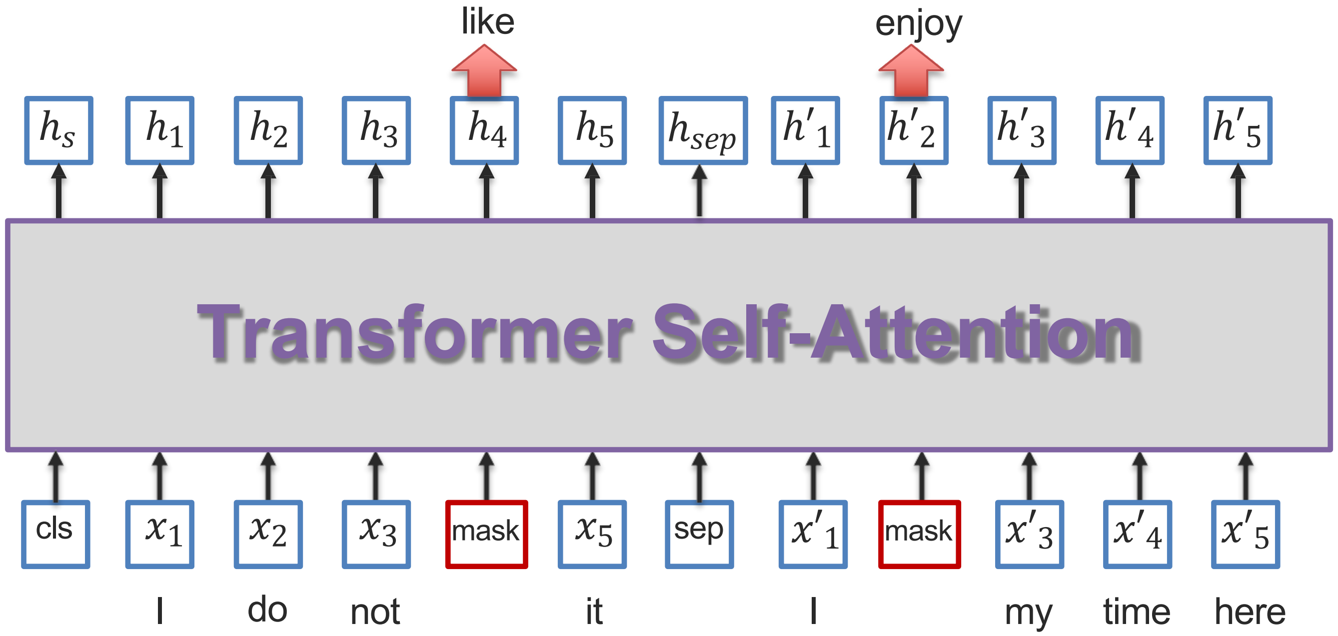
# Fusion with Heterogeneous Modalities

**Goal: Fusion with raw modalities**

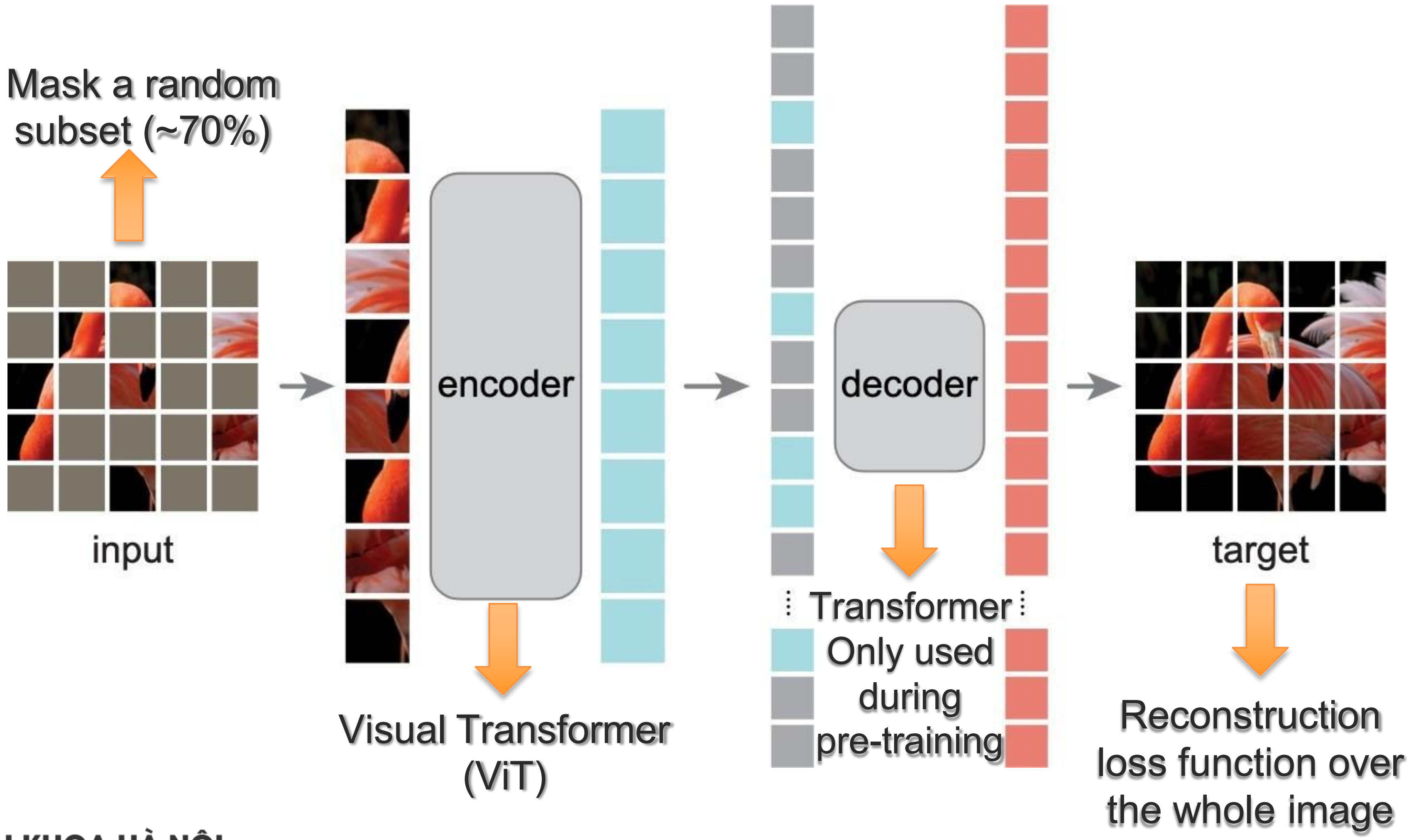



Can the same fusion algorithm handle raw heterogeneous modalities?



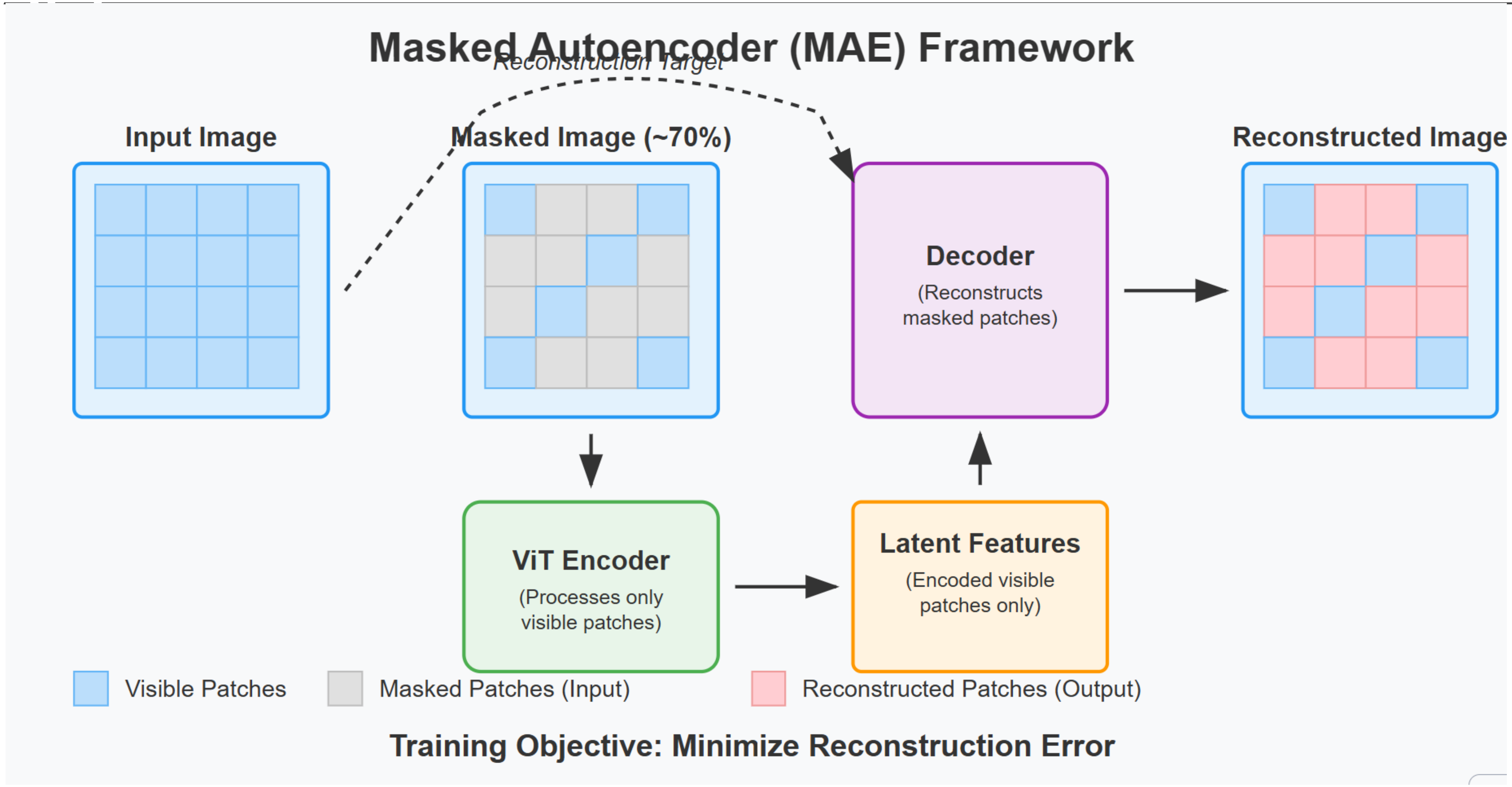
# Image Representation Learning: Masked Auto-Encoder (MAE)






[He et al., Masked Autoencoders Are Scalable Vision Learners, CVPR 2022]

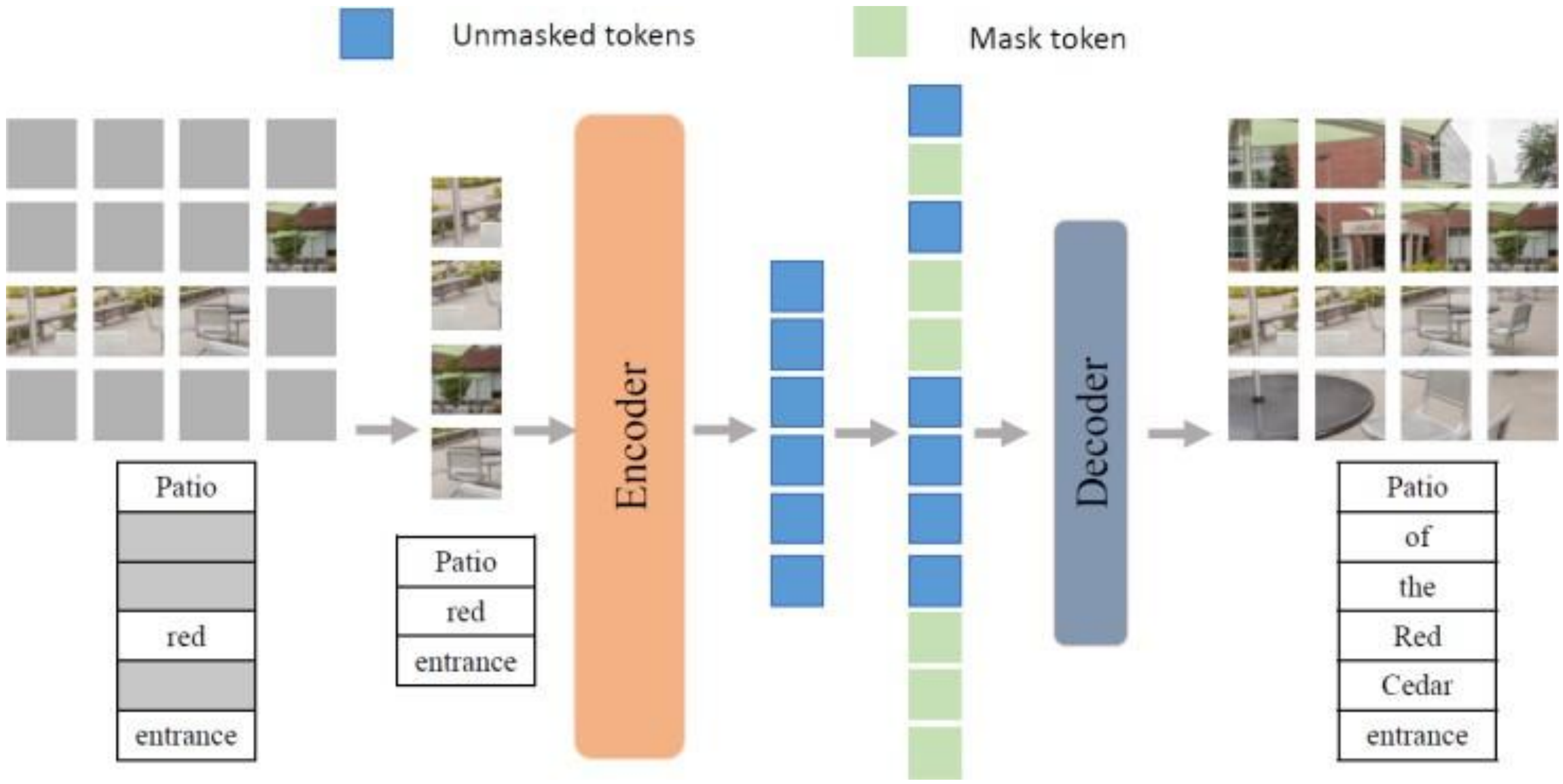
# Image Representation Learning: Masked Auto-Encoder





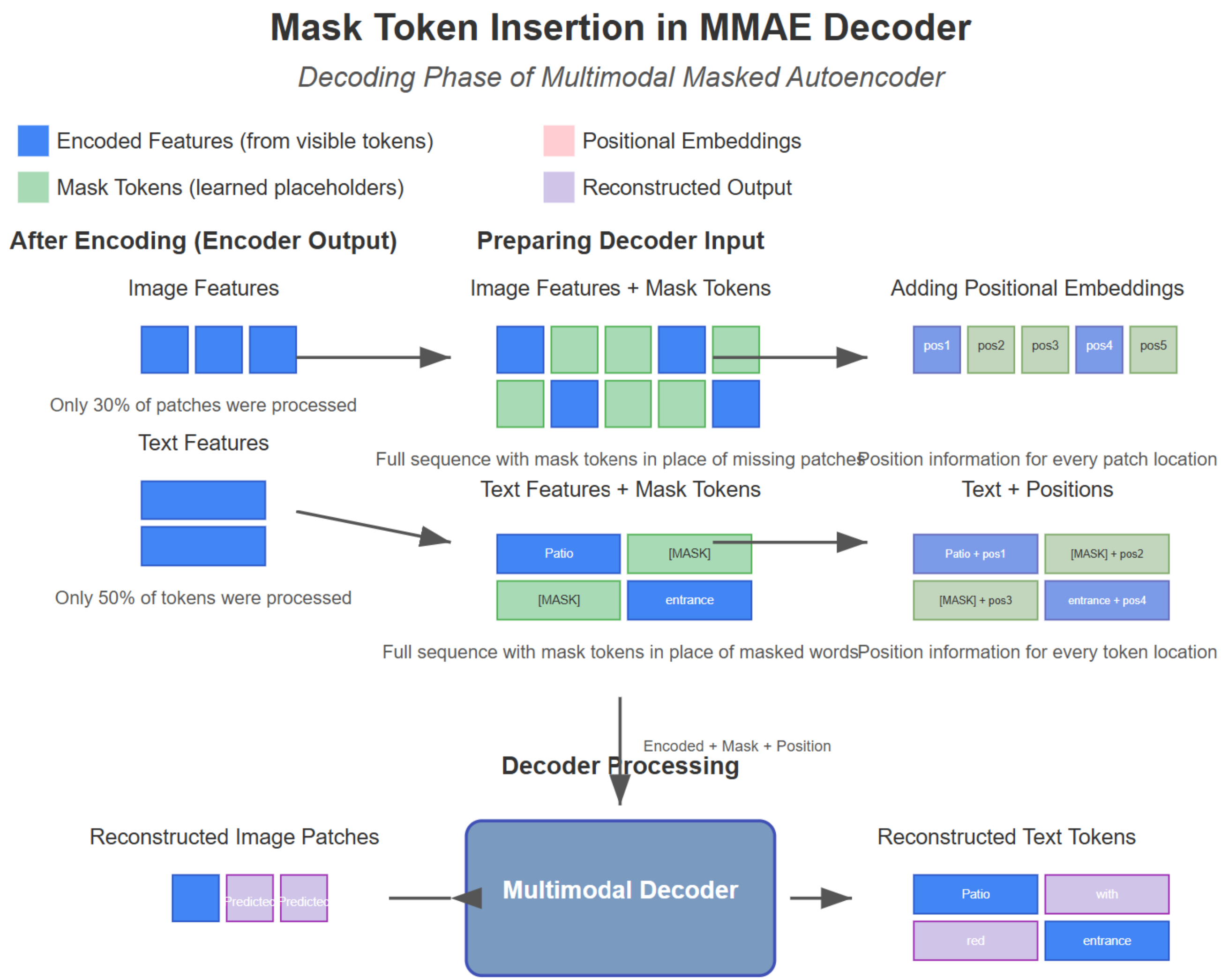

[He et al., Masked Autoencoders Are Scalable Vision Learners, CVPR 2022]

# Multimodal Masked Autoencoder



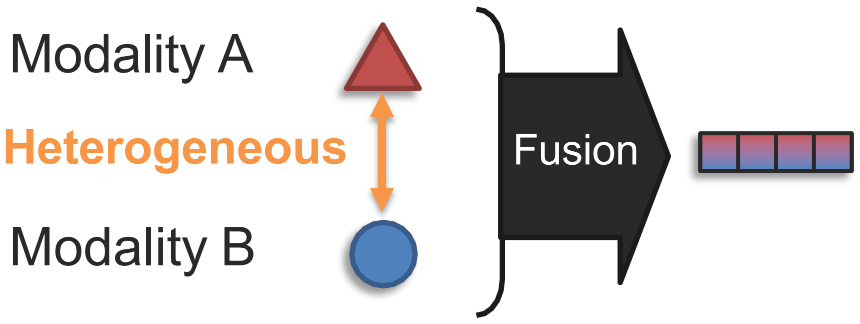

[Geng et al., Multimodal Masked Autoencoders Learn Transferable Representations, 2022]

# Multimodal Masked Autoencoder

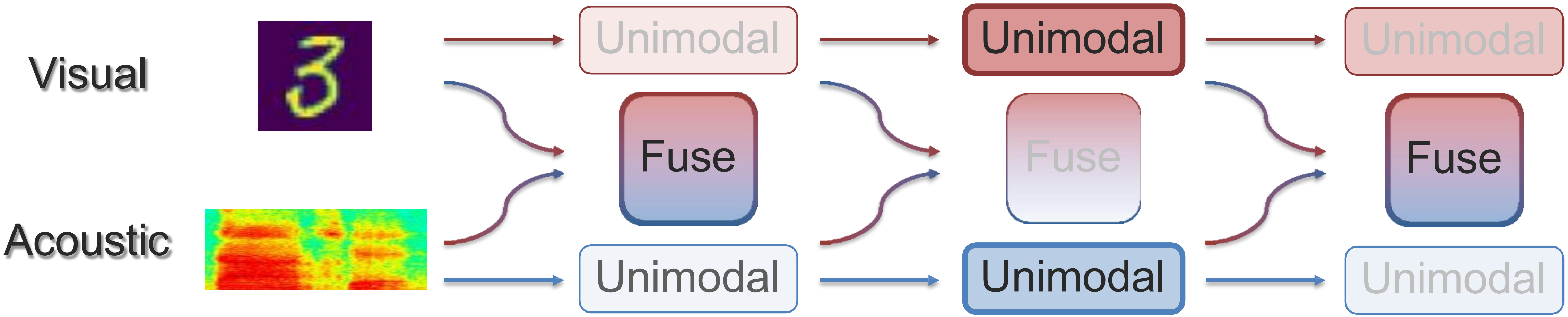



[Based on Geng et al., Multimodal Masked Autoencoders Learn Transferable Representations, 2022]

# Dynamic Early Fusion



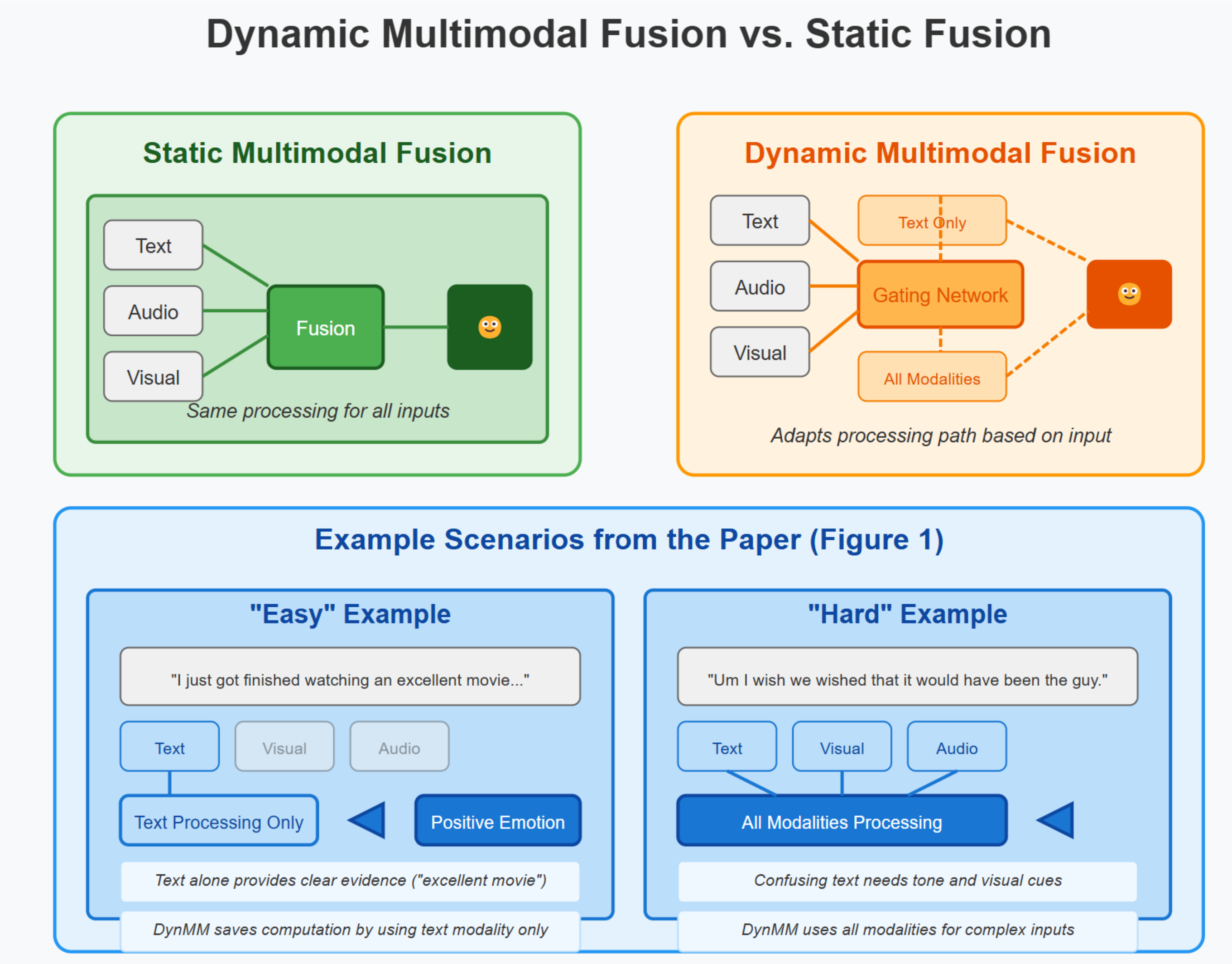

**Idea:** **Deciding when to fuse in early fusion**






[Xue and Marculescu, Dynamic Multimodal Fusion, arxiv 2022]

# Dynamic Early Fusion






[Xue and Marculescu, Dynamic Multimodal Fusion, arxiv 2022]

# Dynamic Early Fusion

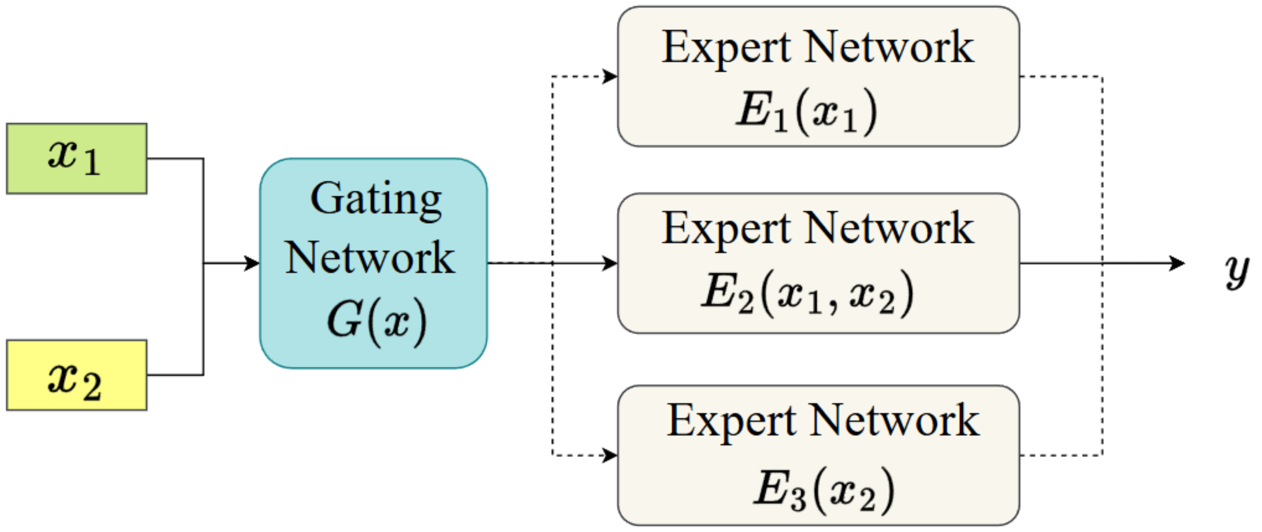



Figure 2. An illustration of modality-level DynMM, where input data has two modalities, denoted by $x_1$ and $x_2$, and the output is denoted by $y$. We design a set of expert networks $\{E_i\}$ that specialize in different subsets of modalities and adopt a gating network $G(\mathbf{x})$ to generate data-dependent decisions on which expert network to select.

[Xue and Marculescu, Dynamic Multimodal Fusion, arxiv 2022]

# Dynamic Early Fusion

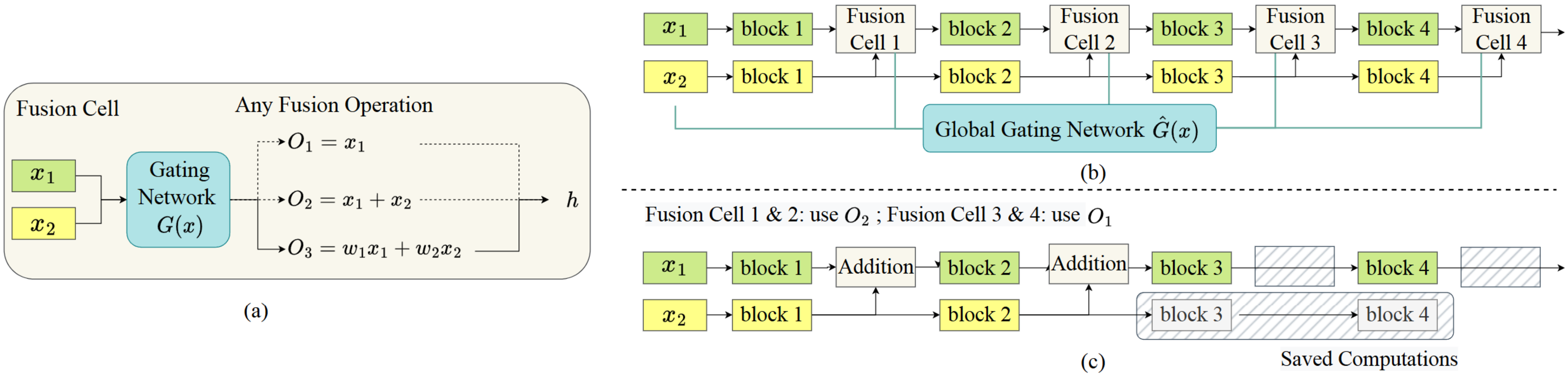
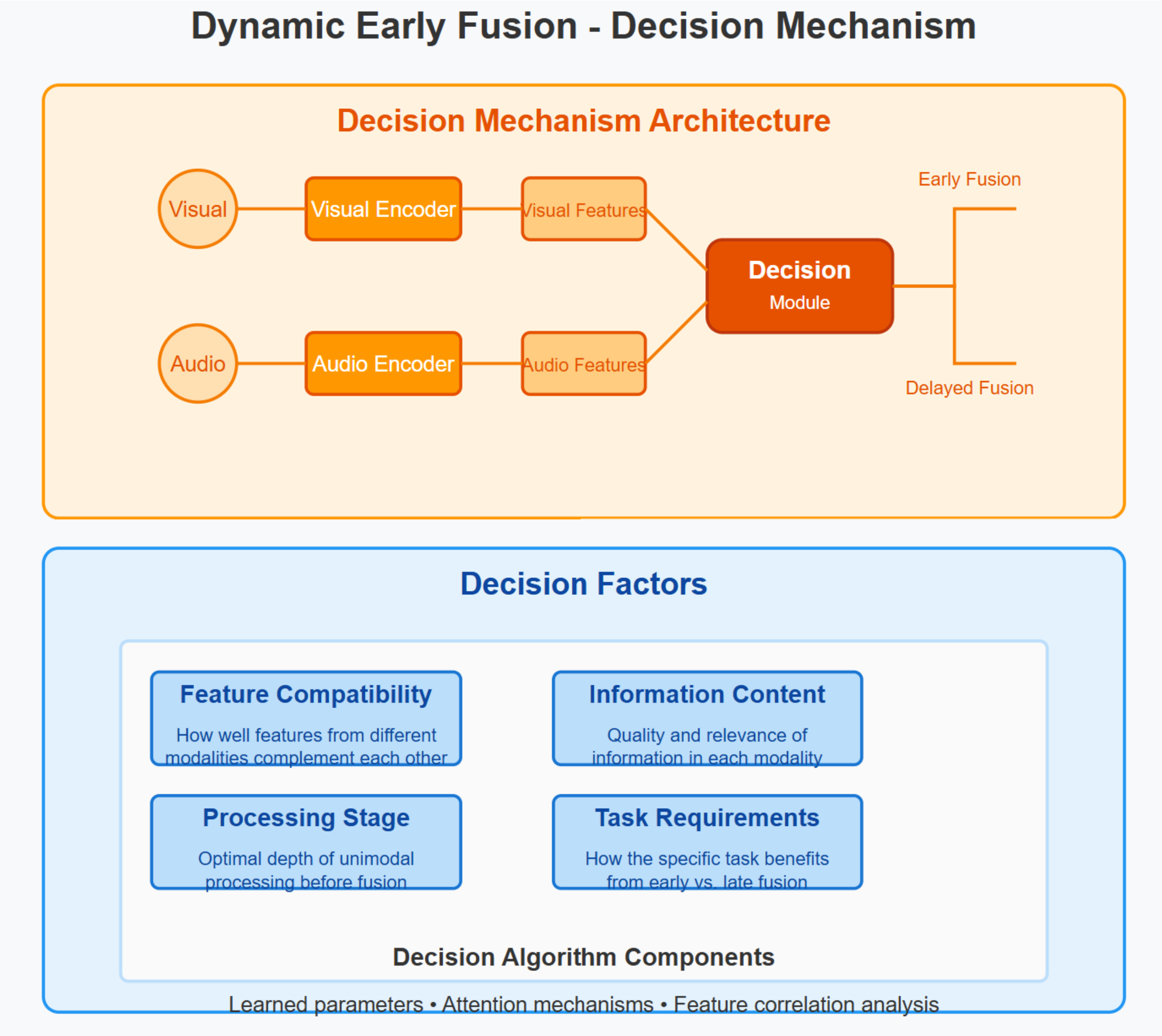



Figure 3. (a) An illustration of fusion-level DynMM, where input data has two modalities, denoted by $x_1$ and $x_2$. We design a fusion cell with a set of candidate operations $\{O_i\}$ and a gating network $G(\mathbf{x})$. $h$ represents output of the cell. (b) A dynamic multimodal architecture with stacked fusion cells, where we interlace static feature extraction blocks (colored with green and yellow) with dynamic fusion cells. Gating network $G(\mathbf{x})$ in four fusion cells are integrated as one global gating network $\hat{G}(\mathbf{x})$ that outputs decisions for four cells at once. (c) An example architecture when the gating network chooses $O_2$ for the first 2 fusion cells and $O_1$ for the last 2 cells. Consequently, computations of fusion cell 3 & 4 and feature extraction cell 3 & 4 for $x_2$ are saved.

[Xue and Marculescu, Dynamic Multimodal Fusion, arxiv 2022]

# Dynamic Early Fusion



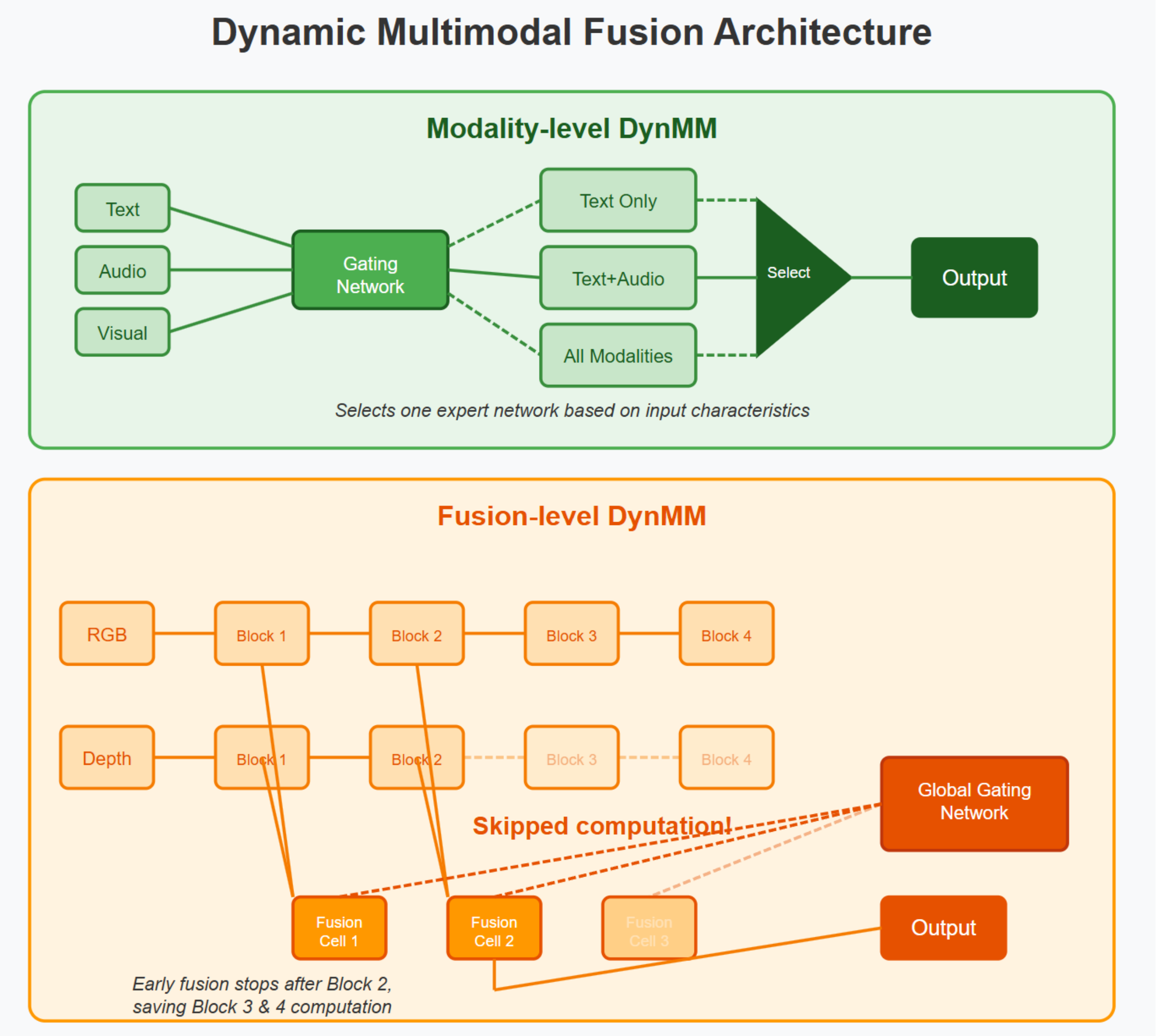

$$\mathcal{L} = \mathcal{L}_{task} + \lambda \sum_{i=1}^{B} g_i C(E_i) \quad \text{(modality-level)} \quad (1)$$

$$\mathcal{L} = \mathcal{L}_{task} + \lambda \sum_{j=1}^{F} \sum_{i=1}^{B} g_i^{(j)} C(O_{i,j}) \quad \text{(fusion-level)} \quad (2)$$



[Xue and Marculescu, Dynamic Multimodal Fusion, arxiv 2022]

# Dynamic Early Fusion




[Xue and Marculescu, Dynamic Multimodal Fusion, arxiv 2022]

# Dynamic Early Fusion

**Fusion fully learned from optimization and data**

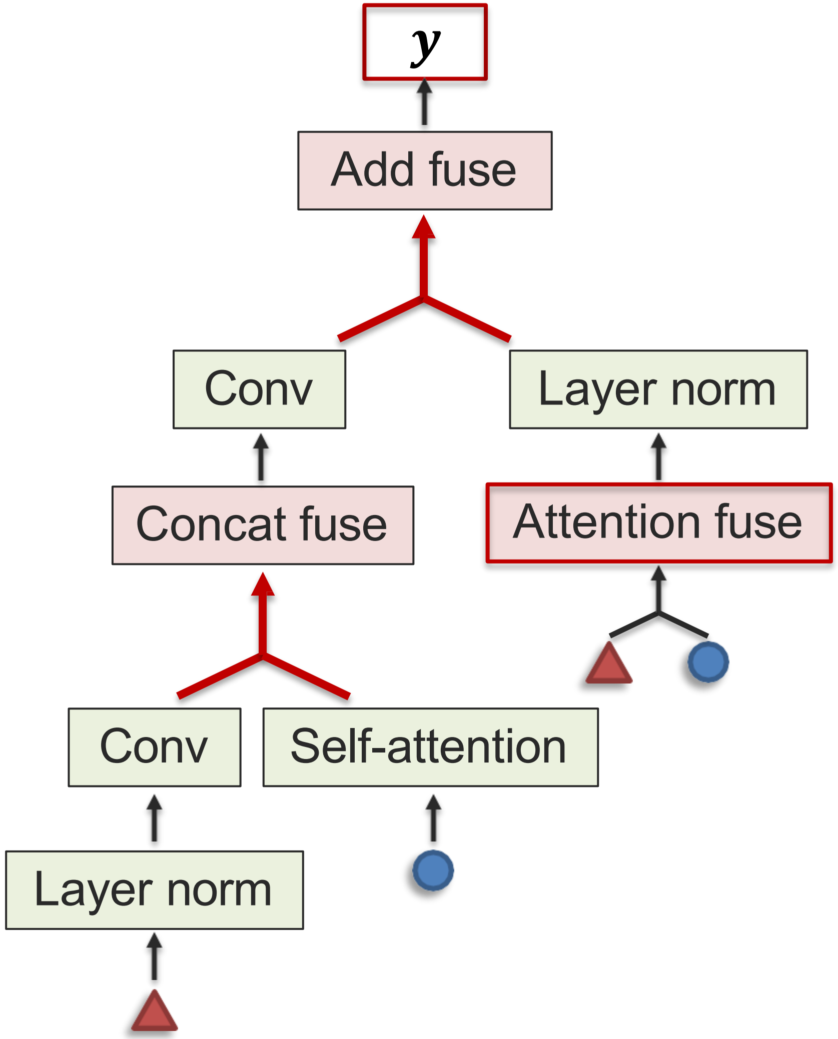
1. Define basic representation building blocks

ReLU | Layer norm | Conv | Self-attention

2. Define basic fusion building blocks

Concat fuse | Attention fuse | Add fuse

3. Automatically search for composition using neural architecture search




[Xu et al., MUFASA: Multimodal Fusion Architecture Search for Electronic Health Records. AAAI 2021]
[Liu et al., DARTS: Differentiable Architecture Search. ICLR 2019]

# Dynamic Early Fusion

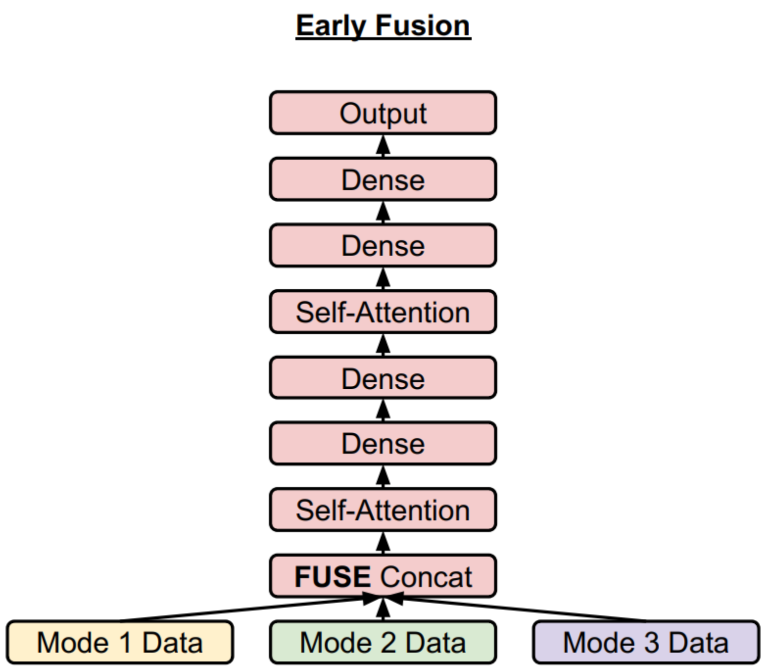



Figure 2: Example of different fusion strategies using a Transformer base architecture. In each depiction, yellow represents modeling only applied to Mode 1, green Mode 2, and purple Mode 3. Red represents modeling applied to all three modes jointly.

[Xue and Marculescu, Dynamic Multimodal Fusion, arxiv 2022]

# Dynamic Early Fusion

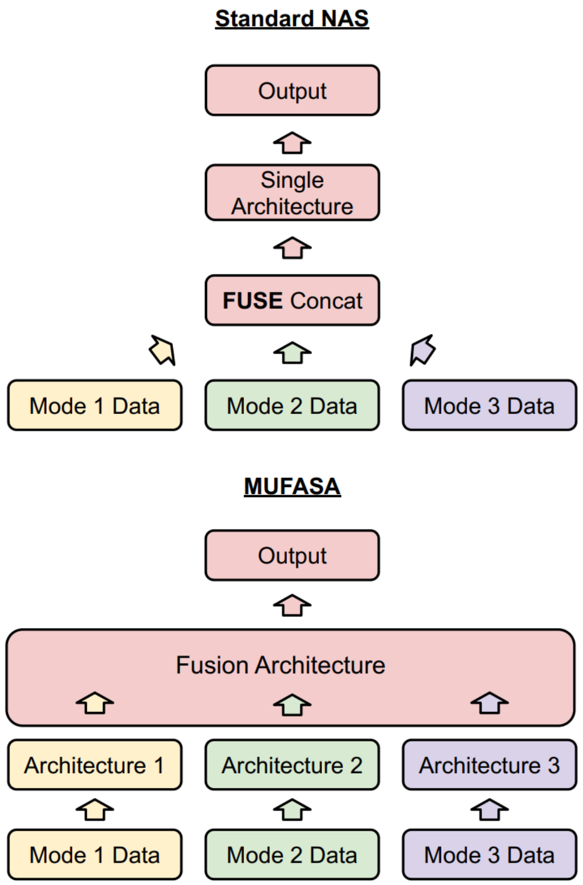



Figure 3: Standard NAS searches for a single unimodal architecture. MUFASA searches for independent architectures for each data modality, as well as a Fusion Architecture to tie those modal architectures together.

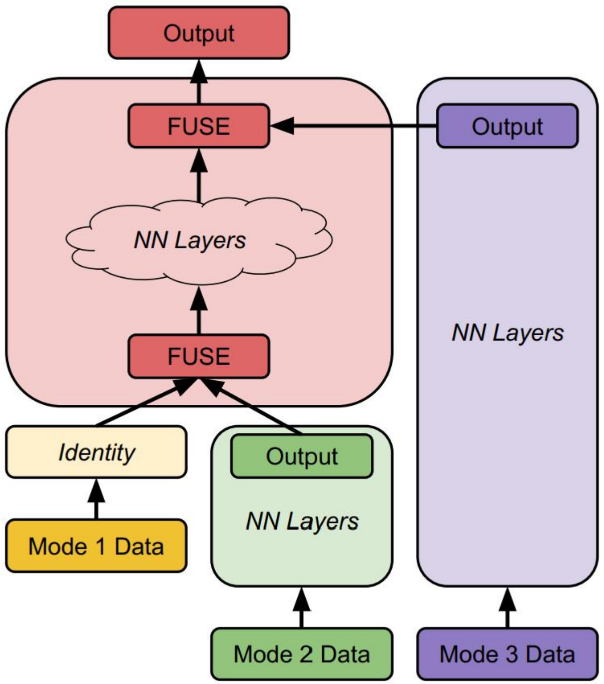


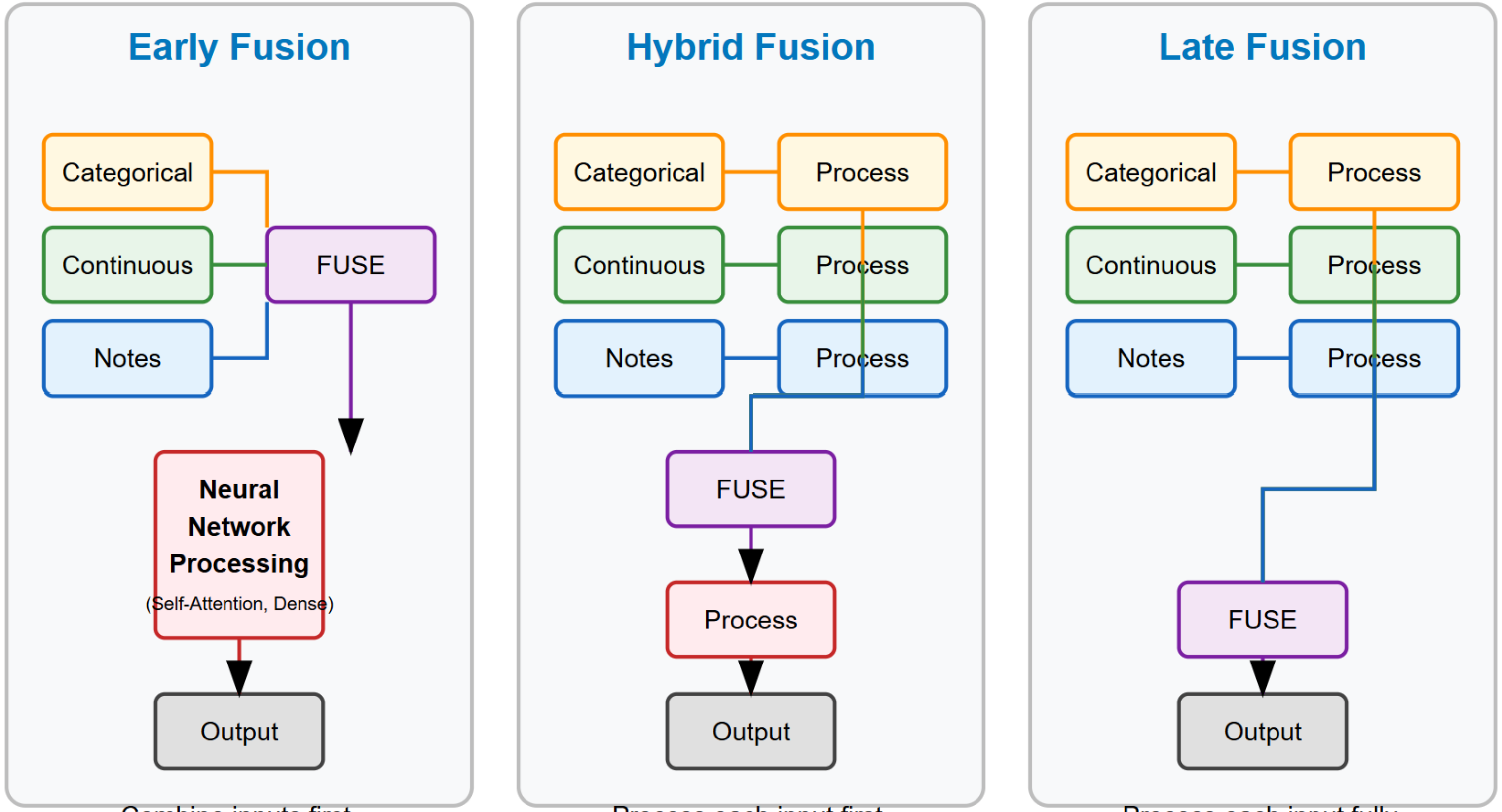

Figure 4: MUFASA example architecture that employs all three fusion types. Mode 1 utilizes early fusion, as its modality architecture is an identity transformation and thus it is fused with Mode 2 before it passes through any neural network (NN) layers. Mode 2 utilizes hybrid fusion, as it first passes through its own non-identity modality architecture before being fused with Mode 1 and transformed by the fusion architecture (in red). Mode 3 utilizes late fusion, as it only receives processing through its independent modality architecture, before being fused with the final output.



[Xue and Marculescu, Dynamic Multimodal Fusion, arxiv 2022]

# Dynamic Early Fusion

**Multimodal Fusion Strategies for EHR Data**




Combine inputs first, then process jointly

Process each input first, then fuse & process jointly

Process each input fully, fuse only at the end

**MUFASA searches for optimal fusion strategy**

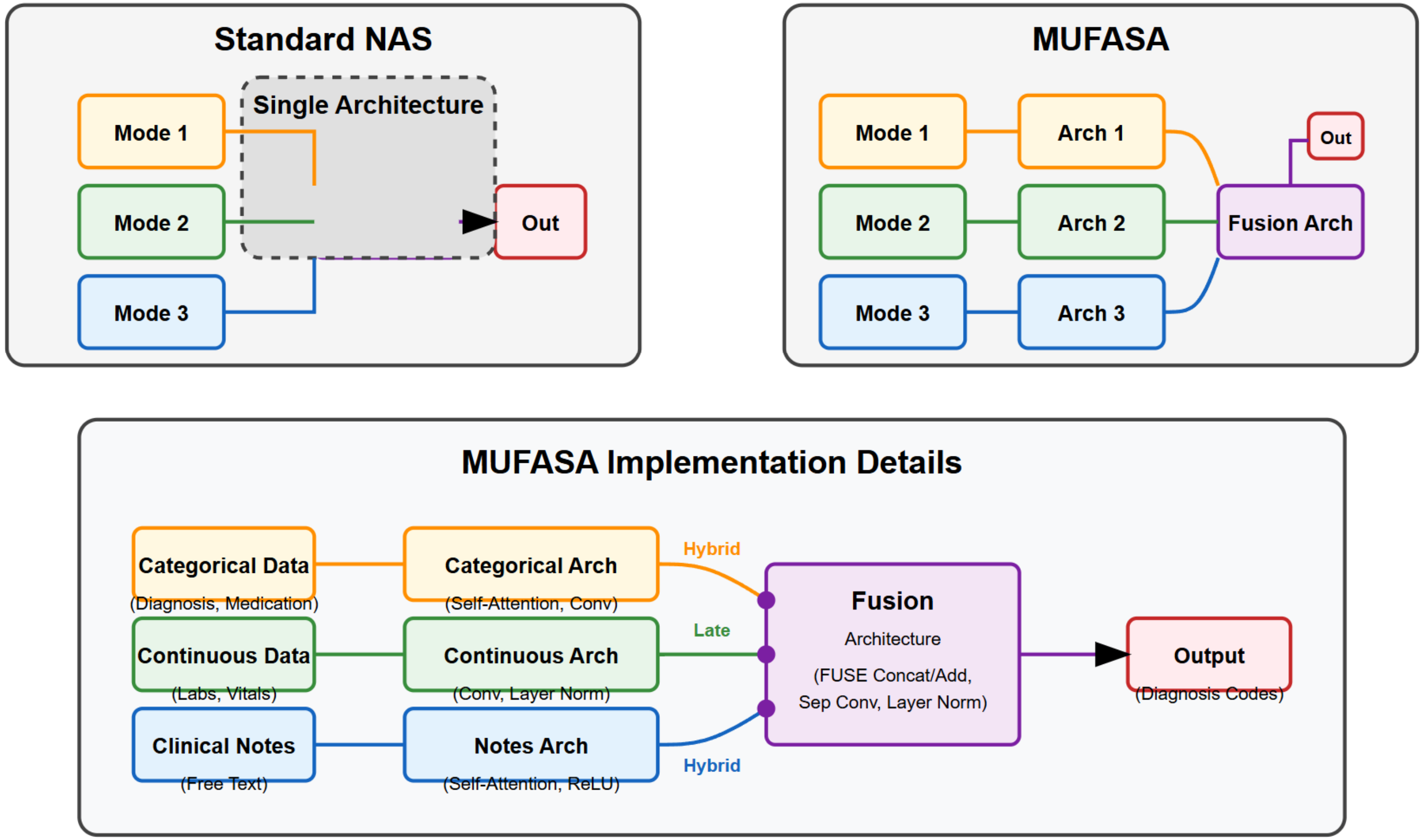


[Xue and Marculescu, Dynamic Multimodal Fusion, arxiv 2022]

# Dynamic Early Fusion

## MUFASA: Multimodal Fusion Architecture Search

Automatically discovering optimal architectures for each modality and fusion strategy

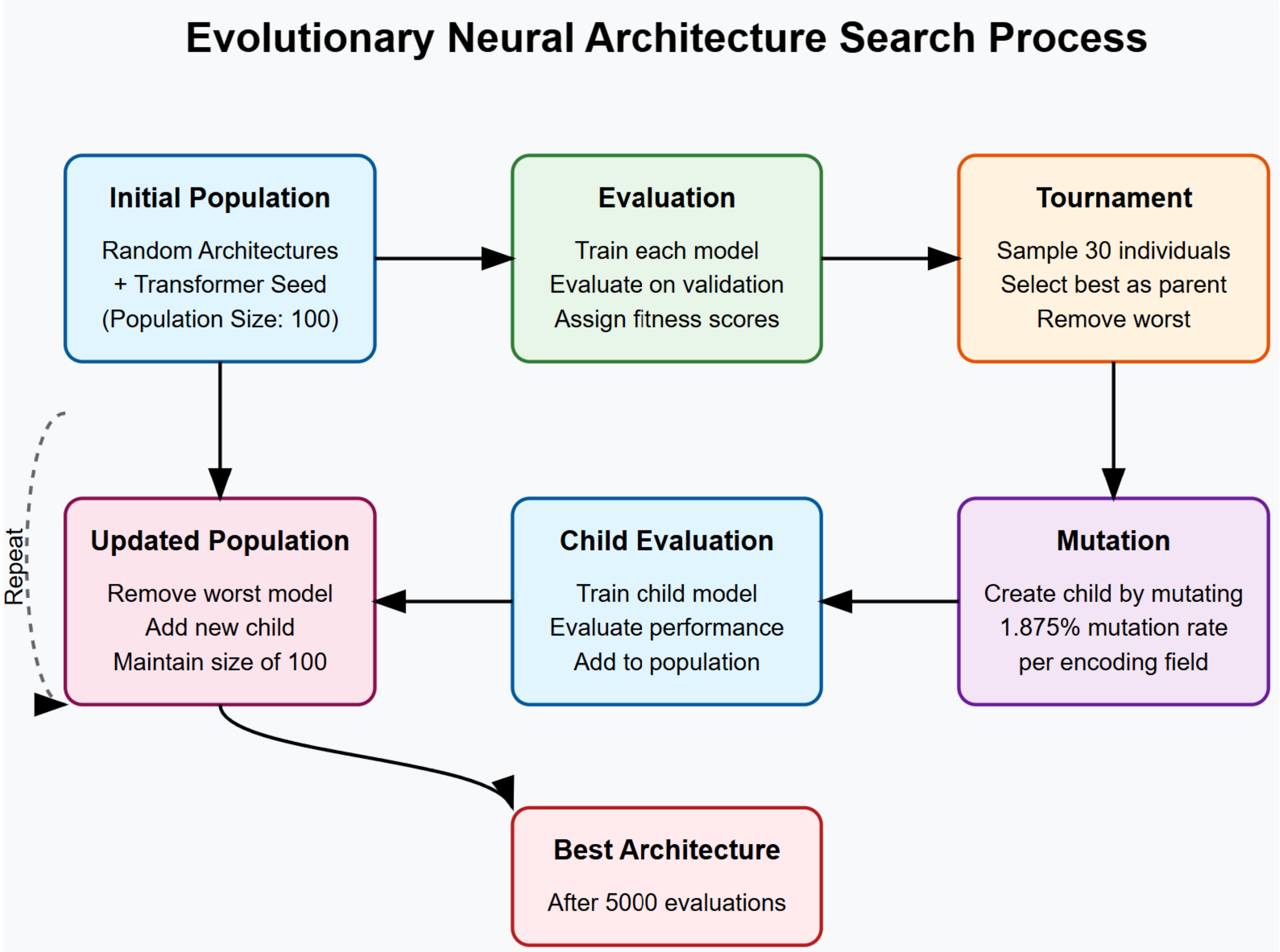



**Key Benefits: Customized modeling per modality + Optimal fusion strategy discovery**



[Xue and Marculescu, Dynamic Multimodal Fusion, arxiv 2022]

# Dynamic Early Fusion



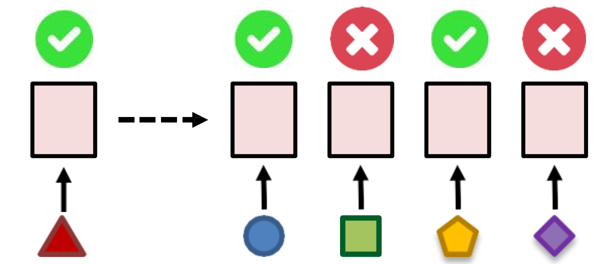



[Xue and Marculescu, Dynamic Multimodal Fusion, arxiv 2022]

# Heterogeneity-aware Fusion

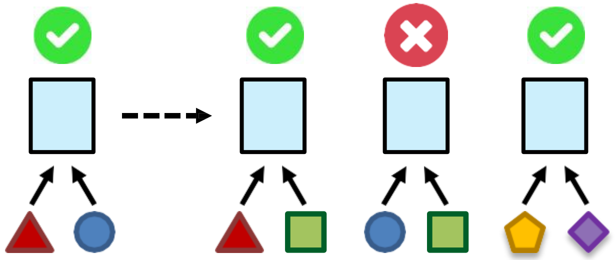
## Information transfer, transfer learning perspective

**1a. Estimate modality heterogeneity via transfer**

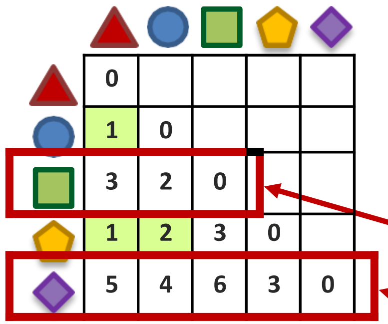


**(Implicitly captures heterogeneity)**

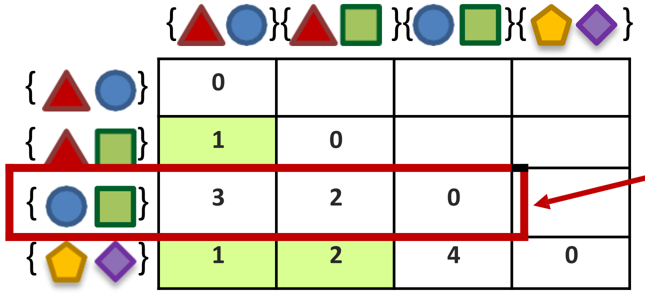
**1b. Estimate interaction heterogeneity via transfer**



**2a. Compute modality heterogeneity matrix**




**2b. Compute interaction heterogeneity matrix**




**3. Determine parameter clustering**

$$\mathbb{U}_1 = \{U_1, U_2, U_4\}$$
$$\mathbb{U}_2 = \{U_3\}$$
$$\mathbb{U}_3 = \{U_5\}$$

$$\mathbb{C}_1 = \{C_{12}, C_{13}, C_{45}\}$$
$$\mathbb{C}_2 = \{C_{23}\}$$

[Zamir et al., Taskonomy: Disentangling Task Transfer Learning. CVPR 2018]



[Liang et al., HighMMT: Quantifying Modality & Interaction Heterogeneity for High-Modality Learning. TMLR 2022]

# Dynamic Early Fusion

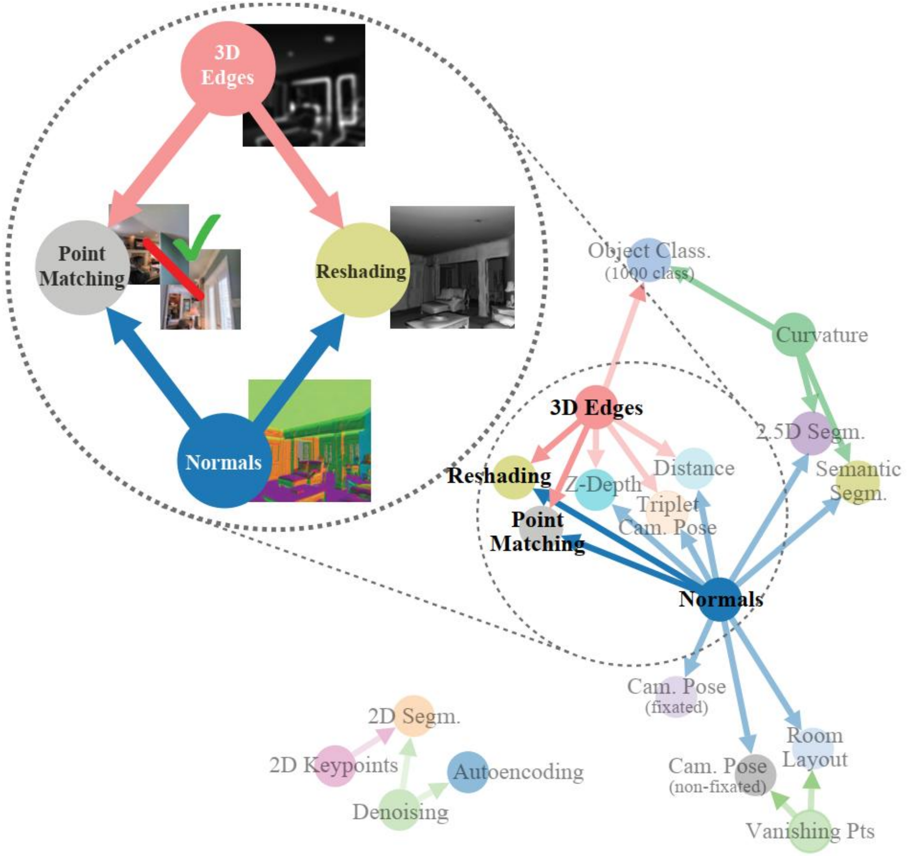



Figure 1: **A sample task structure discovered by the computational task taxonomy (*taskonomy*).** It found that, for instance, by combining the learned features of a surface normal estimator and occlusion edge detector, good networks for reshading and point matching can be rapidly trained with little labeled data.



[Zamir et al., Taskonomy: Disentangling Task Transfer Learning. CVPR 2018]

# Dynamic Early Fusion

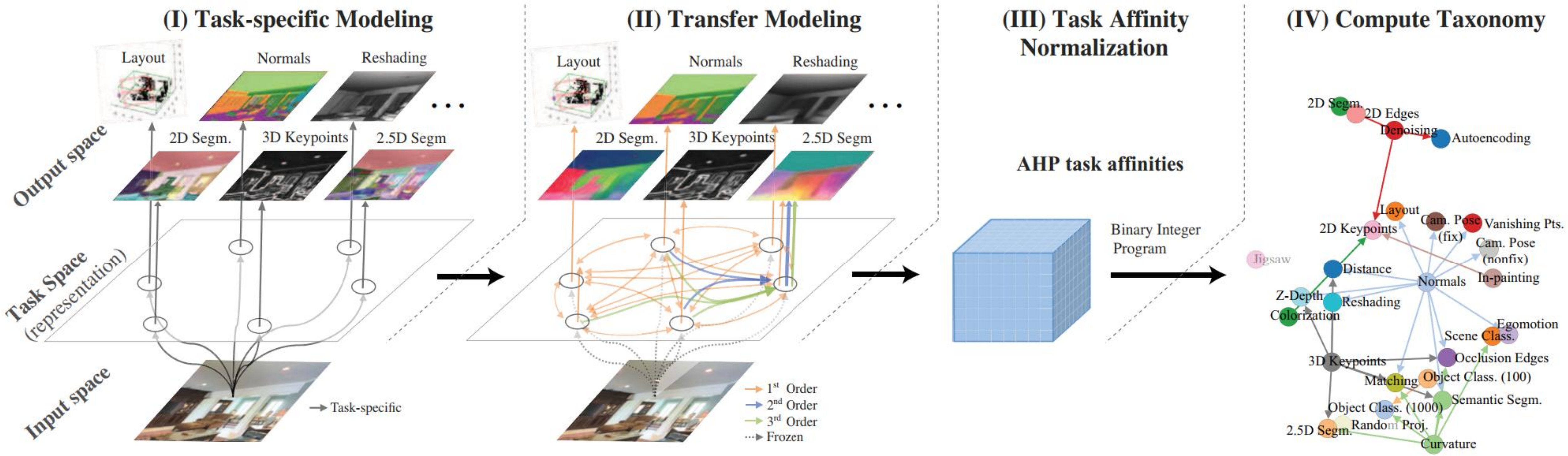



Figure 2: **Computational modeling of task relations and creating the taxonomy.** From left to right: I. Train task-specific networks. II. Train (first order and higher) transfer functions among tasks in a latent space. III. Get normalized transfer affinities using AHP (Analytic Hierarchy Process). IV. Find global transfer taxonomy using BIP (Binary Integer Program).



[Zamir et al., Taskonomy: Disentangling Task Transfer Learning. CVPR 2018]

# Dynamic Early Fusion

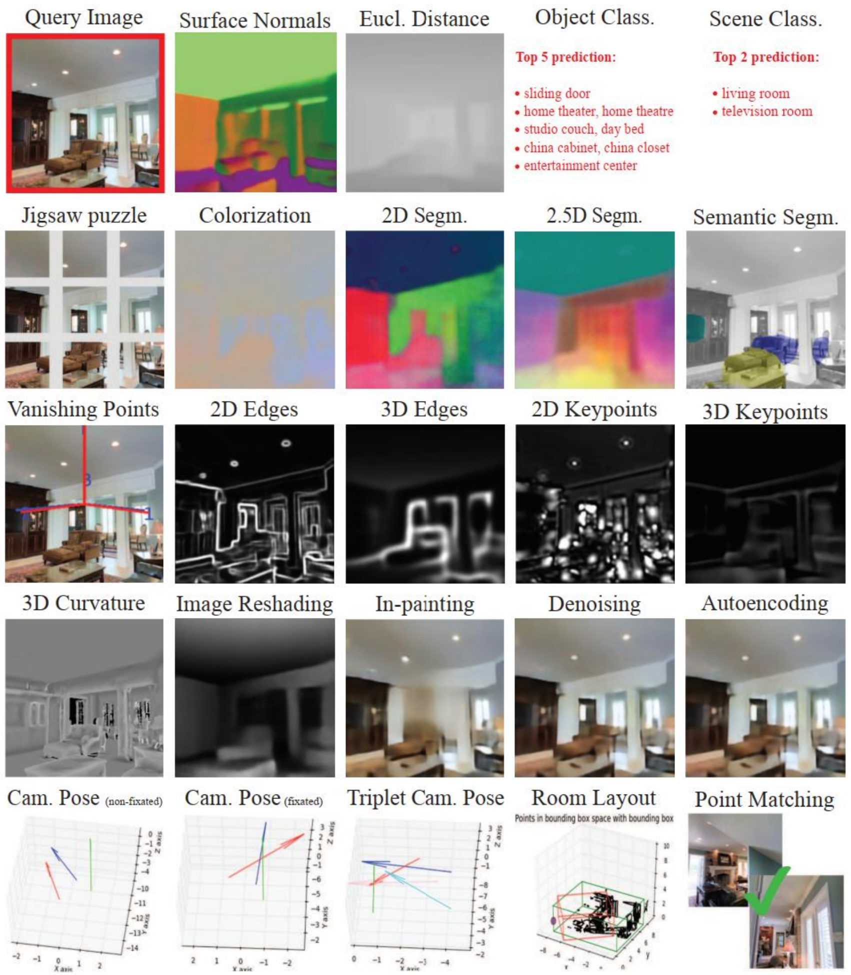



Figure 3: **Task Dictionary.** Outputs of 24 (of 26) task-specific networks for a query (top left). See results of applying frame-wise on a video here.

[Zamir et al., Taskonomy: Disentangling Task Transfer Learning. CVPR 2018]

# Dynamic Early Fusion

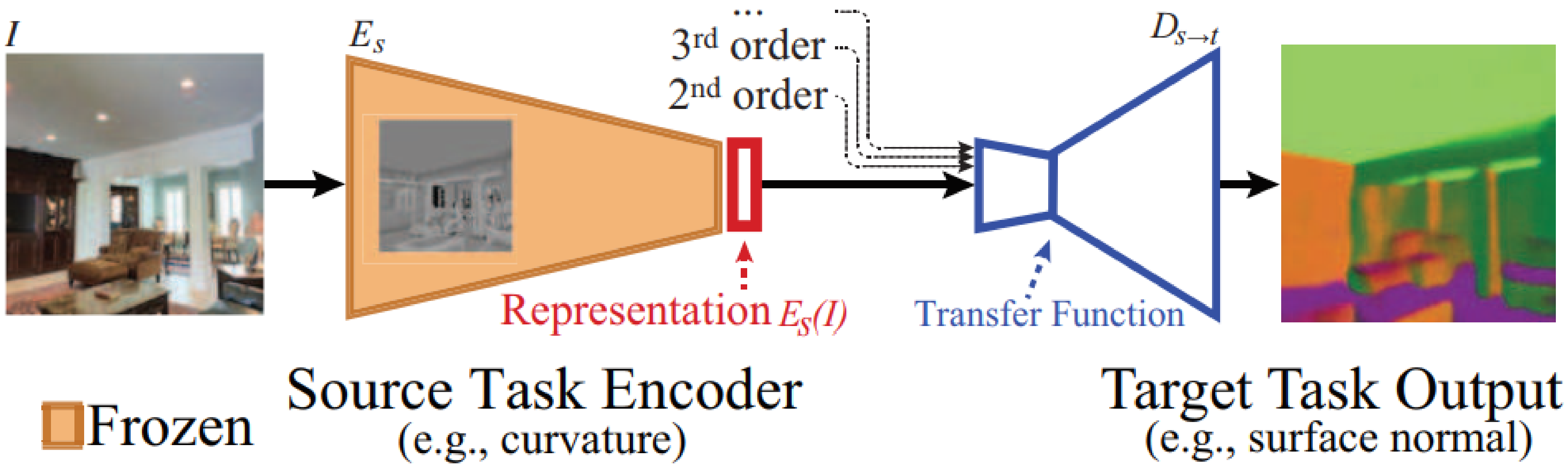



Figure 4: **Transfer Function.** A small readout function is trained to map representations of source task's frozen encoder to target task's labels. If order$>1$, transfer function receives representations from multiple sources.



[Zamir et al., Taskonomy: Disentangling Task Transfer Learning. CVPR 2018]

# Dynamic Early Fusion

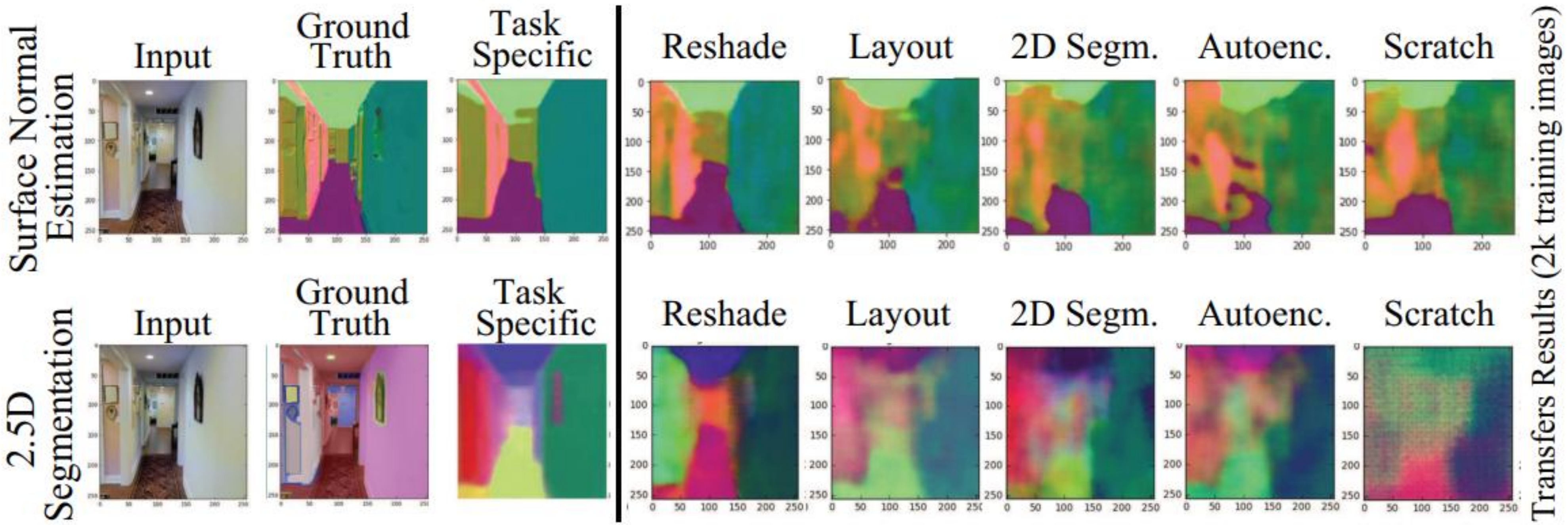



Figure 5: **Transfer results** to normals and 2.5D Segmentation from 5 different source tasks. The spread in transferability among sources is apparent. "Scratch" was trained from scratch without transfer learning.

[Zamir et al., Taskonomy: Disentangling Task Transfer Learning. CVPR 2018]

# Dynamic Early Fusion

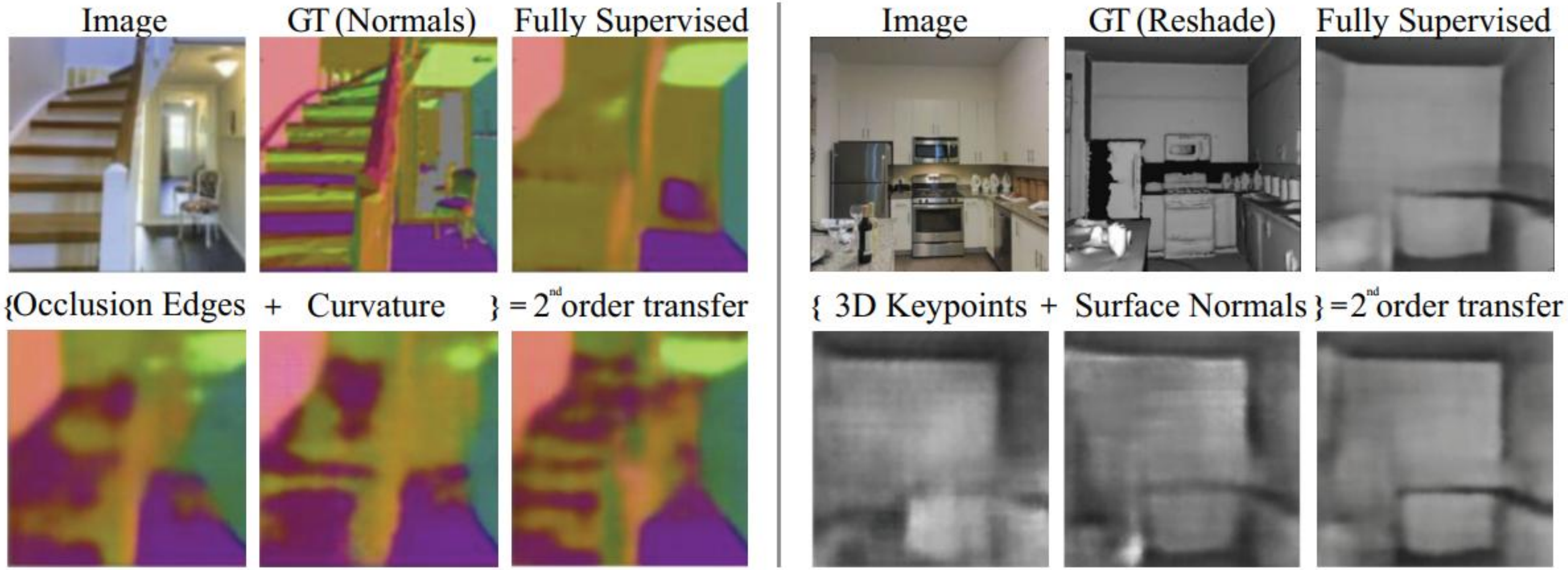



Figure 6: **Higher-Order Transfers.** Representations can contain complementary information. E.g. by transferring simultaneously from 3D Edges and Curvature individual stairs were brought out. See our publicly available interactive transfer visualization page for more examples.

[Zamir et al., Taskonomy: Disentangling Task Transfer Learning. CVPR 2018]

# Dynamic Early Fusion

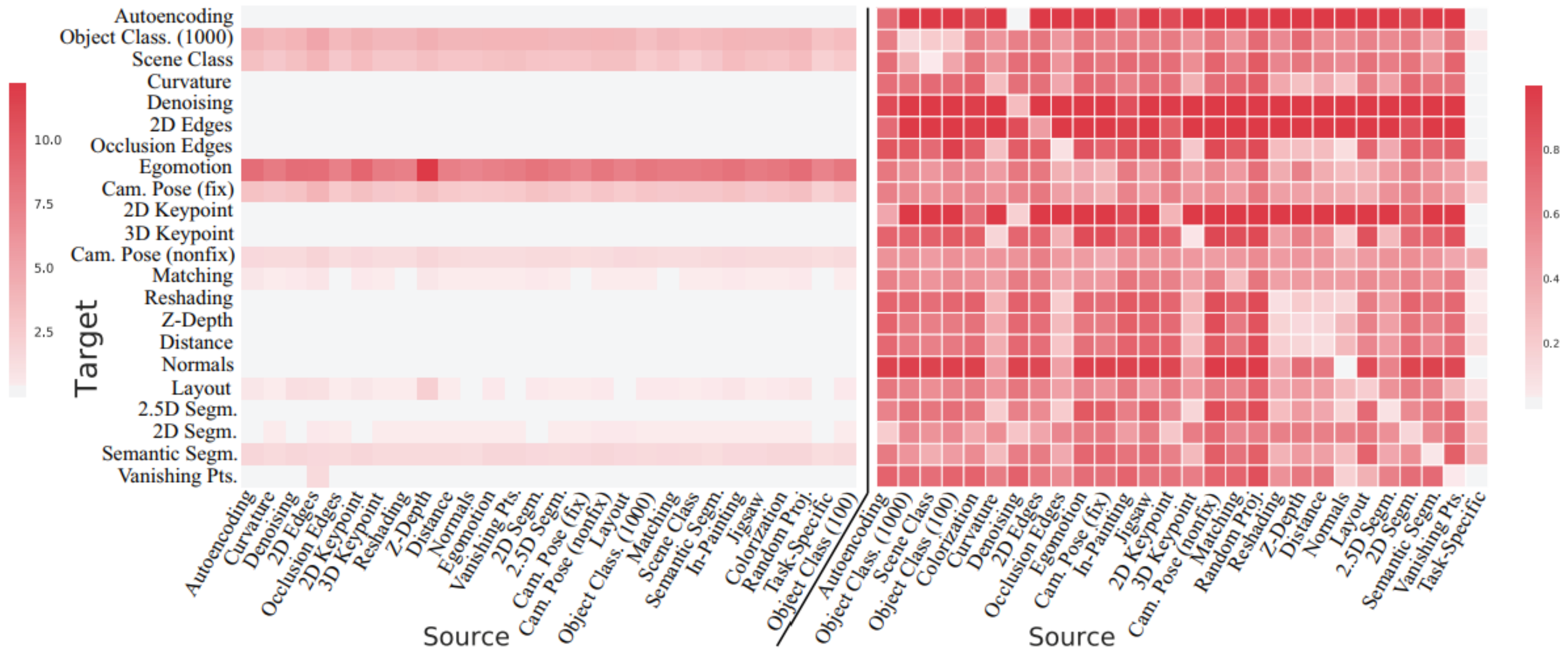



Figure 7: **First-order task affinity matrix** before (left) and after (right) Analytic Hierarchy Process (AHP) normalization. Lower means better transfered. For visualization, we use standard affinity-distance method $dist = e^{-\beta \cdot P}$ (where $\beta = 20$ and $e$ is element-wise matrix exponential). See supplementary material for the full matrix with higher-order transfers.

[Zamir et al., Taskonomy: Disentangling Task Transfer Learning. CVPR 2018]

# Dynamic Early Fusion

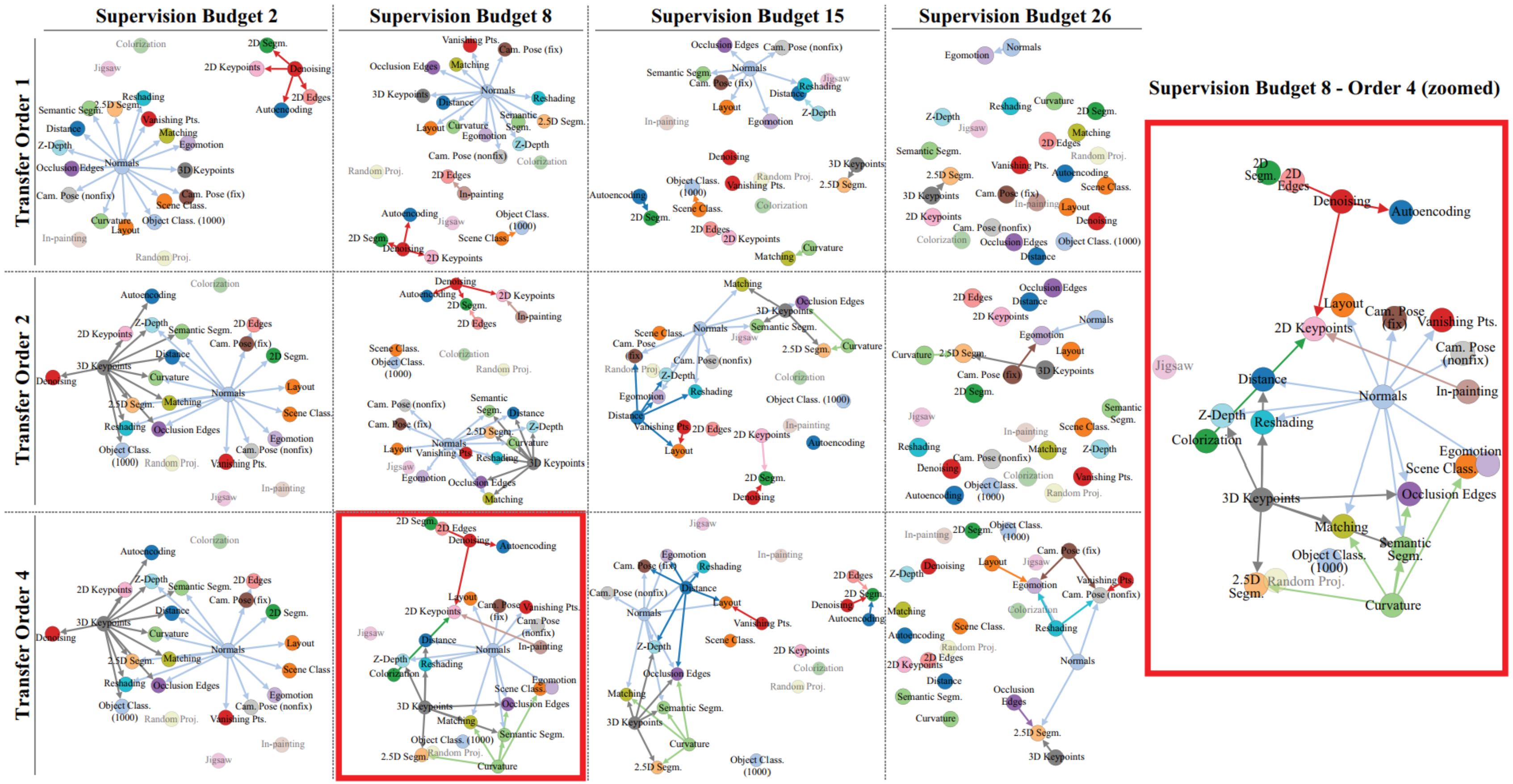



Figure 8: **Computed taxonomies** for solving 22 tasks given various supervision budgets (x-axes), and maximum allowed transfer orders (y-axes). One is magnified for better visibility. Nodes with incoming edges are target tasks, and the number of their incoming edges is the order of their chosen transfer function. Still transferring to some targets when tge budget is 26 (full budget) means certain transfers started performing better than their fully supervised task-specific counterpart. See the interactive solver website for color coding of the nodes by *Gain* and *Quality* metrics. Dimmed nodes are the source-only tasks, and thus, only participate in the taxonomy if found worthwhile by the BIP optimization to be one of the sources.

[Zamir et al., Taskonomy: Disentangling Task Transfer Learning. CVPR 2018]

# Dynamic Early Fusion

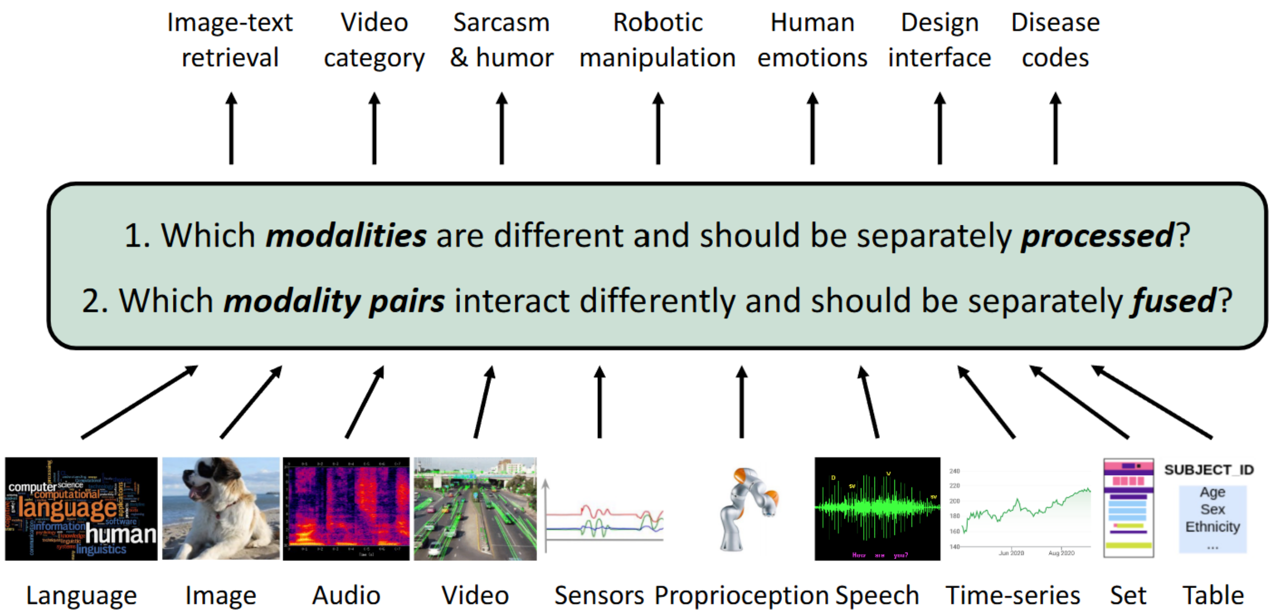
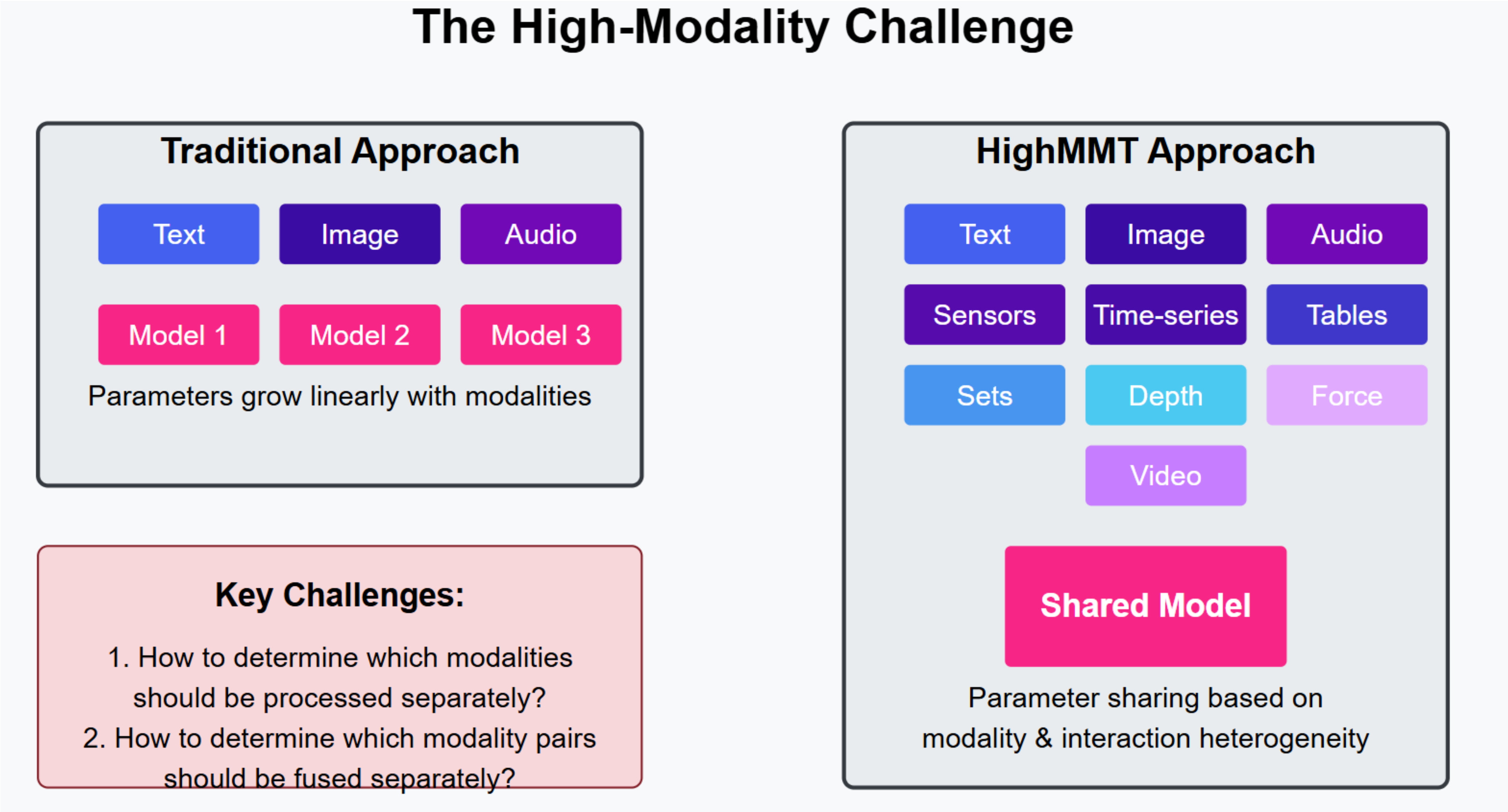



Figure 1: **Heterogeneity quantification**: Efficiently learning from many modalities requires measuring (1) *modality heterogeneity*: which modalities are different and should be separately processed, and (2) *interaction heterogeneity*: which modality pairs interact differently and should be separately fused. HighMMT uses these measurements to dynamically group parameters balancing performance and efficiency.



[Liang et al., HighMMT: Quantifying Modality & Interaction Heterogeneity for High-Modality Learning. TMLR 2022]

# HighMMT




[Liang et al. HighMMT: Quantifying Modality & Interaction Heterogeneity for High-Modality Learning. TMLR 2022]

# Dynamic Early Fusion

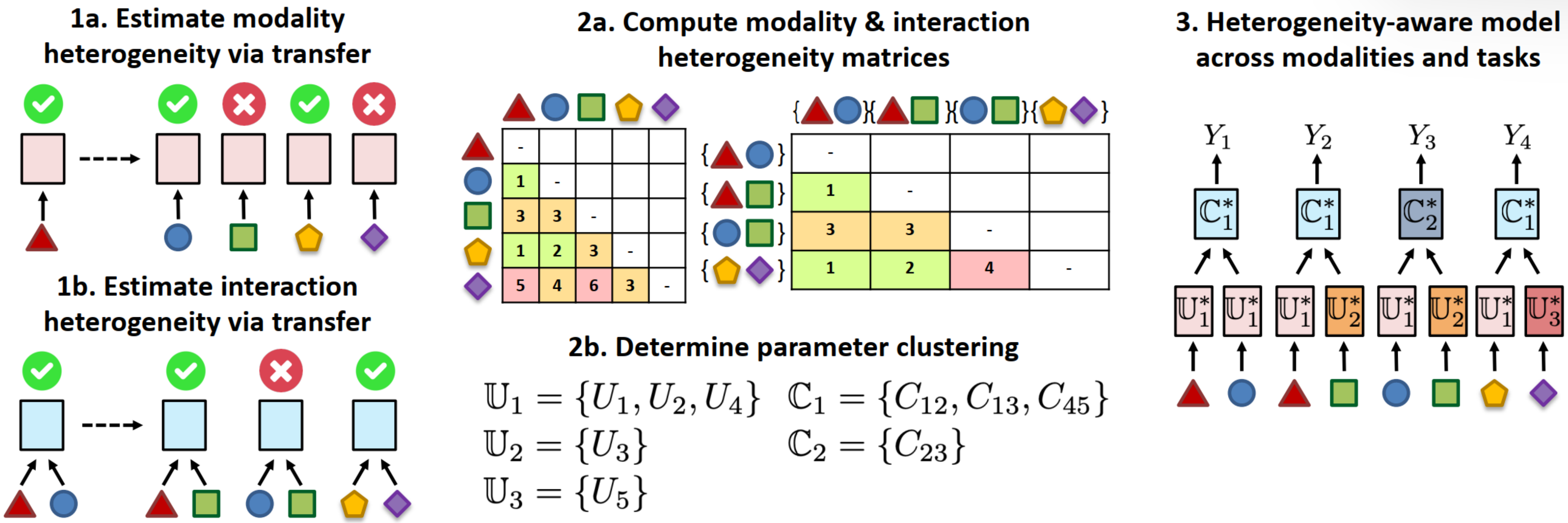



Figure 2: **HighMMT workflow**: (1) We estimate modality and interaction heterogeneity via modality transfer to determine which modalities should be processed and fused differently. (2) Using the inferred heterogeneity, we determine the optimal grouping of parameters balancing both total performance and parameter efficiency, which (3) informs our design of a heterogeneity-aware model with dynamic parameter sharing across many modalities and tasks. HighMMT enables statistical strength sharing, efficiency, and generalization to new modalities and tasks.



[Liang et al., HighMMT: Quantifying Modality & Interaction Heterogeneity for High-Modality Learning. TMLR 2022]

# Dynamic Early Fusion

- **Modality Transfer Loss**:

$$T\left(X_1 \rightarrow X_2 ; Y\right)=\mathcal{L}_{1 \rightarrow 2}^*-\mathcal{L}_2^*$$

This equation measures the difficulty of transferring a model trained on modality $X_1$ to modality $X_2$ for task $Y$. $\mathcal{L}_{1 \rightarrow 2}^*$ is the loss after fine-tuning a model initialized with parameters from $X_1$ on $X_2$, while $\mathcal{L}_2^*$ is the baseline loss of a model trained directly on $X_2$.

- **Modality Heterogeneity Measure**:

$$d\left(X_1 ; X_2\right)=T\left(X_1 \rightarrow X_2 ; Y\right)_{\geq 0}+T\left(X_2 \rightarrow X_1 ; Y\right)_{\geq 0}$$

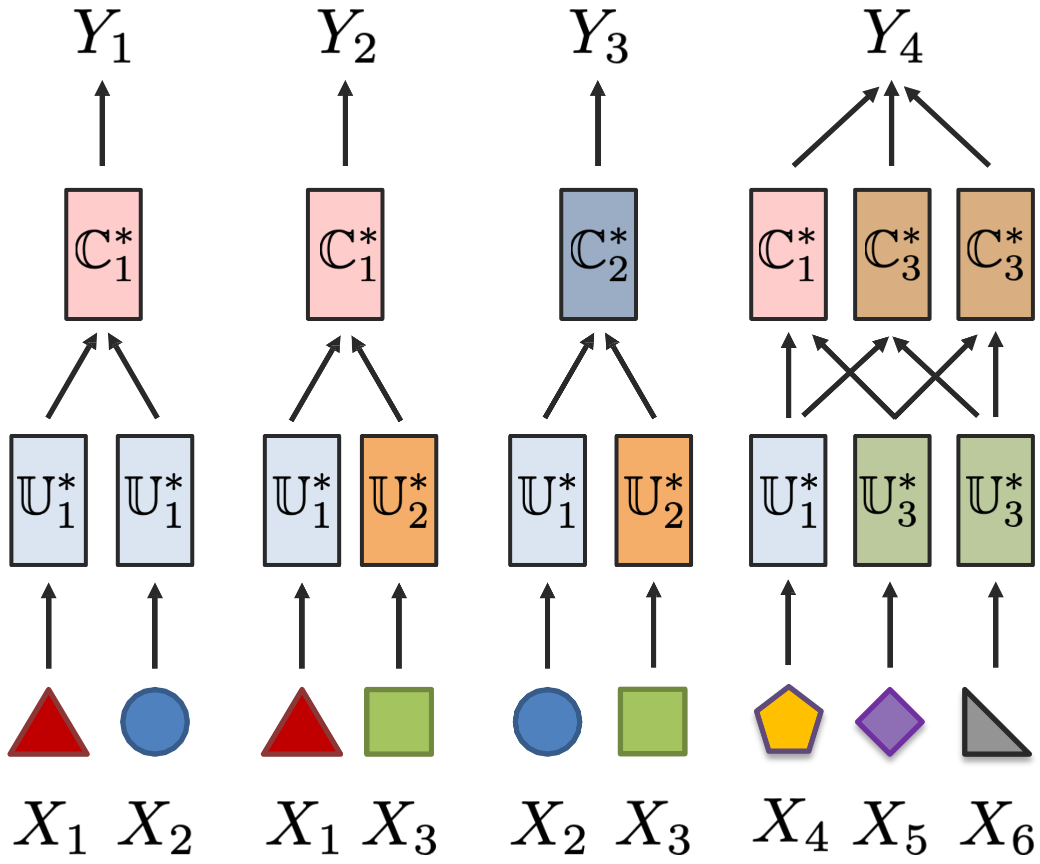
This equation aggregates the non-negative transfer difficulties in both directions between two modalities to quantify their heterogeneity.



[Liang et al., HighMMT: Quantifying Modality & Interaction Heterogeneity for High-Modality Learning. TMLR 2022]

# Heterogeneity-aware Fusion

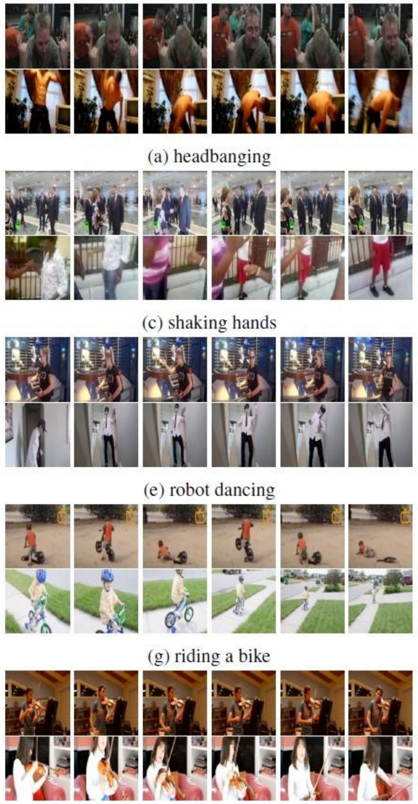
## Information transfer, transfer learning perspective




[Liang et al. High-MMT: Quantifying Modality & Interaction Heterogeneity for High-Modality Learning. TMLR 2022]

# Improving Optimization

Kinetics dataset




**Adding more modalities should always help?**

Modalities: **RGB** (video clips)
**A** (Audio features)
**OF** (optical flow - motion)

| Dataset | Multi-modal | V@1 | Best Uni | V@1 | Drop |
|---|---|---|---|---|---|
| Kinetics | A + RGB | 71.4 | RGB | **72.6** | -1.2 |
| | RGB + OF | 71.3 | RGB | **72.6** | -1.3 |
| | A + OF | 58.3 | OF | **62.1** | -3.8 |
| | A + RGB + OF | 70.0 | RGB | **72.6** | -2.6 |

But sometimes multimodal doesn't help! **Why?**

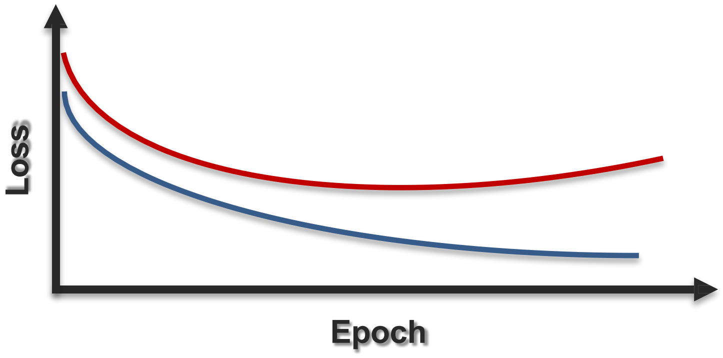
[Wang et al., What Makes Training Multi-modal Classification Networks Hard? CVPR 2020]
[Wu et al., Characterizing and Overcoming the Greedy Nature of Learning in Multi-modal Deep Neural Networks. ICML 2022]

# Improving Optimization

## Relevance heterogeneity

2 explanations for drop in performance:

1. Multimodal networks are more prone to overfitting due to **increased complexity**
2. Different modalities overfit and generalize at **different rates**




**Key idea 1:** compute overfitting-to-generalization ratio (OGR)

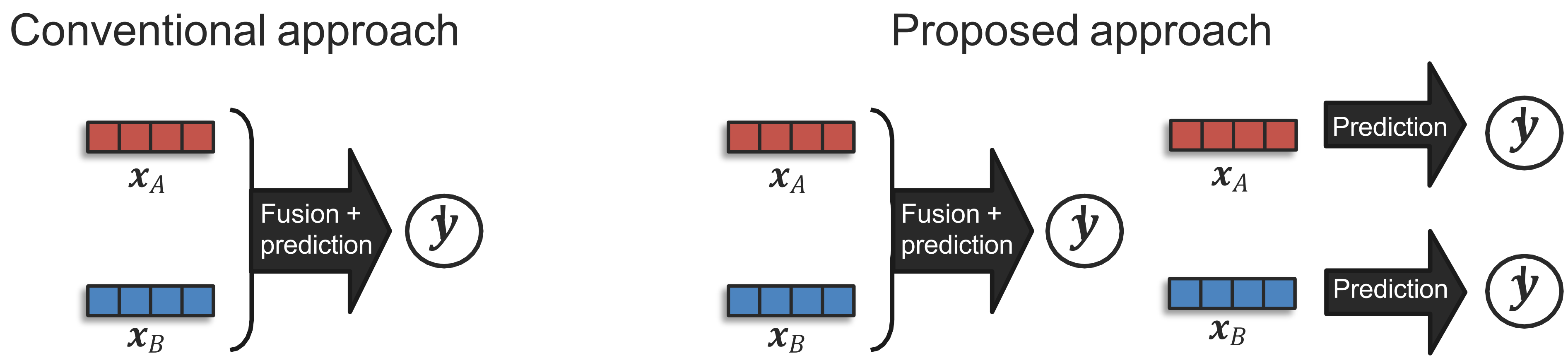
Gap between training and valid loss

OGR wrt each modality tells us how much to train that modality

[Wang et al., What Makes Training Multi-modal Classification Networks Hard? CVPR 2020]
[Wu et al., Characterizing and Overcoming the Greedy Nature of Learning in Multi-modal Deep Neural Networks. ICML 2022]

# Improving Optimization

**Relevance heterogeneity**




**Key idea 2:** Simultaneously train unimodal networks to estimate OGR wrt each modality

- Reweight multimodal loss using unimodal OGR values

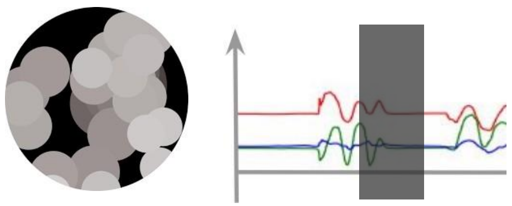
→ Allows to better balance generalization & overfitting rate of different modalities

[Wang et al., What Makes Training Multi-modal Classification Networks Hard? CVPR 2020]
[Wu et al., Characterizing and Overcoming the Greedy Nature of Learning in Multi-modal Deep Neural Networks. ICML 2022]

# Heterogeneity in Noise: Studying Robustness

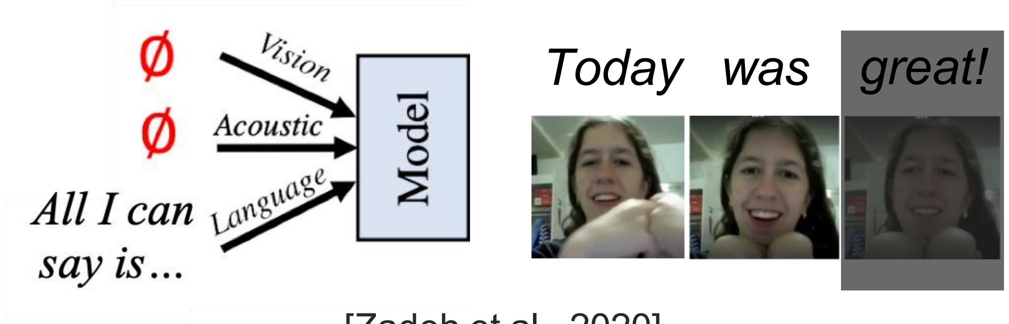
## Noise within Modality

noise ⟶ nosie



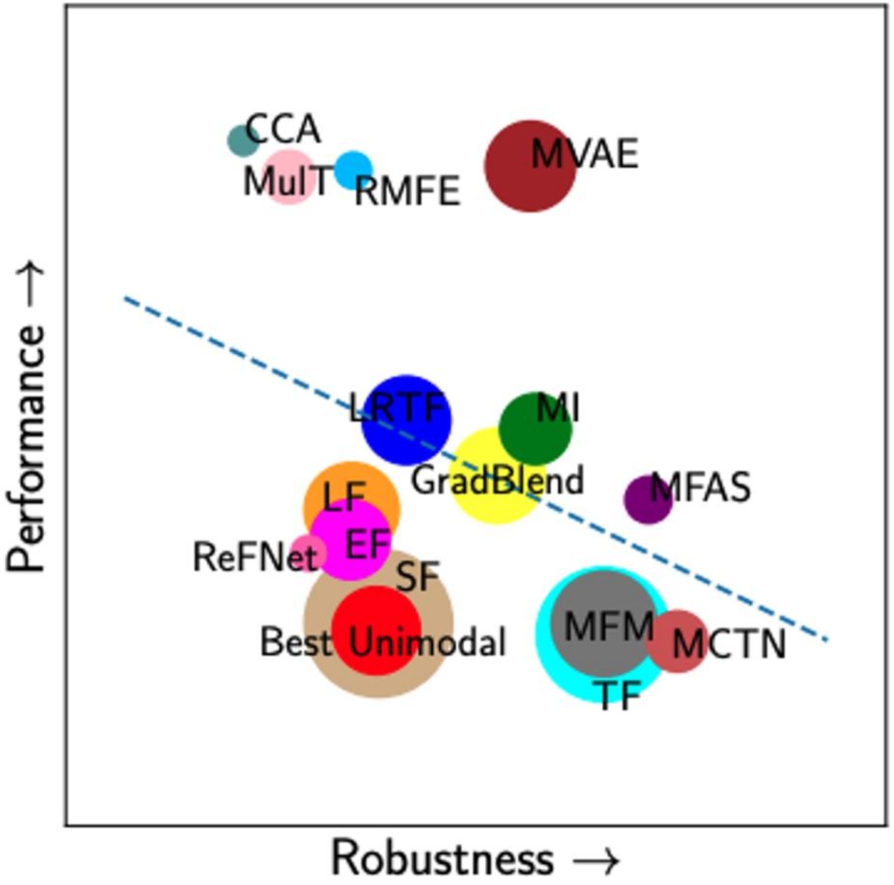
[Belinkov & Bisk, 2018; Subramaniam et al., 2009; Boyat & Joshi, 2015]

## Missing Modalities




[Zadeh et al., 2020]

## Strong tradeoffs between performance and robustness




rate of accuracy drops

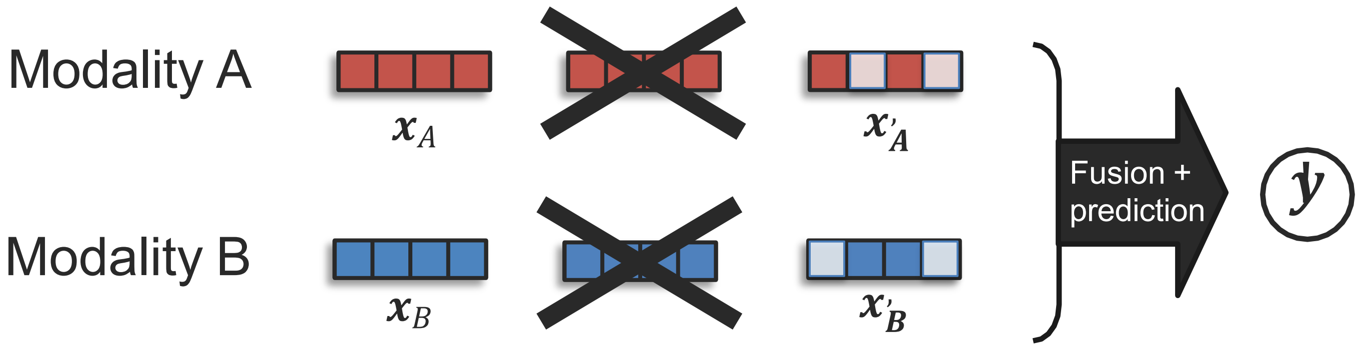


[Liang et al., MultiBench: Multiscale Benchmarks for Multimodal Representation Learning. NeurIPS 2021]

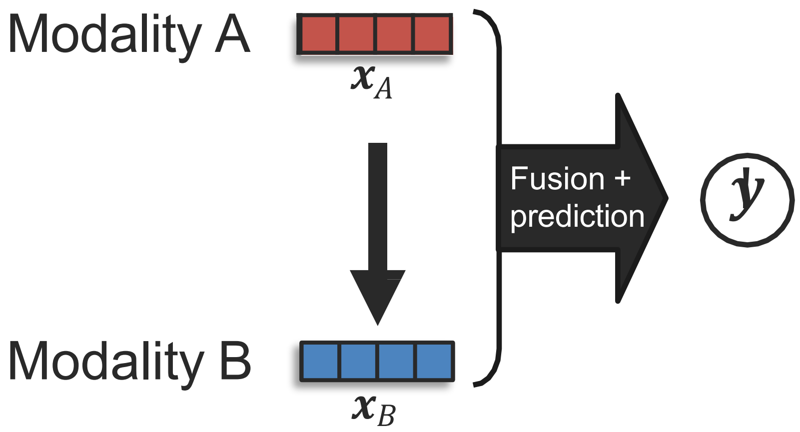
# Heterogeneity in Noise: Studying Robustness

**Several approaches towards more robust models**

Robust data + training



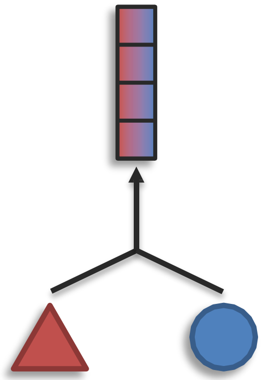

Infer missing modalities



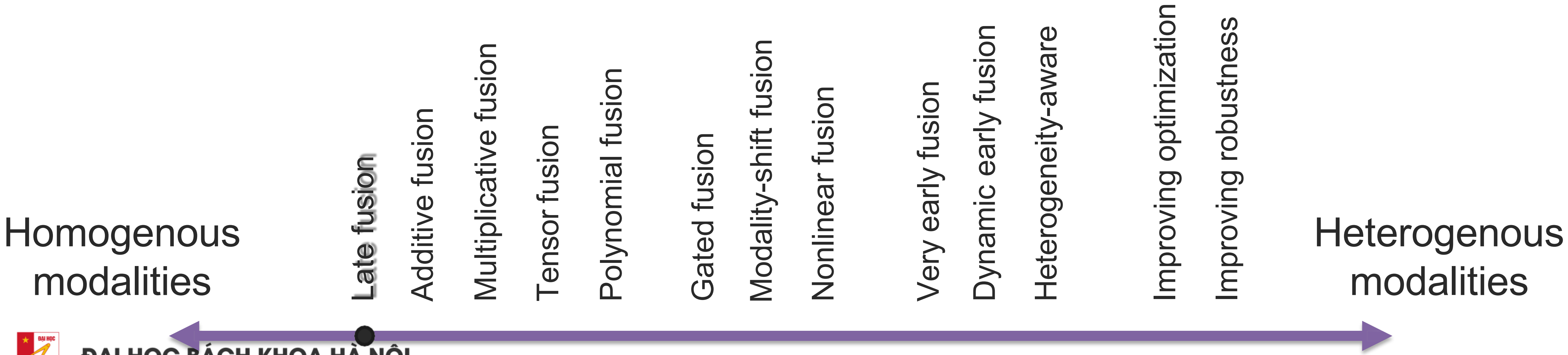

Translation model
Joint probabilistic model

[Ngiam et al., Multimodal Deep Learning. ICML 2011]
[Srivastava and Salakhutdinov, Multimodal Learning with Deep Boltzmann Machines. JMLR 2014]
[Tran et al., Missing Modalities Imputation via Cascaded Residual Autoencoder. CVPR 2017]
[Pham et al., Found in Translation: Learning Robust Joint Representations by Cyclic Translations Between Modalities. AAAI 2019

# Sub-Challenge 1a: Representation Fusion



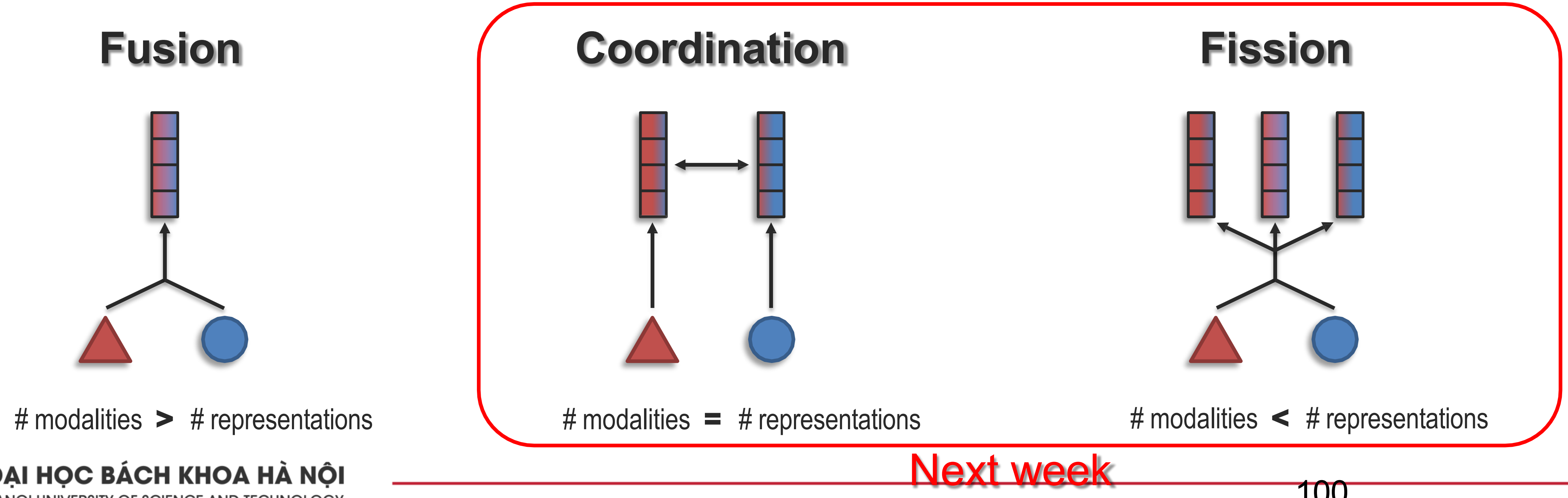
**Definition:** Learn a joint representation that models cross-modal interactions between individual elements of different modalities



# Challenge 1: Representation

**Definition:** Learning representations that reflect cross-modal interactions between individual elements, across different modalities

## Sub-challenges:



# References

- Multi Modal Machine Learning, 11-777 • Fall 2023 • Carnegie Mellon University